

LABORATORY SAMPLE: The sample or sub sample(s) sent to or received by the laboratory. When the laboratory sample is further prepared (reduced) by subdividing, mixing, grinding or by combinations of these operations, the result is called the **test sample**. When no preparation of the laboratory sample is required, the laboratory sample is the test sample. A test portion is removed from the test sample for the performance of the test or for analysis. The laboratory sample is the final sample from the point of view of sample collection but it is the initial sample from the point of view of the laboratory. Several laboratory samples may be prepared and sent to different laboratories or to the same laboratory for different purposes. When sent to the same laboratory, the set is generally considered as a single laboratory sample and is documented as a single sample.

LABRUM: A lip or liplike structure forming the upper lip of some arthropods. It is formed from a plate of cuticle hinged to the head above the mouth and used for biting and chewing.

LABYRINTH: The system of cavities and tubes including the semicircular canals, vestibule and cochlea that comprises the inner ear of vertebrates. It is housed within the temporal bone that is involved in hearing and balance.

LAC OPERON: The DNA sequence in the genome of bacterium *Escherichia coli* (*E. coli*) that regulates lactose metabolism. It explains the control of β -galactosidase synthesis and it is used as a model for the structure of all other operons and regulation of genes in prokaryotes.

LACCOLITH: A dome-like intrusion that has been injected between two layers of sedimentary rock. It is formed by the intrusion of magma, which forces the overlying strata into the shape of a dome.

LACERATE: Having edges that are jagged or torn, as seen in some leaves.

LACERTILIA (SAURIA): A sub-order of the order Squamata (class Reptilia) comprising of the lizards. Lizards usually have well-developed limbs and the nearly 3,800 known species are found mainly in tropical and subtropical regions.



Lacerate leaf

LACHRYMATOR: A substance such as acetyl chloride and acrylaldehyde that produces a flow of tears in a person or animal.

LACINIATE: To divide into deep, narrow, irregular segments or lobes, as seen in some leaves.

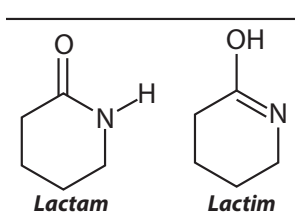


Laciniate leaf

LACRIMAL GLAND (LACHRYMAL GLAND): The gland, present in the eyelids of some vertebrates. It is situated in the upper, outer portion of each orbit. It secretes the aqueous layer of the tear film. The fluid produced by this gland cleanses and lubricates the exposed surface of the eye and drains into the nose through the lacrimal duct.

LACTAMIDE (2-HYDROXYPROPIONAMIDE; LACTIC ACID AMIDE; LACTIC AMIDE; A-HYDROXYPROPIONAMIDE): An amide, which is a white crystalline solid derived from lactic acid. Its molecular formula is $C_3H_5NO_2$, molar mass 89.09 g mol^{-1} and has a melting point of $73-76^\circ\text{C}$.

LACTAMS: Any of a group of cyclic amides containing an amide group $-\text{NHCO}-$ as part of a ring. They are derived from aminocarboxylic acids by the intramolecular elimination of water from the amino and carboxylic groups. Lactams can be formed by the reaction of an $-\text{NH}_2$ group in one part of a molecule with a $-\text{COOH}$ group in the other to give a cyclic



Lactam

Lactim

amide. They can exist in an alternative tautomeric form, the lactim form, in which the hydrogen atom on the nitrogen has migrated to the oxygen of the carbonyl to give $-\text{H}=\text{C}(\text{OH})-$. The pyrimidine base uracil is an example of a lactam.

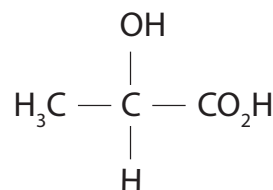
LACTASE: An enzyme secreted in the small intestine in certain yeasts and in the intestinal juices of mammals which breaks down lactose (milk sugar) to glucose and galactose by catalysing the hydrolysis of galactose residues.

LACTATE: 1. Of a woman or female mammal, to produce milk. 2. A salt or ester of lactic acid ($C_3H_5O_3$). It occurs in blood and muscle as a product of glycolysis. Insufficient oxygen during strenuous exercise causes reduction of pyruvic acid to lactate by lactate dehydrogenase, which causes tiredness, sore muscles and muscle cramps. Lactates also increase during fasting.

LACTATION: The secretion of milk from the mammary glands of female mammals after birth. It is stimulated by the sucking action of the new-born offspring and is under the control of the hormones, prolactin and oxytocin.

LACTEAL: 1. Also called **lacteal lymph**, it is a central, blind-ended lymph vessel that occurs in each villus of the small intestine, responsible for absorbing fatty acids and glycerol in the small intestine. 2. Of or relating to any of numerous minute intestinal lymph-carrying vessels that convey chyle from the intestine to lymphatic circulation and thereby to the thoracic duct.

LACTIC ACID (2-HYDROXYPROPANOIC ACID): An alpha hydroxy carboxylic acid, with a sour taste and the chemical formula $C_3H_5O_3$. It is produced from pyruvic acid in active muscle tissue during anaerobic respiration, when there is shortage of oxygen and subsequently removed for conversion to glucose by the liver. It can build up in muscle to give cramp-like pains. It is also produced by fermentation in certain bacteria and present in sour milk, koumiss, leban, yogurt and cottage cheese.

Chemical structure of lactic acid
(2-Hydroxypropanoic Acid)

LACTIC ACID BACILLUS: A member of the genus *Lactobacillus*.

LACTIC ACID BACTERIA: A group of gram-positive anaerobic bacteria that use pyruvic acid as the acceptor for the hydrogen produced in glycolysis, so forming lactic acid. Such organisms such as *Streptococcus lactis* are used on a large scale in the food industry, especially as starter cultures in cheese making. Most lactic acid bacteria are rod-shaped organisms belonging to the family Lactobacillaceae of the Eubacteriales.

LACTIC ACID DEHYDROGENASE: An enzyme that is present in almost all of the tissues in the body. Its levels rise in response to cell damage. LDH levels are measured from a sample of blood taken from a vein to help diagnose lung disease, lymphoma, anaemia and liver disease. They also help determine how well chemotherapy is working during treatment for lymphoma.

LACTIC ACID FERMENTATION: A type of fermentation carried out by lactic acid bacteria in which sugar, for example, lactose, glucose and pentose are converted either entirely or almost entirely to lactic acid, called **homolactic fermentation** or to a mixture of lactic acid and other products, called **heterolactic fermentation**.

LACTIC ACID OXIDATIVE DECARBOXYLASE: A flavoprotein oxidoreductase catalyzing oxidation (with O_2) of L-lactate to acetate plus CO_2 and water.

LACTIC ACIDEMIA (LACTACIDEMIA): The presence of dextrorotatory lactic acid in the circulating blood.

LACTIC ACIDOSIS: Acidosis caused by accumulation of lactic acid more rapidly than it can be metabolised. It may occur spontaneously or in association with diseases such as diabetes mellitus, leukaemia or liver failure.

LACTIDE: A white, crystalline substance with the molecular formula $C_6H_8O_4$, obtained from lactic acid by distillation and regarded as an anhydride. Most lactides are relatively low melting solids and are easily hydrolysed by base to form salts of the parent acid, such as sodium lactate. The term also refers to any similar substance.

LACTIFEROUS: Bearing or containing milk or a milky fluid, such as the lactiferous vessels, cells or tissue of various vascular plants.

LACTIFEROUS AMPULLA (SINUS LACTIFERI; AMPULLA LACTIFERA; AMPULLA OF MILK DUCT; LACTIFEROUS AMPULLA): A circumscribed spindle-shaped dilation of the lactiferous duct just before it enters the nipple. In nursing mothers this dilatation stores a droplet of milk which is expressed by compression as the infant begins to suckle; this is thought to encourage continual suckling while the let-down reflex ensues.

LACTIFEROUS DUCTS (CANALICULAR DUCTS; GALACTOPHORE; GALACTOPHOROUS CANALS; GALACTOPHOROUS DUCTS; MAMILLARY DUCTS; MAMMARY DUCTS; MILK DUCTS): Fifteen or twenty ducts of the mammary gland that drain the lobes of the mammary gland and open at the nipple.

LACTIFEROUS GLAND: Milk producing gland of female mammals. An adapted sweat gland, it is made up of milk producing alveolar cells, surrounded by contractile myoepithelial cells, together with considerable numbers of fat cells. Milk production is hormonally controlled.

LACTIMS: Tautomeric forms of lactams, having an endocyclic carbon-nitrogen double bond.

LACTIVORE: An animal with a diet consisting mainly of milk or other dairy products. **Lactivorous** means needing or subsisting on milk.

LACTO-: A prefix that denotes a relationship to milk, for example, lactoalbumin.

LACTO-N-BIOSIDASE: An enzyme that hydrolyses oligosaccharides containing a type 1 structure at the nonreducing terminus and produces lacto-n-biose.

LACTOABUMIN: The albumin present in milk, identical with ordinary serum albumin. It is distinct from the casein of milk. It includes alpha and beta lactalbumins.

LACTOBACILLARY MILK: Milk inoculated with a culture of *Bacillus acidophilus*, *B. Bulgaricus* or other lactic acid-forming microorganism.

LACTOBACILLUS (LACTOBACILLI): Any one of rod-shaped gram-positive anaerobic bacteria of the genus *Lactobacillus*, widely distributed in animal feeds, manure, milk and milk products that ferment sugars to form lactic acid. They also inhabit animal and human intestinal tracts but do not damage to it. Commercial preparations of lactobacilli are used to restore normal intestinal flora after antibiotic therapy. They are often used in starter cultures to produce lactic acid during fermentation. Lactobacilli are important in food technology for the production of cheese, yoghurt and other foods from milk.

LACTOBACILLUS ACIDOPHILUS: Bug that produces acidophilus milk.

LACTOBACILLUS CASEI: A rod-shaped bacterium isolated from milk and cheese, dairy products and dairy environments, sour dough, cow dung, silage and human mouth, human intestinal contents and stools and the human vagina.

LACTOBUTYROMETER: An instrument for determining the amount of butter fat contained in a given sample of milk.

LACTOCELE (GALACTOCELE): A cystic tumour containing milk or a milky substance that is usually located in the mammary glands. It

is caused by a protein plug that blocks off the outlet. Once lactation has ended the cyst should resolve on its own without intervention. A galactoceles does not cause infection as the milk within is sterile and has no outlet for which to become contaminated.

LACTOCHROME (LACTOFLAVIN; OVOFLAVIN; RIBOFLAVIN; VITAMIN B2): The flavin in milk. It is a yellow crystalline compound with the molecular formula $C_{17}H_{20}N_4O_6$ that is a growth-promoting member of the vitamin B complex and occurs both free as in milk and combined as in liver.

LACTOCOCCUS: A genus of gram-positive, coccoid bacteria mainly isolated from milk and milk products. These bacteria are also found in plants and nonsterile frozen and dry foods. Previously thought to be a member of the genus streptococcus, it is now recognised as a separate genus.

LACTOCOCCUS LACTIS: A non-pathogenic lactococcus found in dairy products and responsible for the souring of milk and the production of lactic acid.

LACTOCRIT: An instrument used to estimate the amount of butterfat in milk.

LACTODENSIMETER: A form of hydrometer, specially graduated, for finding the density of milk and thus discovering whether it has been mixed with water or some of the cream has been removed.

LACTOFERRIN: An iron-binding protein found in milk, saliva, tears and intestinal and respiratory secretions that interferes with the iron metabolism of bacteria; in conjunction with antibodies, it plays an important role in resistance to certain infectious diseases.

LACTOGEN: An agent such as prolactin that stimulates milk production or secretion.

LACTOGENESIS: The secretion of milk by the mammary glands.

LACTOGENIC: Inducing lactation.

LACTOGENIC HORMONE (PROLACTIN, PRL): A hormone produced and secreted into the bloodstream by the anterior pituitary gland. It stimulates the development and growth of the mammary glands after the glands have been prepared by oestrogen, progesterone, thyroxine, insulin, growth hormone, glucocorticoids and human placental lactogen. After parturition, prolactin, together with glucocorticoids, is essential for the initiation and maintenance of milk production. Prolactin synthesis and release from the pituitary are mediated by the central nervous system in response to suckling by the infant. When suckling or its mechanical equivalent ceases, prolactin secretion slows and milk production ceases.

LACTOGLOBULIN: The globulin present in milk, comprising 50 to 60% of bovine whey protein.

LACTOL: Any cyclic hemiacetal formed by the intramolecular addition of a hydroxyl group to a carbonyl group.

LACTOLYSIS: The hydrolysis of lactose typically by galactosidase.

LACTOMETER: A type of hydrometer used to measure the specific gravity and therefore the purity or richness of milk.

LACTONE: A cyclic compound containing the group $-COO-$ in a ring, formed by heating hydroxycarboxylic acids. They often occur as the odour-bearing component of a plant product. Lactones can be formed by reaction of an $-OH$ group in one part of a molecule with a $-COOH$ group in the other to give a cyclic ester. This type of reaction occurs with hydroxyl group on the third carbon from the carboxyl group. The resulting lactone has a five-membered ring. Similarly, lactones have six-membered ring. Lactones, with a four-membered ring, are not produced directly from hydroxyl acids, but can be synthesised by other means.

LACTOSCOPE: An instrument for estimating the amount of cream contained in milk by ascertaining its relative opacity.

LACTOSE: A sugar (disaccharide), with the molecular formula $C_{12}H_{22}O_{11}$, density 1.525 g/cm^3 , melting point 202.8°C (475.95 K) and

boiling point 668.9 °C (942 K). It is found most notably in milk. It is made up of one molecule of glucose and one molecule of galactose. It forms 5% of cow's milk.

LACTOSE INTOLERANCE: A genetic trait in which the body is unable to metabolise lactose, the main sugar in milk and other dairy products. It is caused by the absence of the enzyme lactase, which breaks down lactose. Such people suffer from diarrhoea and bloating when they eat foods containing lactose.

LACTOVEGETARIAN (OVO-LACTO VEGETARIAN): A person whose diet includes dairy products, eggs, vegetables, fruits, grains and nuts.

LACUNA: A small cavity in the tissues of an organism, such as the hollow centre of certain plant stems or any of the small cavities in bone in which the bone-forming cells are found.

LACUNARY SERIES: A series in which the terms with non-zero coefficients are widely separated; especially, a power series in which there is a subsequence of the powers $\{\lambda_n\}$ corresponding to non-zero coefficients that satisfies $\lambda_{(n+1)} > (1+\epsilon)\lambda_n$ for some $\epsilon > 0$, so that infinitely often there is a geometric jump in the size of the power.

LACUNARY VALUE: A value not taken by an analytic function in its domain of definition. By Picard's theorem, a non-constant entire function has at most one lacunary value.

LACUSTRINE: Growing, living or formed in or at the edge of a lake such as soil deposited by lakes. **Lacustrine deposits** are fine particles such as fine sand, silt, and clay, devoid of coarse particles such as coarse sand or gravels that form deposits in the bottom of ancient lakes that a river or stream has transported there.

LADDER CHAIN: A double-strand chain, consisting of an uninterrupted sequence of rings, with adjacent rings having two or more atoms in common. Alternatively, a ladder chain is a double-strand chain with adjacent constitutional units joined to each other through four atoms, two on one side and two on the other side of each constitutional unit.

LADDER MACROMOLECULE: A double-strand macromolecule consisting of an uninterrupted sequence of rings, with adjacent rings having two or more atoms in common. Alternatively, a ladder macromolecule is a double-strand macromolecule with constitutional units joined to each other through four atoms, two on one side and two on the other side of each constitutional unit.

LADENBURG BENZENE (PRISMANE): A saturated hydrocarbon, with the chemical formula C_6H_6 , in which the six carbon atoms are arranged at the corners of a triangular prism. It was proposed by Albert Landenburg in 1869 as a possible structure for benzene. It is far less stable than benzene.

LADY NIPPLES (COW'S UDDER; LADY NIPPLES FRUIT; *Solanum mammosum*): A herbaceous perennial tree in the family Solanaceae that is native to the tropical regions of America. It grows to a height of about a metre tall. The large, hairy leaves are simple, alternate, purple-veined, broad-ovate with toothed margins. It produces showy flowers whose petals are purple with white stripes. The poisonous fruits develop central lobes at the tips, resembling nipples and other lobes at the bases. The plant does well in sunny loamy soils. Lady nipples is usually cultivated for its ornamentally attractive nature.

LADY OF THE NIGHT (NIGHT-BLOOMING JASMINE; NIGHT CESTRUM; NIGHT JASMINE; NIGHT-SCENTED JESSAMINE; QUEEN OF THE NIGHT; *Cestrum nocturnum*): An evergreen, perennial woody shrub in the plant family Solanaceae (the potato family). It is native to the West Indies, but has adapted to South Asia. It can grow up to 4 metres tall. The glossy, dark green leaves are simple, narrow lanceolate with entire leaf margin with a pointed tip. The flowers are greenish-white, the corolla is a slender tubular which opens at night and occurs as cymose inflorescences. The

fruits are rounded white berries. The flowers release sweet perfume which is released at night and are therefore, the plant is grown as an ornamental plant. The entire plant, especially the berries are poisonous to humans and other mammals, however, birds eat the berries.

LADYBIRD (LADYBUG): Small round flying beetles making up the family Coccinellidae, which have red or orange outer wings with black spots. They are useful in biological control because many species feed on aphids or scale insects, which are crop pests.

LAEVOROTATORY: Describing a chemical compound that rotates the plane of plane-polarised light to the left or anticlockwise direction.

LAEVULOSE (L-FRUCTOSE; FRUIT SUGAR; 1,3,4,5,6-PENTAHYDROXY-2-HEXANONE): A simple sugar with the molecular formula $C_6H_{12}O_6$ that is found in honey, most sweet fruits and starches that gives fruits their sweet taste. It is the sweetest of all naturally occurring carbohydrates and has low glycaemic value. Fructose, together with glucose and galactose are absorbed directly into the bloodstream during digestion. Crystalline fructose obtained from processing corn or sugar is used in food and beverages as a nutritive.

LAG: In physics, it is the time delay between a specified maximum value of one wave or vibration and the maximum value of another wave or vibration expressed in degrees.

LAG PHASE: The time period when there is no increase in the number of bacteria population (the population remains constant) after inoculation of fresh growth medium.

LAGERSTREMIA (CRAPE MYRTLES; CREPE MYRTLES): A genus of about 50 species of woody deciduous and evergreen trees or shrubs in the family Lythraceae (Loosestrife), native to tropical Asia and Africa. The height of the species ranges from less than 1 metre to about 33 metres. The leaves are simple, oppositely arranged with entire leaf margins. The red, pink, purple or white conical flower clusters have crinkled petals. The fruits are capsules which are green and succulent but turn dark brown or dark and dry when ripe. Some of the species are grown ornamentally for their showy flower colours and also for their exfoliating bark; whilst the timber of some are used to manufacture bridges, furniture and railway sleepers.

LAGGING: 1. Insulating material used to cover, for example, hot water pipes and containers to prevent loss of heat. 2. Planking used especially for preventing cave-ins in earthwork or for supporting an arch during construction.

LAGGING CURRENT: An alternating current that reaches its maximum value of up to 90° behind the voltage that produces it in a cycle. In other words, the waveform of the current lags the voltage by a certain angle (called phase angle), as the current rises more less slowly than the voltage. This usually happens in inductive circuits due to the presence of capacitors or inductors. In a purely resistive circuit, where there is no capacitor or inductor the current passing through a load rises or decreases at the same time as that of the voltage across it and there is no phase angle between them and are said to be **in phase**.

LAGGING STRAND: A discontinuously synthesised DNA strand that elongates in a direction away from the replication fork in the mandatory 3' – 5' prime direction.

LAGOMORPHA: An order of mammals that includes the hares, rabbits and pikas. They are differentiated from rodents by two pairs of upper incisors covered by enamel, vertical or transverse jaw motion, three upper and two lower premolars, fused tibia and fibula and a spiral valve in the cecum. Lagomorpha are native to many parts of the world and have been introduced by humans to Australia, where they are significant crop pests.

LAGOON: 1. Coastal body of relatively shallow salt water usually with limited access to the sea. 2. A human-made pool of shallow water or other liquid created near an industrial site or by a waste site.

LAGPHASE: A stage in the bacterial growth curve where the rate of growth rate is slow and the microbes acclimatise to the food and nutrients in the new habitat.

LAGRANGE FORM OF THE REMAINDER: The expression of the remainder or the error, of a Taylor series that is the difference between a function and its n th Taylor polynomial. For example, $R_n(f, a)$ is the difference $f(a+h) = \left\{ f(a) + \frac{f'(a)}{1!}h + \dots + \frac{f^{(n)}(a)}{n!}h^n \right\}$ expressed as

$$R_n(f, a) = \frac{f^{(n+1)}(c)}{(n+1)!} h^{n+1} \text{ for some } c \text{ between } a \text{ and } a+h.$$

LAGRANGE INTERPOLATION FORMULA: The formula for obtaining a polynomial of at most degree n that interpolates a given real function $f(x)$ at $n+1$ distinct points: $P_n(x) = \sum_{k=0}^n L_n(x) f_n(x_k)$, where $L_n(x) = \prod_{k=0, k \neq i}^n \frac{x - x_k}{(x_j - x_k)}$. The Hermite interpolation produces the unique

polynomial of degree at most $2n-1$ that takes prescribed function and first derivative values at n distinct points.

LAGRANGE METHOD OF MULTIPLIERS: The method of reducing a constrained optimisation problem to an unconstrained problem by the addition of Lagrange multipliers. The method is valid if the gradients of the constraints are linearly independent at x_0 . If, in addition, the objective function is convex and the constraints are affine, then any such stationary point is optimal.

LAGRANGE MULTIPLIERS: Parameters, unusually denoted by λ introduced to assist in finding the maximum or minimum value of several variables $x_1, x_2, x_3, \dots$ subject to some constraint that connects the variables. An important application of the method of Lagrangian multipliers is its use in the derivation of the Boltzmann distribution in statistical mechanics, in which one of the Lagrangian multipliers is $-\frac{1}{kt}$, where k is the Boltzmann constant and t is the thermodynamic temperature.

LAGRANGE PROBLEM: The special case of a control theory Bolza problem in which there is no extra term depending on the endpoints.

LAGRANGE, JOSEPH LOUIS (1736-1813): Italian-born French mathematician and astronaut, who made important contributions to number theory and to classical and celestial mechanics. He is considered one of the greatest mathematicians of the 18th century. He created the calculus of variations, systematised the field of differential equations and worked on the theory of numbers. He also investigated calculations of the liberation of the Moon and motions of the planets. Many mathematical terms including Lagrange form of the remainder, Lagrange interpolation formula, Lagrange multipliers, etc., have been named after him.

LAGRANGE'S EQUATIONS: Equations devised for solving dynamic problems with constraints. They are partial differential equations of the form $\frac{d}{dt} \left(\frac{\partial L}{\partial \dot{q}_i} \right) - \frac{\partial L}{\partial q_i} = 0$. This equation results in the differential equations that describe the equations of motion of the system.

LAGRANGE'S FOUR-SQUARE THEOREM (FOUR SQUARES THEOREM; BACHET'S CONJECTURE): The theorem that every positive integer can be written as the sum of the squares of four integers.

LAGRANGE'S IDENTITY: The identity $vLu - uL^*v = \langle k(v), P \cdot k(u) \rangle$,

where $\langle \cdot, \cdot \rangle$ is the inner product, $Lu = \sum_{k=0}^n p_{n-k} u^{(n-k)}$ and $L^*v = \sum_{k=0}^n (-1)^k [p_{n-k} u]^{(k)}$ are linear differential operator and its scalar adjoint, u' and $u^{(k)}$ indicate differentiation, $k(u)$ is the vector

of derivatives $(u, \dots, u^{(n-1)})$ and the concomitant matrix, P , has

$$\text{coefficients } p_{i,j} \text{ defined by } p_{i,j} = \sum_{h=1}^{n-j+1} (-1)^{h-1} \binom{h-1}{i-1} [p_{n-h-j+1}]^{(h-1)}$$

for $i \leq n-j+1$ and is zero otherwise. If one obtains a solution w to the adjoint equation $L^*v = 0$, Lagrange's identity allows one to use this solution as an integrating factor in the sense that any solution of $Lu = 0$ must satisfy $\langle k(w), P \cdot k(u) \rangle = c$ for an arbitrary constant c and this equation is of lower order.

LAGRANGE'S LINEAR EQUATION: The partial differential equation

of the form $\sum_{i=1}^n P_i(x_1, x_2, \dots, x_n) \frac{\partial z}{\partial x_i} = R(x_1, x_2, \dots, x_n)$, where P_i and R are differentiable functions. If the equation is integrable, its general solution is $\phi(u_1, u_2, \dots, u_n) = 0$, where ϕ is an arbitrary function and the u_i are independent solutions of the simultaneous differential equations $\frac{dx_1}{P_1} = \frac{dx_2}{P_2} = \dots = \frac{dx_n}{P_n}$. A Lagrange's linear equation may also have a special integral.

LAGRANGE'S THEOREM: The theorem that the order of each subgroup of a finite group is a factor of the order of the group.

LAGRANGIAN (KINETIC POTENTIAL; LAGRANGE FUNCTION):

The difference between the kinetic energy and the potential energy of a system of particles, expressed as a function. It is used to define a dynamical system in terms of functions of coordinates, times and velocities given by: $L = T - V$, where L is the Lagrange function, T is the kinetic energy of the system and V is the potential energy of the system. The function deals with only two scalar functions T and V and not vector quantities.

LAGRANGIAN FLUID DYNAMISM: A formulation of fluid mechanics in which the trajectories of small regions of the moving fluid are analysed, in contrast to Eulerian fluid dynamics.

LAGRANGIAN POINT (LAGRANGE POINTS; L-POINTS; LIBRATION POINTS): Any one of five points in a system of two bodies orbiting a common centre of gravity such as a satellite with respect to the Earth and Moon at which a body of very low mass can be in equilibrium. The Lagrange points mark positions, where the combined gravitational pull of the two large masses provides precisely the centripetal force required to rotate with them.

LAIRAGE: A shed or outdoor enclosure such as holding pens, covered housing, water points and feed bunkers for the temporary housing of animals such as those on the way to market or when being transported for export.

LAKE: 1. A body of relatively large still water lying in a depressed ground without direct outlet to the sea. It is usually fed by melting ice and snow, springs, rivers, runoff from the land surface and direct precipitation. A lake has good supply of light, heat, oxygen and nutrients on the surface and harbours a large number of diverse aquatic organisms. 2. An insoluble pigment made by combining an organic dyestuff with an inorganic compound such as an oxide, hydroxide or salt. Absorption of the organic compound on the inorganic substrate yields coloured complex. The combination of a dyestuff with mordant lakes produces colours which are not as fast as many inorganic dyes, but their colours are more brilliant. It is used in paints and printing inks.

LAMARCK, JEAN BAPTISTE PIERRE ANTOINE DE MONET, CHEVALIER (1744-1829): French natural historian who published a flora of France in 1778, which included a dichotomous identification key and credited with the first use of the word "biology". He later worked on the classification of invertebrates, published in a seven-volume natural history (1815-1822). In 1809 he put forward the theory of evolution that has become known as Lamarckism.

LAMARCKISM (LAMARCKIAN INHERITANCE): One of the earliest superficially plausible theories of evolution proposed by Jean-Baptiste de Lamarck in 1809. He suggested that changes in an individual are acquired during its lifetime, controlled by three biological laws: environmental influence on organ development, change in body structure based on use and disuse of parts and the inheritance of acquired characteristics; and that these changes are inherited by its offspring. Thus, the long neck and limbs of a giraffe, for example, are explained as having evolved by the animal stretching its neck to browse on the foliage of trees.

LAMB: A young sheep or the meat of a young sheep.

LAMB DYSENTERY: A bacterial disease which enters the lamb from the pasture caused by *Clostridium perfringens* type B. It is a severe, highly fatal enteritis characterised by diarrhoea and dysentery.

LAMB-DIP SPECTROSCOPY: Named after Willis Eugene Lamb (1913-2008), it is a spectroscopic technique enabling the centres of absorption lines to be determined very precisely using Doppler shift associated with very rapidly moving molecules. An intense monochromatic beam of radiofrequency electromagnetic radiation is passed through a sample of a gas with the frequency being slightly higher than that of maximum absorption. Only certain molecules moving at a certain specific speed can absorb radiation. The beam is then reflected back through the sample so that radiation is absorbed by molecules moving at exactly this same speed, except that they are moving away from the mirror. If the radiation is exactly at the absorption peak, only molecules moving perpendicular to the line of the beam (which hence have no Doppler shift) absorb both in the initial path and the reflected path of the radiation. Since molecules being excited in the initial path leave fewer molecules to be excited in the return path this causes a less intense absorption peak to be found very accurately.

LAMBDA: 1. Also known as **lambda operator**, the Greek letter, λ , defined by the rules of lambda abstraction and lambda conversion and is studied by lambda calculus. 2. A Greek numeral with a value of 30. 3. A symbol in scientific literature. For example, in electrochemistry, it denotes the equivalent conductance of an electrolyte. In physics and engineering, it represents the wavelength of a wave. In probability theory, it represents the density of occurrences within a time interval in Poisson distribution. 4. It signifies, with its superscript, the bonding number, i.e., the sum of the number of skeletal bonds to an atom and the number of hydrogen atoms associated with it in a parent hydride, real or hypothetical. Its usage in names of compounds is generally restricted to non-standard valence states.

LAMBDA ABSTRACTION: The operation, in lambda calculus, that forms an expression denoting a function (or, equivalently, a class or predicate) from any expression whatever, by prefixing to it the abstraction operator, λ and a bound variable; the notation $\lambda x[\dots]$ can be most generally read as 'is an x such that $\dots$ '. For example, lambda abstraction applied to a binary relation such as ' x is to left of y ' (here written Lxy), yields $\lambda x[Lxy]$, the predicate ... is to the left of y ; $\lambda y[Lxy]$, the predicate x is to the left of ...; and $\lambda xy[Lxy]$, the relation ... is to the left of ...

LAMBDA CALCULUS (λ -CALCULUS): A branch of mathematical logic for the study of abstraction in terms of the operator, lambda (λ), developed by Alonzo Church in the 1940s, dealing with the application of functions to their arguments. This formulation is equivalent to combinatorial logic. It has been important in the development of the theory of computer languages.

LAMBDA CONVERSION: The rule, $\lambda x[Fx](a) = F(a)$ for the application of the lambda operator, λ ; that is, the application to a of the abstracted predicate ... 'is an x such that Fx ', is equivalent to Fa .

LAMBDA PARTICLE: A type of elementary particle containing strange, charm or bottom quarks with no electric charge. The lambda particle has a mass of 1115.60 MeV and an average lifetime of 2.6×10^{-10} s.

LAMBDA PHAGE (LAMBDA BACTERIOPHAGE): A temperate bacteriophage or virus particle infecting cells of the bacterium *Escherichia coli*, where it can either exist as a quiescent prophage or undergo replication leading to lysis of the host cell and releasing new phage particles. It was discovered by Esther Lederberg in 1950. Lambda phage consists of a head, containing double-stranded linear DNA as its genetic material and a tail that can have tail fibres. Modified forms are also used as vectors in gene cloning, especially for packaging relatively large amounts of foreign DNA.

LAMBDA POINT: The temperature of approximately 2.180 K (-270.97°C) below which helium (helium I) becomes super fluid helium II. The specific heat capacity tends towards infinity as the temperature approaches the lambda point.

LAMBDA SERIES: A series of the form $F(x) = \sum_{n=1}^{\infty} f(n) \frac{x^n}{1-x^n}$, where $F(x)$ is said to be the generating function of $f(n)$.

LAMBERT: A unit of luminance or brightness in the centimetre-gram-second system. It is equal to one lumen per square centimetre.

LAMBERT'S LAWS: Laws of light proposed by Johann H. Lambert (1728-77). The first law states that the illuminance of a surface illuminated by light perpendicularly from a point source is inversely proportional to the square of the distance between the surface and the source. The second law states that if the rays make an angle θ with the normal to the surface, the illuminance is proportional to $\cos \theta$. The third law (Bouguer's law) states that the luminous intensity (I) of light or other electromagnetic radiation decreases exponentially with the distance d as it enters an absorbing medium. It is expressed as: $I = I_0 \exp(-\alpha d)$, where I_0 is the intensity of the radiation that enters the medium and α is its linear absorption coefficient.

LAMBLING: 1. The work of assisting ewes to give birth to lambs. 2. The season when lambs are born or the action of giving birth to lambs. A **lambling tunnel** is a covered enclosure for ewes and lambs.

LAMBSHIFT: A small energy difference between two energy levels ($^2S_{1/2}$ and $^2P_{1/2}$) in the hydrogen spectrum, named after Willis Lamb (1913-2008) resulting from the quantum interaction between the atomic electromagnetic radiation. It was calculated and explained using renormalisation by Feynman and Julian Seymour Schwinger (1918-94) and the Japanese physicist Sinitiro Tomonaga (1906-79). Lamb won the Nobel Prize in Physics in 1955 for his discoveries related to the Lamb shift.

LAMELLA: 1. Any of the paired folds of membranes seen between the grana in a plant chloroplast. 2. A spore-bearing gills on the underside of the cap of many mushrooms and toadstools. 3. Any of various thin layers of membranes, especially the thin layers of tissue of which compact bone is formed.

LAMELLAR SOLID: A solid substance in which the crystal structure has distinct layer composed of fine, alternating layers of different materials in the lamellae. Intercalation compounds are lamellar compounds formed by interposition of atoms, ions, etc., between the layers of an existing element or compound. Examples are mica and graphite.

LAMELLAR STRUCTURES (MICROSTRUCTURES): Fine, alternating layers of different materials in the form of lamellae. They are often observed in cases where a phase transformation front moves quickly, leaving behind two solid products, as in rapid cooling of eutectic such as solder or eutectoid such as pearlite systems.

LAMELLIPODIUM: A flat, leaf-like cytoplasmic extension of a motile cell, such as a migrating skin cell (keratinocyte) or connective tissue fibroblast, that develops in the direction of movement. It contains bundles of actin microfilaments, which polymerise and extend at the leading edge and become cross-linked to form a supportive network to maintain the shape of the lamellipodium.

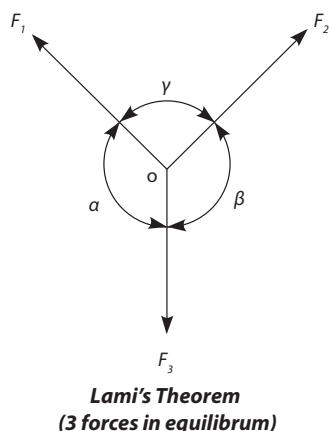
LAMI'S THEOREM: A relationship between the magnitudes and orientations of three concurrent forces in equilibrium. It is a consequence of the sine rule for triangles. For example, as indicated in the diagram, if three forces F_1 , F_2 and F_3 acting at a point O are in equilibrium, then

$$\frac{F_1}{\sin \beta} = \frac{F_2}{\sin \alpha} = \frac{F_3}{\sin \gamma}$$

LAMIN (NUCLEAR LAMINS; CLASS V INTERMEDIATE FILAMENTS):

Any of a group of proteins that are components of intermediate filaments, which interact with membrane-associated proteins to form the nuclear lamina on the inner surface of the nuclear envelope. They are involved in the breakdown and reformation of the nuclear envelope during mitosis, as well as the positioning of nuclear pores. They are classified into A, B and C. Some lamins are involved in the positioning of chromosomal regions inside the nucleus and control of gene expression. Mutations in lamin genes have been implicated in certain cancers.

LAMINA: 1. The blade or flat part of a leaf or petal on either side of the midrib, in which photosynthesis and transpiration occurs. 2. Any of the several parallel layers of thin scale-like or plate-like sensitive tissue just inside the hard exterior part of the hoof of, for example a horse. 3. A thin layer of bone, membrane or other tissue. 4. The leaf-like part of the thallus of certain algae, notably kelps. 5. Thin sedimentary layer that is generally less than 5 cm thick. 6. In mathematics, a surface with infinitesimal or uniform positive but negligible, thickness. Alternatively, a two-dimensional planar closed surface L which has a mass M and a surface density $\sigma(x,y)$ (in units of mass per areas squared) such that

$$M = \int_L \sigma(x,y) dx dy.$$


LAMINAR FLOW: The streamline flow of a fluid in which the fluid moves smoothly in regular paths without fluctuations or turbulence so that velocity, pressure and other flow properties at each point in the fluid remain constant. Over a horizontal surface the flow consists of thin layers, all parallel to each other, that slide over each other. It is commonly observed, where the flow channel is relatively small, the fluid is moving slowly and its viscosity is relatively high. Examples include the flow of oil through a thin tube and blood flow through capillaries.

LAMINARIN: The main assimilation product of algae belonging to the Phaeophyta. It is composed chiefly of glucose units though mannitol may also be present.

LAMINATE PLACENTATION: A form of placentation in which the placentae arise from all over the inner surface of the ovary. It is seen in the flowering rush (*Bulomus umbellatus*) and the white water lily (*Nymphaea alba*).

LAMINATED: A term used to describe materials which are made up of two or more layers of material bonded together usually with a resin such as epoxy. Examples of laminated materials are plywood and vehicle windshields, which are commonly made by laminating a tough plastic film between two layers of glass. In electrical engineering, lamination is used in the construction of components such as the magnetic cores of transformers to reduce unwanted heating effects due to eddy currents.

LAMINATED CORE: The core of a transformer or other electromagnetic device such as armature in which the ferromagnetic material such as iron or steel is made into thin sheets of laminations, which are oxidised or varnished to provide insulation between them.

LAMININ: A family of major protein components of the basal lamina, the dense fibrous layer of the basement membrane underlying most epithelial and endothelial cells. Laminins are large cross-shaped molecules that contain α -chain, a β -chain and a γ -chain, found in five and three genetic variants, respectively. They bind the collagen fibres of the lamina to adjoining cells and to the looser surrounding connective tissue, thus helping to create and stabilise the basic fibrous network. They are important and biologically active part of the basal lamina, influencing cell differentiation, migration, adhesion as well as phenotype and for the maintenance and survival of tissues.

LAMINITIS (FOUNDER): The inflammation of the lamina in a hoof of a horse or dairy cattle, characterised by swelling, often leading to deformed hooves. It may be caused by overfeeding on grain, toxemia, overweight especially in pregnant mares, retention of the placenta in a mare after foaling, foot injury due to excessive exercising over hard ground and some other diseases.

LAMP BLACK (CARBON BLACK): A special type of carbon black produced by incomplete combustion of a fuel rich in aromatics that is burned in flat pans. It is characterised by a relatively broad particle size distribution. It is used as a black pigment and filler in plastic products and also to make ink.

LAMPBRUSH CHROMOSOME: A type of chromosome found in the oocytes of amphibians; occurring during the diplotene stage of meiosis I. It has numerous loops arising from the long axis of the chromosomal arms. Seen under a microscope it appears to be organised into a series of large chromatin loops emanating from a linear chromosomal axis. The loops are regions where the chromosome is uncoiled and the DNA exposed for active transcription by an associated matrix of RNA and proteins.

LAN (LOCAL AREA NETWORK): A computer network limited to a small area such as an office building, university or even a residential home. A WAN (wide-area network) is a LAN connected to other LANs over any distance via telephone lines and radio waves. LANs are capable of transmitting data at very fast rates, much faster than data can be transmitted over a telephone line; but the distances are limited and there is also a limit on the number of computers that can be attached to a single LAN. Currently, the most common type of LANs are Ethernet-based and use software from Novell or Oracle. However, with the emergence of wireless networking, wireless LANs have become a popular alternative.

LAN CARD (NETWORK ADAPTER; NETWORK INTERFACE CARD; NIC): A printed circuit board that plugs into the bus of both user machines and servers in a local area network (LAN). The network adapter transmits data onto the network and receives data from the network. It works at the data link protocol level.

LANATE: Covered with or having fine, woolly hairs or hair-like filaments.

LANCEOLATE: Tapering from a rounded base toward an apex, as found in some leaves. Examples are found in white willow, Spanish chestnut, crack willow and common osier.



LAND: The solid dry part of the Earth's surface not covered by any body of water. Its characteristics embrace all reasonably stable, or predictably cyclic, attributes of the biosphere vertically above and below, including those of the atmosphere, the soil, and the underlying geology, the hydrology, the plant and animal populations, and the results of past and present human activities, to the extent that these attributes exert a significant influence on the present and future uses of land by man.

LAND BREEZE: A gentle wind that blows from the land towards the sea developing in coastal areas usually in the night. Land breeze occurs where the pressure is relatively higher above the land than the sea as the land cools more rapidly in the evening. The air then moves towards the sea in order to even up the pressure difference.

LAND CLASSIFICATION: The arrangement of land units into various categories based on the properties of the land or its suitability for some particular purpose such as its potential for agricultural use.

LAND COVER: The observed biophysical characteristics of the land surface of the earth. Land covers include vegetation, asphalt, trees, water, man-made constructions such as bridges and buildings, etc. Surfaces consisting of bare rock or bare soil describe land itself rather than land cover.

LAND EROSION CONTROL: A method, such as planting trees to prevent the soil from being worn away by erosion.

LAND RECLAMATION (RECLAMATION): Conversion of natural phenomena (such as the sea, wetlands and other water bodies) or areas that have been degraded by human activities (such as sand and gravel extraction or open-cast mining) into new land. Some of the methods used include filling the area with large amounts of materials such as waste, rock, cement, clay and soil; or in of case wetlands, draining it.

LAND TENURE: The way in which land is owned and used. Common forms of land tenure include owner-occupied farms which range from large farms using hired labour to peasant plots and tenancies which vary very widely but basically involve payment, in one form or another, to the landlord from the tenant. This payment can be in the form of labour, in the form of a portion of the crop (share-cropping) or in the form of cash.

LAND USE: The classification of land according to different human purposes or economic activities, such as agricultural, timber extraction, conservation, industrial, residential, urban, rural, or commercial.

LAND-USE CHANGE: A change in the use or management of land by humans, which may lead to a change in land cover. Land cover and land-use change may have an impact on the albedo, evapotranspiration, sources, and sinks of greenhouse gases, or other properties of the climate system, and may thus have an impact on climate, locally or globally.

LAND USE, LAND USE CHANGE AND FORESTRY (LULUCF): The United Nations Framework Convention on Climate Change (UNFCCC) Secretariat defines it as 'a greenhouse gas inventory sector that covers emissions and removals of greenhouse gases resulting from direct human-induced land use, land-use change and forestry activities.'

LANDAU DAMPING: A damping mechanism for collective oscillations in plasmas, named after Soviet physicist Lev Davidovich Landau (1908-68), who discovered it. It is the effect of damping or attenuation of longitudinal space charge waves in plasma or such similar environment. If the velocity of the wave associated with the collective oscillation is comparable to the velocities of electrons in

plasmas, with the velocities of the electrons being determined by their thermal motion, there is a transfer of energy from the wave motion to the electrons. This gives rise to damping or attenuation and eventually, destruction of the wave motion. An example of this mechanism occurs in the damping of collective excitations in the quantum theory of many-body systems by energy transfer to individual quasi-particles.

LANDAU GHOST (MOSCOW ZERO): The possible inconsistency in the renormalisation procedure that appears at very high energies in quantum electrodynamics and other quantum field theories in which there is no asymptotic freedom. It was named after Soviet physicist, Lev Davidovich Landau (1908-68) who together with others showed it in the 1950s. It was shown in approximation techniques that, in the limit of extremely high energies the relation between the unrenormalised and actual electric charges is such that the observed charge falls to zero, thus making the procedure of renormalisation impossible. Asymptotic freedom that takes place in quantum electrodynamics (a theory that describes the strong interaction in terms of quarks and antiquarks and the exchange of massless elementary particles between them) is the opposite situation to the Landau ghost.

LANDAU LEVELS: The energy levels found in quantum mechanics of free electrons in a uniform magnetic field at very low temperatures and which are quantised because of the quantisation of the electron motion perpendicular to the field. It was named after the Soviet physicist Lev Davdovich Landau (1908-68). These energy levels, have discrete values, which are integer multiples of heB/m , where h is the Planck constant, e is the charge of the electron, m is the mass of the electron and B is the magnetic flux density. Each Landau level is a highly degenerate level, with each level being filled by $2eB/h$, where the factor 2 is due to the spin of the electron.

LANDAUER'S PRINCIPLE: The principle put forward by Rolf Landauer of IBM in the 1960s that energy has to be expended to erase information. Each bit of the erased information will lead to the release of an amount of $kT \ln 2$ of heat, where k is the Boltzmann constant and T is the absolute temperature. This principle links thermodynamics and information theory.

LANDÉ INTERVAL RULE: A rule in atomic spectroscopy that if the spin-orbit coupling is weak in a given multiplet, an energy level having definite spin angular momentum and orbital angular momentum is split into levels of differing total angular momentum, so that energy differences between two successive j levels are proportional to the larger of the two values of j , where j is the total resultant angular momentum of the coupled electrons. The rule can be deduced from the quantum theory of angular momentum. The rule also assumes that the interactions between spin magnetic moments can be ignored, but this assumption does not hold for very light atoms, such as helium.

LANDER: A spacecraft, particularly a probe, designed to set down on the surface of another celestial body.

LANDFILL SITE (LANDFILL; SANITARY LANDFILL): A large deep hole in the ground in which used household and commercial waste is dumped and sometimes used to build up low-lying land. Although landfill sites may be the most cost-efficient way to dispose of waste, they may be dangerous, as toxins and other leachates may be released into the soil; it is also wasteful because of the materials dumped and the size of land it occupies. Alternatives to disposal of waste at landfills include waste to energy (converting some waste items into heat, electricity or fuel through a variety of processes); anaerobic digestion (it produces gas that may be used for purposes such as to generate electricity); composting (natural bio-degradation that converts organic wastes into nutrient rich food, which may be used for agricultural or gardening purposes); incineration or combustion (solid wastes are burned at high temperatures to convert them into residue and gaseous products); recovery and recycling; and plasma gasification (converting carbon-based materials into organic matter into synthetic gas, electricity and slag using plasma).

LANDFORM: The shape, size, form, nature and characteristics of a specific feature such as an ocean, continent, mountain, waterfall, peninsula, swamp or valley of the planet's surface that is located above ground or underwater. Landforms are classified by processes by which they are created as follows: (i) **structural landforms** such as fold mountains, rift valleys and volcanoes that are created by massive earth movements due to plate tectonics under the Earth that can push up mountains and hills. (ii) **weathering landforms** in which rocks and sediments are decomposed and disintegrated through weathering. In time, this can turn into soil, karst (limestone bedrock) and patterned ground with features that resemble circles, polygons, nets, steps and stripe. (iii) **erosional landforms** such as river valleys, glacial valleys and coastal cliffs that are created by the removal of weathered and eroded surface materials by wind, water, glaciers and gravity. (iv) **depositional landforms** such as beaches, deltas and flood plains that are created from the deposition of weathered and eroded surface materials. The highest landform on Earth is Mount Everest in Nepal with a height of 8,850 metres above sea level. **Geomorphology** is the study of landforms.

LANDSCAPE: 1. The scenery, general shape, structure and features of the surface of an area of land or a wide view of country scenery or an expanse of scenery of a particular type, especially as much as can be seen by the eye. 2. A painting, drawing or photograph of scenery, especially rural scenery. 3. The branch of art dealing with the painting, drawing or photographing of scenery. 4. In printing or page setting, it is a horizontal orientation mode in which content is printed for reading on the longer length of the sheet of paper. It is also used to display wide-screen content, such as a Web page, image, document or text. Portrait mode is similar to landscape.

LANDSLIDE: 1. A sudden downward movement of a mass of soil, rock and other weathered materials from a cliff or steep slope. It occurs when the force of gravity acting on the materials within a slope overcomes the material's resistance to shearing. Steepening of a slope by erosion or excavation and motion caused by an earthquake can cause landslide. 2. The part of a plough which takes the sideways turn as the furrow is turned.

LANGERHANS' CELLS: A type of immune cells in the epidermis produced in the bone marrow that participate in the inflammatory response by engulfing microorganisms and releasing chemicals that mobilise immune system cells. It is a means by which the immune system becomes sensitised to a substance causing contact allergy. It was named after Paul Langerhans (1847-1888), a German anatomist.

LANGEVIN EQUATION: A type of random equation of motion used to study Brownian movement. It was named after the French physicist Paul Langevin (1872-1946). The Langevin equation can be best solved by the use of statistical methods and the theory of probability. The

Langevin equation is given by, $m \frac{du}{dt} + \gamma u = F(t)$, where m is the mass of the particle, $F(t)$ is the random component and μ is the velocity of the particle.

LANGLANDS PROGRAM: A set of conjectures relating different branches of mathematics. It is a way of organising number theoretic data in terms of analytic objects. It was proposed by Robert Langlands in 1967.

LANGMUIR ADSORPTION ISOTHERM (LANGMUIR EQUATION; LANGMUIR ISOTHERM; HILL-LANGMUIR EQUATION): An equation used to describe the coverage or adsorption of molecules on a solid surface to gas pressure or concentration of a medium above the solid surface at a fixed temperature. It was derived by and named after Irving Langmuir (1881-1957), an American chemist and Nobel laureate, using the kinetic theory of gases. The Langmuir with the isotherm can be written as: $\theta = \frac{ap}{1 + ap}$, where θ is the fraction of the surface covered by the adsorbate, p is the pressure of the gas and a is a constant called the adsorption coefficient, which is the equilibrium constant for the process of adsorption. The equation has the following

assumptions: (i) the adsorption consists entirely of a monolayer at the surface; (ii) there is no interaction between molecules on different sites and each site can hold only one adsorbed molecule; and (iii) the heat of adsorption does not depend on the number of sites and is equal for all sites. The Langmuir adsorption isotherm is of limited application since for real surfaces the energy is not the same for all sites and interactions between adsorbed molecules cannot be ignored.

LANGMUIR FREQUENCY (PLASMA FREQUENCY): The frequency of plasma oscillations in an equilibrium charge distribution, derived by combining Poisson's equation with Newton's law of motion, named after Irving Langmuir (1881-1957), an American chemist and Nobel laureate. The Langmuir frequency can also be derived using quantum mechanics for the electron gas.

LANGMUIR-BLODGETT FILM: A film of molecules on a surface that can contain multiple layers of an organic material, deposited from the surface of a liquid onto a solid by dipping the solid substrate into a liquid. Multiple layers can be made by dipping a plate into a liquid so that it is covered by a monolayer and the process is repeated. It is named after Irving Langmuir and Katharine B. Blodgett, who invented the technique. Langmuir-Blodgett films have many practical applications, including insulation for optical and semiconductor devices and selective membrane in biotechnology.

LANGMUIR-HINSHELWOOD MECHANISM: A possible mechanism for a bimolecular process catalysed at a solid surface. It is assumed that two molecules are adsorbed on adjacent sites and that a reaction takes place via an activated complex on the surface to yield the products of the reaction. This mechanism was suggested by Irving Langmuir (1881-1957), an American chemist and Nobel laureate in 1926. Some bimolecular reactions at surfaces are in agreement with the predictions of this model.

LANGMUIR-RIDEAL MECHANISM: A possible mechanism for a bimolecular process catalysed by one of the face. It is envisaged that one of the molecules is adsorbed and then reacts with a second molecule that is not adsorbed. This mechanism was suggested by US chemist, Irving Langmuir (1881-1957) in 1992 and developed by the British chemist Sir Eric Rideal in 1939. The Langmuir-Rideal mechanism is not common but certain reactions involving hydrogen atoms probably occur by this mechanism.

LANGUAGE: An interpreted system; a formal calculus endowed with a semantic interpretation.

LANGUAGE PROCESSOR: A program that performs tasks, such as translating and interpreting, required for processing a specified programming language. Examples of language processors include a Fortran processor and a COBOL processor.

LANOLIN (ADEPS LANAE; WOOL WAX; WOOL GREASE): A yellow waxy substance secreted by the sebaceous glands of sheep and present in sheep wool to protect the wool and skin against climate and the environment. It contains cholesterol and certain terpene alcohols and ester extracted from sheep's wool. It is used in the cosmetics industries for making soaps, skin cream, shampoos and in leather preparation.

LANSFORDITE: A mineral form of magnesium carbonate pentahydrate, with the chemical formula $\text{MgCO}_3 \cdot 5\text{H}_2\text{O}$, which occurs as stalactites.

LANTHANIDS (LANTHANIDE SERIES): The series of rare-earth metallic elements in the periodic table with atomic numbers 21, 39 and 57-71. It includes, in order of increasing atomic number, scandium, yttrium, lanthanum, cerium, praseodymium, neodymium, promethium, samarium, europium, gadolinium, terbium, dysprosium, holmium, erbium, thulium, ytterbium and lutetium. The lanthanoids all have two outer s-electrons (a $6s^2$ configuration), follow lanthanum and are classified together because an increasing proton number corresponds to an increase in the number of 4f electrons.

LANTHANUM: A silver-white metallic element, the first element of the Lanthanide series with the chemical symbol La, atomic number 57 and atomic weight 138.91. It occurs naturally, made up of the isotopes ^{138}La , 0.089% with a half-life of 1.1×10^{11} years and ^{139}La (stable), 99.91%. Its principal ore is bastnasite, from which it is separated by an ion-exchange process. It is also the most basic of the rare earths and can be separated rapidly from other members of the rare-earth series by fractional crystallisation. The metal, being pyrophoric, is used in alloys for lighter flints and the oxide is used in some optical glasses. The largest use of lanthanum, however, is as a catalyst in cracking crude oil.

LANUGINOSE (LANUGINOUS): Covered with down or soft hairs. **Lanugo** means a covering of soft downy hairs.

LAPAROSCOPY: A medical procedure in which a small incision is made, usually in the abdominal wall through which a laparoscope is inserted to diagnose conditions or to perform certain types of operations. **Laparoscope** is a viewing tube with a small camera attached to the eyepiece that is inserted into an incision to permit examination of structures within the abdomen and pelvis on a video monitor connected to the tube. If necessary, certain treatments such as gallbladder and gallstones can be removed by the same procedure by attaching the necessary instruments.

LAPIS LAZULI: An azure-blue to greenish-blue rock that is widely used as a semiprecious stone. It is composed of lazurite, a sodium aluminium sulfur silicates and calcium aluminium sulfur silicates, mixed with small amounts of calcite, sodalite and pyrite. It is found in Afghanistan and Argentina in crystalline limestones as a contact metamorphic mineral. It was formerly used to make the artists pigment ultramarine.

LAPLACE DISTRIBUTION: The probability distribution, $f(x)$ given by $f(x) = \frac{1}{2} \theta^{-1} \exp[-(\frac{|x-\theta|}{\theta})]$ $\theta > 0$, $-\infty < x < \infty$. It can be derived as the distribution of the difference of two independent random variables each having an identical exponential distribution.

LAPLACE EQUATION: A partial differential equation: $\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} + \frac{\partial^2 u}{\partial z^2} = 0$ useful for investigating physical problems in three dimensions involving gravitational, electrical and magnetic fields. It states that the sum of the second partial derivatives (the Laplace operator or Laplacian) of an unknown function is zero. It may also be written in the form $\nabla^2 u = 0$, where ∇^2 is called the **Laplace operator**. It is named after Pierre Simon de Laplace (1749-1827), French astronomer and mathematician.

LAPLACE OPERATOR (LAPLACIAN): The differential operator given by the divergence of the gradient of a function on Euclidean space. It is of the form $\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} + \frac{\partial^2}{\partial z^2}$. Its symbol is ∇^2 , for example,

the Laplacian of f is written $\nabla^2 f$ or Δf and called nabla squared, del squared or grad squared.

LAPLACE TRANSFORM: An integral transform related to the Fourier transform that transforms a function of time into a function of complex frequency. It is of the form $g(y) = \int_0^\infty e^{-xy} f(x) dx$. It is used in the solution of differential equations.

LAPLACE, PIERRE SIMON, MARQUIS DE (1749-1827): French mathematician and physicist, who formulated Laplace's equation and pioneered the Laplace transform which appears in many branches of mathematical physics. Many mathematical terms including Laplacian differential operator, Laplace equation, Laplace transform, etc., have been named after him. In physics, he restated and developed the nebular hypothesis of the origin of the solar system and was one of the first scientists to postulate the existence of black holes and the notion of gravitational collapse. He is regarded as one of the greatest scientists of all time.

LAPLACE'S CORRECTION: A correction to the calculation of the speed of sound in a gas. Newton assumed that the pressure-volume changes that occur when a sound wave travels through the gas are isothermal. Laplace was subsequently able to obtain agreement between theory and experiment by assuming that pressure-volume changes are adiabatic.

LAPORTE SELECTION RULE: A selection rule in atomic spectra stating that spectral lines associated with electric-dipole radiation must arise from transitions between states of opposite parity. It states that if a molecule has a centre of symmetry, transitions within a given set of p or d orbitals (i.e. those which only involve a redistribution of electrons within a given subshell) are forbidden.

LAPSE RATE: The variation of an atmospheric variable with height; unless otherwise stated the variable is temperature.

LAPTOP (NOTEBOOK): General name for a portable computer that can sit on a lap. It includes a screen, keyboard and a trackpad or trackball, which serves as the mouse, a battery which allows it to operate without being plugged into a power outlet, a power adapter that allows it to use power from an outlet and recharges the battery and several other features.

LARD: Pork fat, melted and strained from the tissues. It is an important by-product of adenosine triphosphate (ATP). Good lard melts quickly and is free from disagreeable odour. Pure lard (99% fat) is highly valued as cooking oil because it smokes very little when heated. Lard is also used in the meat packing industry.

LARGE DEVIATIONS THEORY: A branch of mathematics concerned with the rate of decay of the probabilities a family of probability measures assign increasingly unlikely events.

LARGE HADRON COLLIDER (LHC): A proton collider at CERN, which started operating in 2008 and the world's largest and highest-energy particle accelerator. It is located in a tunnel 27 kilometres in circumference and 175 metres beneath the Franco-Swiss border near Geneva, Switzerland. It accelerates two beams of protons to very high energies and then causes the beams to collide with each other, giving effective energies of about 14 TeV. It is hoped that the LHC will address some of the most fundamental questions of physics and discover the Higgs boson and provides information on physics beyond the standard model, such as super symmetry.

LARGE INTESTINE: The posterior region of the vertebrate intestine, where no digestion of food takes place. It is about 1.5 m long compared to the small intestine, which is about 6 m, but has a larger diameter than the small intestine and consists of the caecum, colon and rectum. Its function is to extract water from the indigestible materials and form faeces. Many symbiotic bacteria present in the large intestines synthesise amino acids and some vitamins, especially vitamin K.

LARGE PARTICLE: In radiation scattering, a particle with dimensions comparable with the wavelength of the radiation in the medium or larger. In practice a particle must be treated as large if its largest dimension exceeds about one-twentieth of the wavelength employed.

LARGE SCALE FARMING: The production of crops in commercial quantities.

LARGE-SCALE INTEGRATION, LSI: The technology that enables thousands of electronic components to be etched into a single integrated circuit that is just a few square millimetres.

LARGE-SCALE STRUCTURE: The structure of the distribution of visible matter in the universe at very large scales, about 100 million light-years or greater, which is observed from Earth because light or other signals from those objects has had time to reach us since the beginning of the cosmological expansion. This structure includes galaxies, clusters of galaxies, superclusters and the Shapley Concentration separated by void.

LARMOR PRECESSION: The precession of the motion of charged particles in a magnetic field. For an electron in an orbital motion around the nucleus of an atom in a magnetic field of flux density the frequency of precession is given by $\omega = \frac{eB}{4\pi m\mu}$, where e and m are the electronic charge and mass respectively, μ is the permeability and v is the velocity of the electron. This frequency, ω is known as the Larmor frequency, which is numerically equal to $2\pi \times 1.40$ MHz per oersted or $2\pi \times 111$ MHz per SI unit of magnetic field strength (ampere-turn per metre).

LARVA: A feeding stage to fuel transition into the adult form in the life cycle of insects and other invertebrates; or independent, immature animal that undergoes a profound change or metamorphosis, to assume the typical adult form. It is an intermediate stage in their life history, between hatching from the egg and becoming adult. Examples of larvae are caterpillar, which develops into butterfly; a tadpole, which may develop into a frog, toad or salamander; and the ciliated planktonic larvae of many marine animals. It is unlike the adult in form and is usually incapable of sexual reproduction.

LARYNX: A hollow chamber at the front or upper part of the trachea or wind pipe of mammals, frogs and toads containing the vocal cords that make sound when air passes over them. Movement of the cartilage in the walls of the larynx alters the tension of the vocal cords, changing the pitch of the sound emitted by the vocal cords when they vibrate. The skeletal structure of the larynx is made up of three large cartilaginous structures: the epiglottis, thyroid cartilage and cricoid cartilage and of several pairs of small cartilages. The epiglottis, a leaf-shaped elastic cartilage with its stem inserted into the thyroid notch, forms a lid to the larynx.

LASER (LIGHT AMPLIFICATION BY STIMULATED EMISSION OF RADIATION): Laser is an acronym for light amplification by stimulated emission of radiation, which works on the same principle as the maser. It is a light amplifier that uses the principle of amplification of electromagnetic waves by stimulated emission of radiation to produce monochromatic coherent radiation in the infrared, visible and ultraviolet regions of the electromagnetic spectrum. Laser light is very pure in colour, can be extremely intense and can be directed with great accuracy. Lasers are used in many modern technological devices including bar code readers, compact disc (CD) players and laser printers.

LASER COOLING: A technique, in which atomic and molecular samples are cooled through the interaction with one or more laser light fields. They include Doppler cooling, Sisyphus cooling, Resolved sideband cooling, Velocity selective coherent population trapping (VSCPT), Anti-Stokes inelastic light scattering (typically in the form of fluorescence or Raman scattering), Cavity mediated cooling and Sympathetic cooling. The basic method is to direct a set of crossed laser beams at a sample of gas, with the wavelength set so that photons are absorbed by the atoms. One atom moving towards the photons beam will lose momentum on absorbing a photon and be cooled.

LASER DYE: Organic dyes typically used dissolved in an organic solvent. The dye solution is circulated continuously through the laser chamber of the dye laser, in which it is excited (pumped) by flash lamps or by another laser.

LASER IONISATION: It occurs when a sample is irradiated with a laser beam. In the irradiation of gaseous samples, ionisation occurs via a single- or multi-photon process. In the case of solid samples, ionisation occurs via a thermal process.

LASER MICRO EMISSION SPECTROSCOPY (LAMES): Any technique in which a specimen is bombarded with a finely focused laser beam with a diameter less than $10 \mu m$ in the ultraviolet or visible range under conditions of vaporisation and thermal excitation of electronic states of sample material and in which the photon emission spectrum is observed.

LASER MICRO MASS SPECTROMETRY (LAMMS): Any technique in which a specimen is bombarded with a finely focused laser beam with a diameter less than $10 \mu m$ in the ultraviolet or visible range under conditions of vaporisation and ionisation of sample material and in which the ions generated are recorded with a mass spectrometer.

LASER PRINTER: A computer printer that rapidly produces high quality text and graphics on plain paper using a focused laser beam to place an image on a photosensitive drum, which uses electrostatic charge to print the image. In place of reflected light from an image as is used in xerography, a laser printer uses data sent from a computer to turn a laser beam on and off rapidly as it scans a charged drum. The drum then attracts toner powder to the areas not exposed to the light. Finally, the toner is fused to paper over a belt by heated rollers. They are faster, quieter and capable of producing more attractive results than standard printers and used in printing business documents.

LASER RAMAN MICROANALYSIS (LRMA): Any technique in which a specimen is bombarded with a finely focused laser beam with a diameter less than $10 \mu m$ in the ultraviolet or visible range and the intensity versus wavelength function of the Raman radiation is recorded yielding information about vibrational states of the excited substance and, therefore, also about functional groups and chemical bonding.

LASER SPECTROSCOPY: Spectroscopy that makes use of laser light; or the study of the interaction between laser radiation and matter. The beams of coherent monochromatic radiation produced by lasers have several significant advantages compared with other spectroscopic techniques, particularly in those that employ the Raman Effect. The use of lasers can greatly increase the resolution and sensitivity of conventional spectroscopic techniques, such as absorption spectroscopy, fluorescence spectroscopy or Raman spectroscopy.

LASER-JET PHOTOCHEMICAL TECHNIQUE: Technique by which high concentrations of excited states are generated in a jet, permitting the ready determination of spectroscopic properties of those states.

LASER-MARKED KEYCAP: A keycap on a keyboard that has its legend burned into the surface with a laser. It is one of the main printing technologies used for keyboard keycap printing

LASING: The process of light amplification by stimulated emission of radiation.

LASSAIGNE'S TEST: A method of testing for the presence of a halogen, nitrogen or sulfur in an organic compound. A sample is heated in a test tube with a pellet of sodium. The hot tube is dropped into pure water and the fragments ground up in a mortar. The presence of a halogen now in the form of a sodium halide is detected by precipitation with silver nitrate solution. Nitrogen is revealed by the formation of a precipitate of Prussian blue on heating part of the solution with iron(II) iron(III) ions. Lead ethanoate or sodium nitroprusside gives a precipitate with any sulfur present.

LASO: A rope with a noose at one end for catching cattle.

LASSO CELL (COLLOBLAST): A type of cell in comb jellies that is similar to the thread cells of cnidarians. Lasso cells are embedded in the tentacles of the organism; with a vesicle containing a gluey substance, which ruptures on contact, entangling a prey in the tentilla, the fine threads attached to the tentacle. Each has two protruding filaments, one straight and one spiraled. Unlike nematocytes of cnidarians, lasso cells do not penetrate the prey.

LAST COMMON ANCESTOR: The most recent known and shared common individual between two species, as well as individuals from which all others are directly descended.

LAST IN, FIRST OUT (LIFO): A method of processing data, which assumes that the newest data received, is the first to be sent out after processing.

LATCH: A circuit that stays switched on even when the original signal is switched off, used for example, in burglar alarm circuit, when a bell is kept ringing, until it is reset.

LATENCY: The time needed for an electronic message to travel across a communications system and for the remote system to respond. When a message is sent, there is latent time, for the computer that sent the packet waits for confirmation that the packet has been received. **Network latency** is the time taken for a packet of data to get from one designated point to another. Latency is inherent in all computer systems because nothing happens instantly, but usually things happen so quickly that the delay does not matter. Latency and bandwidth are the two factors that determine the network connection speed.

LATENT HEAT: The heat absorbed or radiated by a substance as it changes state such as the melting of ice or the boiling of water at constant temperature and pressure. In other words, it is the heat energy given out or taken in as the state of a substance changes at a constant temperature or the heat that is absorbed or emitted when a substance undergoes a physical phase change at constant temperature. Two of the more common forms of latent heat (or enthalpies or energies) encountered are latent heat of fusion (melting) and latent heat of vapourisation (boiling). For example, the **latent heat of vaporisation** is the energy a substance absorbs from its surrounding in order to overcome the attractive forces

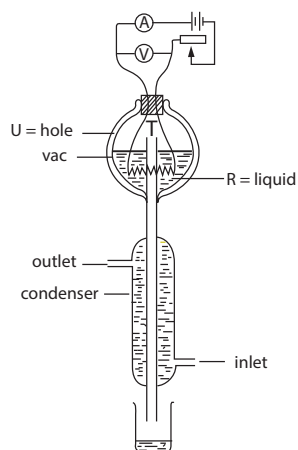
between its molecules as it changes from a liquid to gas and in order to do work against the external atmosphere as it expands. In thermodynamic terms the latent heat is the enthalpy of evaporation (ΔH), i.e. $L = \Delta H = \Delta U + p\Delta V$, where ΔU is the change in the internal energy, p is the pressure and ΔV is the change in volume. **Latent heat of fusion or enthalpy of fusion**, is the change in its enthalpy (total heat content of a system which is equal to the internal energy of the system plus the product of the volume and pressure) resulting from providing energy (heat) to a given quantity of the substance to change its state from a solid to a liquid at constant pressure. More specifically, it is the amount of heat required to change 1 g of a substance at the temperature of its melting point from the solid state to the liquid state without changing temperature. Thus, the total heat gained or released, $Q = m \times L_f + m \times C \times \Delta t$; where Q is Heat, L_f is specific latent heat of fusion, m is Mass, C is Specific heat and Δt ($t_2 - t_1$) is change in temperature. Note: Specific heat is the amount of heat required by a given mass of a material to produce a unit change in its temperature; its unit is joule per kelvin. Latent heat of fusion is associated with melting a solid or freezing liquid and it is the phase change between states of solid and liquid.

LATENT IMAGE: The primary result of radiation absorption in a photoimaging system which is susceptible to development.

LATENT INFECTION: An infection in which no symptoms are visible during the first phase of the infection. Growth of the pathogen stops soon after penetration but resumes at some later stage.

LATENT LEARNING: A form of non-associative learning in which there is apparently no immediate reward for the animal and the knowledge is not immediately expressed. An example is an animal exploring its surroundings, when learning about the geography of its home area may bring an animal no immediate benefits, but can prove vital in the future when fleeing a predator or searching for food.

LATENT PERIOD: 1. The short time that elapses between the



Electrical method for latent heat of evaporation

reception of a stimulus and the start of the response in an irritable tissue. For a contraction muscle the latent period lasts about 0.02 seconds, during which time there is initiation of an action potential, its propagation, release of calcium ions from the sarcoplasmic reticulum, activation of myosin cross-bridges which binds to actin and the generation of force. 2. The interval between exposure to an infectious organism or a carcinogen and the clinical appearance of a disease.

LATENT ROOT (CHARACTERISTIC NUMBER/ROOT/VALUE; EIGENVALUE): In a characteristic polynomial, for a square matrix A , in which x is a nonzero vector called the eigenvector that does not change its direction under the associated linear transformation and satisfies the equation $Ax = \lambda x$, the scalar, λ is the eigenvalue. In other words, it is a root, λ , of the characteristic equation $\det(A - \lambda I)$. For a linear operator A , it is a solution, λ , of the equation $AX = \lambda X (X \neq 0)$.

LATENT VECTOR (CHARACTERISTIC VECTOR; EIGENVECTOR): For a square matrix or linear operator, A , it is a nonzero vector, x that does not change its direction under the associated linear transformation and satisfies the equation $Ax = \lambda x$, where λ is a scalar of A called an eigenvalue.

LATENT VIRUS: Any virus that remains in its host organism without undergoing replication (dormant). Latent viruses, which include herpes simplex virus, may be reactivated by conditions such as reduced host defences, toxins or irradiation, to cause disease.

LATERAL: 1. Of, at, from or towards the side(s), such as a lateral vein, artery, limbs, buds, shoots, branches, etc. 2. A shoot which branches off from the main branch of a tree or shrub.

LATERAL AREA: In a solid, it is the area of its faces other than its bases. For example, the lateral area of a cube is the total area of squares forming the sides of the cube.

LATERAL FACE: Any side of a solid geometrical figure other than the base.

LATERAL INVERSION (PERVERSION): The apparent left to right reversal of an image in a mirror. For example, a person with a mark on his left cheek sees an image in a plane mirror of himself as a person with a mark on his right cheek.

LATERAL MERISTEM: A meristem arranged parallel to the sides of the organ in which it occurs and responsible for increase in girth usually by the formation of secondary tissues.

LATERAL MORaine: A deposition of sand and rocks of varying sizes and shapes along the sides of a glacier.

LATERAL ROOT: Any root that originates endogenously from the pericycle of another root.

LATERAL SQUARE: A group shown as number of elements arranged in columns and rows, where every element occurs only once in each row and each column.

LATERAL UNDULATION (SERPENTINE LOCOMOTION): A type of movement in snakes characterised by waves of lateral bending along the body from head to tail.

LATERAL VELOCITY: Component of the velocity of a celestial body that is at right angle to its line of sight velocity.

LATERAL-LINE CANAL: A sense organ in some fish and larvae of aquatic amphibians consisting of a groove on the head and along each side of the body that contains sensory hair cells. These are sensitive to variations in water pressure and are used to detect moving objects, such as predators and to monitor progress through water during swimming. The canals and the hairs within them collectively form the lateral-line system.

LATERITE: A type of red soil occurring in tropical regions, rich in iron and aluminium and widely used for making roads. They develop by intensive and long-lasting weathering of the underlying parent rock. Laterites are a source of aluminium ore; the ore exists largely in clay minerals and the hydroxides, gibbsite, boehmite and diaspore, which resembles the composition of bauxite.

LATEX: A fluid of mixed composition found in some herbaceous plants and trees or a fluid of some plants such as the rubber tree with a characteristic milky juice or an emulsion of resins and proteins suspended in a watery medium with dissolved salts, sugars, tannins, alkaloids, enzymes and other substances used as the basis for making rubber. Synthetic latexes made by emulsion polymerisation from styrene-butadiene copolymer, acrylate resins, polyvinyl acetate or other materials are used to make articles from rubber or plastics by such techniques as dipping, for example, rubber gloves; spreading as in waterproof cloth; and electrodeposition as in plastic-coated metal. They are also employed in paints and adhesives.

LATH CRYSTAL: A lamellar crystal prevailing extended along one lateral dimension.

LATICIFER: A type of elongated secretory cell found in the leaves and or stems of plants. Laticifer produces a milky fluid known as latex which penetrates various tissues in certain plants. A laticifer which consists of single cell is known as simple, whereas a joined complex is termed, compound. It may be further divided into articulated laticifers and non-articulated laticifers. **Articulated laticifers** are composed of a series of cells joined together, whereas **non-articulated laticifers** consist of one long syncytial cell. A laticifer serves as a defence against herbivores and is also involved in wound healing. Examples of plants that contain laticifers are the rubber tree and the *Hevea brasiliensis*. **Laticiferous** means having a milky fluid known as latex.

LATIMER DIAGRAM: A simple diagram summarising the standard electrode potentials for an element in different oxidation state. The different species are written in a horizontal line in order of decreasing oxidised species. The most highly oxidised form of the element is on the left, with successively lower oxidation states to the right. For example, for oxygen, the species would be in the order O_2 (0), H_2O_2 (-1), H_2O (-2). Arrows are written between adjacent species with the standard potential in volts indicated above the arrow. Often the oxidation number is included in the diagram. The standard electrode potential for nonadjacent species can be calculated, but values for common couples are often indicated by additional arrows. Latimer diagrams depend on pH and are commonly given for both acidic and alkaline conditions.

LATIN SQUARE: An $n \times n$ square array of n different symbols, each symbol appearing once in each row and once in each column. These symbols prove useful in ordering the observations of an experiment. An example of a 3×3 latin square is shown in the diagram.

A	B	C
C	A	B
B	C	A

An example of a 3×3 Latin square

LATITUDE AND LONGITUDE: Coordinate system or imaginary lines on the Earth's surface enabling any point or location of any place on the Earth's surface to be determined and described in terms of two angles subtended at its centre or imaginary lines used to locate positions on the globe. **Latitude** is a measurement of location north or south of the Equator. **Longitude** in contrast, is a measurement of location east or west of the prime meridian, which passes through Greenwich. The combination of longitude and latitude establishes a grid by which exact positions can be determined: for example, a point described as 45° N, 60° W is located 45° north of the Equator and 60° west of the Greenwich meridian.

LATORSE: This relates to anthers that split longitudinally and laterally but not towards the centre of the flower.

LATTICE: 1. In mathematics, especially in geometry and group theory, a lattice in R^n is a discrete subgroup of R^n which spans the real vector space R^n . Every lattice in R^n can be generated from a basis for the vector space by forming all linear combinations with integer coefficients. 2. In materials science and physics, the 3-dimensional

array regular arrangement of atoms, ions or molecules within a crystalline solid. They contain large numbers of particles, which are arranged in very particular ways and in a repeating pattern.

LATTICE DISTORTION: Structural disorder resulting from misalignment of the unit cells within the crystals.

LATTICE ENERGY: A measure of the strength of bonds or stability of a crystal lattice; defined as the enthalpy of formation of one mole of ionic compound from gaseous ions and as such is invariably exothermic. It is given by the energy that would be released per mole, if atoms, ions or molecules of the crystal were brought together from infinite distances apart to form the lattice.

LATTICE ENTHALPY: The enthalpy change when 1 mole of a crystal structure is formed from gaseous ions.

LATTICE GAUGE THEORY: A formulation of gauge theories in which space and time are taken to be discrete, rather than continuous. In this theory, it is necessary to take the continuum limit at the end of calculations in lattice gauge theory. The theory is used to make calculations for gauge theories with strong coupling, such as quantum chromodynamics, in which many of the important features of the theory cannot be obtained by perturbation theory. Lattice gauge theory is particularly suitable for numerical and computational calculations. Statistical mechanics techniques can be used in the theory.

LATTICE MULTIPLICATION: A method of multiplication that uses a lattice to multiply two multi-digit numbers. It requires only basic multiplication facts and addition of 1-digit numbers in a lattice diagram. It is mathematically identical to long multiplication algorithm, but it breaks the process into smaller steps, which may be easier to use.

LATTICE POINTS: Points of intersection of two or more grid lines in a lattice.

LATTICE STRUCTURE: The orderly arrangement of atoms (ions) in a crystalline material.

LATTICE VIBRATIONS: The periodic vibrations of the atoms, ions or molecules in a crystal lattice about their equilibrium positions. Because the atoms are bound not to their average positions but to the neighbouring atoms, vibrations of neighbours are not independent of each other. On heating, the amplitude of the vibrations increases until they are so energetic that the lattice breaks down. The temperature at which this happens is the melting point of the solid and the substance becomes a liquid. On cooling, the amplitude of the vibrations diminishes. At absolute zero, a residual vibration persists, associated with the zero point energy of the substance. The increase in the electrical resistance of a conductor is due to increased scattering of the free conduction electrons by the vibrating lattice particles.

LATUS RECTUM (LATERA RECTA): A chord of an ellipse passing through a focus and perpendicular to the major axis of the ellipse.

LAUE, MAX THEODOR FELIX VON (1879-1960): German physicist and professor at University of Berlin in 1919-43. In 1943, he joined the Max Planck Institute at Göttingen. He is best known for his discovery in 1912 of x-ray diffraction, demonstrating that x-rays are electromagnetic radiation similar to light and that the molecular structure of crystals is a regularly repeating arrangement; for which he was awarded the 1914 Nobel Prize for Physics.

LAUGHING GAS (NITROUS OXIDE; DINITROGEN MONOXIDE; NITROGEN(I) OXIDE): A colourless non-flammable, sweet-smelling and sweet-tasting gas with the chemical formula N_2O , density 1.977 g/L (gas), melting point -90.86°C (182.29 K) and boiling point -88.48°C (184.67 K). It is used as an anaesthetic. The gas is prepared commercially by the thermal decomposition of ammonium nitrate, NH_4NO_3 , at about 240°C (513 K) to produce nitrous oxide and water. The reaction must be carefully controlled to prevent explosive decomposition of the nitrous oxide.

LAUNCH VEHICLE: A rocket system used to launch a satellite, spacecraft, etc., into space. The velocity V needed to launch a satellite

into space is given as: $V = \sqrt{\frac{GM}{R_E}} = \sqrt{gR_E} = 7.9 \times 10^3 \text{ ms}^{-1}$.

LAURASIA: The hypothetical landmass in Northern Hemisphere formed 200 million years ago by the splitting of the hypothetical supercontinent Pangaea to form North America, Greenland, Europe and Asia excluding the Indian subcontinent. The southern landmass is called Gondwanaland.

LAURENT EXPANSION (LAURENT SERIES): Of a complex function $f(z)$, representing a power series $f(z) = \sum_{n=-\infty}^{\infty} a_n(z-a)^n$ which

includes terms of negative degree. It may be used to express complex functions in cases where a Taylor series expansion cannot be applied.

LAURIC ACID (DODECANOIC ACID; N-DODECANOIC ACID; DODECYLIC ACID; DODECOIC ACID; LAUROSTEARIC ACID; VULVIC ACID; 1-UNDECANECARBOXYLIC ACID; DUODECYLIC ACID): A white crystalline, long chain carboxylic or fatty acid with the chemical formula $\text{CH}_3(\text{CH}_2)_{10}\text{COOH}$, density 0.880 g/cm^3 , melting point 43.2°C (316.35 K) and boiling point 298.9°C (572). Glycerides of the acid are present in natural fats and oils. Lauric acid is the main acid in coconut oil and in palm kernel oil but not in palm oil. It is also found in human milk (6.2% of total fat), cow's milk (2.9%) and goat's milk (3.1%). It is used in making soaps and detergents, insecticides, cosmetics and lauryl alcohol.

LAVA: Molten rock material (magma) that has forced its way to the surface of the Earth or erupts from a volcano. It cools and solidifies to form extrusive igneous rock.

LAVENDER: *Lavandula officinalis*, a small shrub with small grey-green leaves with sweetly scented pale mauve flowers from which an essential used in perfumes is derived. The lavenders are members of the family Lamiaceae (formerly Labiatae).

LAVES PHASE: Intermetallic phases that occur for alloys of the type AB_2 , in which the atoms of the element A are arranged in a diamond-like structure and the atoms of the element B are arranged in tetrahedral around the atoms of A. It was named after Fritz Laves who first described them. There are three different classification classes: cubic, hexagonal MgZn , and hexagonal MgNi_2 . Laves phases are of particular interest in modern metallurgy research because of their abnormal physical and chemical properties.

LAVOISIER, ANTOINE LAURENT (1743-94): French chemist, who studied law but went into science and best known for proving the law of conservation of mass. In the 1770s, he discovered oxygen and nitrogen in air and demolished the phlogiston theory of combustion by demonstrating the role of oxygen in the process. In 1783, he made water by burning hydrogen in oxygen. He also devised a rational nomenclature for chemical compounds.

LAW: A statement of a scientific fact or phenomenon that is invariable under given conditions, which can predict the outcome of observations before they are made or a statement expressing what has been seen always to happen in certain conditions. For example, Ohm's law predicts that the voltage across any conductor, is proportional to the electric current flowing through it and inversely proportional to the resistance provided the temperature is constant: $V = IR$.

LAW OF AVERAGES: The supposed expectation that a possible event is bound to occur regularly with a frequency approximating to its probability. For example, in tossing a coin and getting five heads in a row, the law of averages would make tails more likely to appear in the next toss of the coin, but this is in fact false.

LAW OF CONSERVATION OF CHARGE (CHARGE CONSERVATION): The law which states that the net charge of an isolated system remains constant during any physical process, such as two charge objects making contact and separating. In other words, any system that is not exchanging mass or energy with its surroundings will never have a different total charge at any two times.

LAW OF COMBINING VOLUMES (GAY-LUSSAC'S LAW): It states that at constant temperature and pressure, the volumes of reacting gases or any gaseous products, can be expressed as ratios of small whole numbers.

LAW OF CONSERVATION OF ENERGY: It states that energy cannot be created nor destroyed but may be changed from one form to another.

LAW OF CONSERVATION OF LINEAR MOMENTUM (CONSERVATION OF LINEAR MOMENTUM): It states that if no external force such as friction acts on a system of colliding objects, the total momentum of the object in a given direction before collision equals total momentum after collision in the same direction.

LAW OF CONSERVATION OF MATTER (CONSERVATION OF MATTER; LAW OF CONSERVATION OF MASS): A fundamental principle of classical physics which states that in a chemical reaction, the total amount of matter/mass in a closed system remains constant and that matter cannot be created nor destroyed in a chemical reaction.

LAW OF CONSERVATION OF MOMENTUM: It states that the momentum of an object remains constant. For example, the total momentum of two objects before collision is equal to the total momentum of the two objects after the collision.

LAW OF COSINE: An equation relating the lengths of the sides of a triangle and one of its angles. It can be used to calculate the third side of a triangle if the other two sides and the angle between them are known: $c^2 = a^2 + b^2 - 2ab \cos C$. In this formula a, b and c are the three sides of the triangle and C is the angle opposite the side. The other sides can also be found by $a^2 = b^2 + c^2 - 2bc \cos A$ and $b^2 = a^2 + c^2 - 2ac \cos B$.

LAW OF DEFINITE PROPORTIONS (LAW OF CONSTANT COMPOSITION; LAW OF CONSTANT PROPORTION; PROUST'S LAW): Established by Joseph Proust (1754-1826), French chemist, it states that the proportions of the amounts of the elements in a pure compound are always the same and are independent of the method by which the compound was produced or that all pure samples of the same chemical compound always contain the same elements combined in the same proportion by mass, no matter how they are made.

LAW OF EXCLUDED MIDDLE (EXCLUDED MIDDLE LAW; PRINCIPLE OF EXCLUDED MIDDLE): In logic, one of the three classic laws of thought, which states that for any proposition, either that proposition is true or its negation is true: there is neither a third truth value nor do any statements lack truth value.

LAW OF FLOTATION: It states that a floating body sinks into a fluid until it has displaced its own weight of the fluid, in which it floats; at which point weight is equal to upthrust and the body is in equilibrium: weight of floating object = weight of fluid displaced; and mass of floating object = mass of fluid displaced.

LAW OF INDEPENDENT ASSORTMENT: It states that each of the two alleles of a gene (gene pair) may combine randomly with either of the two alleles of another gene. It was formulated by Johann Mendel Gregor Mendel (1822-1884), Austrian botanist and monk. It can also be stated that each member of a pair of alleles (genes) combine randomly with either of another pair in a gamete.

LAW OF LARGE NUMBERS: In statistics, it is the law which states that by performing the same experiment a large number of times, the average of the results obtained should be closer to the expected value.

LAW OF MASS ACTION (LAW OF CHEMICAL EQUILIBRIUM): Fundamental law of chemical kinetics formulated in 1864-79 by the Norwegian scientists Cato M. Guldberg (1836-1902) and Peter Waage (1833-1900). It states that the reaction rate of any simple chemical reaction is proportional to the product of the molar concentrations of the reacting substances, each raised to the power corresponding to the number of molecules of that substance in the reaction.

LAW OF MULTIPLE PROPORTIONS: One of the fundamental laws of stoichiometry, first discovered by the English chemist John Dalton in 1803. It states that if two elements form more than one compound, then there is a simple relationship between the masses of

one of the elements and a fixed mass of the other element. In other words, when two elements A and B combine to form more than one compound, the different masses of one element that combine with a constant mass of the other are in a simple whole number ratio.

LAW OF OCTAVES (NEWLANDS' LAW): An attempt at classifying elements made by John Newlands (1837-98) in 1863, which predated Mendeleev's periodic table. He arranged 56 elements in order of increasing atomic mass in horizontal groups of eight, with each new row directly beneath the previous one. He pointed out that each element resembled the element eight places from it in the list. He named this periodic repetition the law of octaves, by analogy with the musical scales. Examples are oxygen and sulfur; nitrogen and phosphorus and fluorine, chlorine, bromine and iodine. In some cases it was necessary to put two elements in the same position. The proposal was rejected at the time.

LAW OF PARTIAL PRESSURES (DALTON'S LAW): Formulated by John Dalton (1766-1844), British chemist and physicist, it states that the total pressure exerted by a mixture of gases is the sum of the partial pressures of the individual gases. The law of partial pressures may be expressed as $P = P_A + P_B + \dots P_n$. This expression is valid only for mixtures of ideal gases.

LAW OF SEGREGATION: Mendel's first law, stating that a character is controlled by a pair of genes (alleles) and of such a pair only one is represented in a gamete. Allele pairs separate during gamete formation and then randomly re-form (re-combine) during the fusion of gametes at fertilisation.

LAW OF SINES: It expresses a relationship involving the sides and angles of a triangle as follows: $\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C}$, where a , b and c are the lengths of the sides of a triangle and A , B and C are the angles opposite these three sides, respectively.

LAW OF SUPERPOSITION: The principle that a sedimentary rock layer is older than the layers above it and younger than the layers below it if the layers are not disturbed.

LAW OF TANGENTS: If a , b and c are the lengths of the sides of a triangle and A , B and C are the angles opposite these three sides, respectively, then the law of tangents states that the following

relations will be true: $\frac{a-b}{a+b} = \frac{\tan\left[\frac{1}{2}(A-B)\right]}{\tan\left[\frac{1}{2}(A+B)\right]}$, $\frac{b-a}{b+c} = \frac{\tan\left[\frac{1}{2}(B-C)\right]}{\tan\left[\frac{1}{2}(B+C)\right]}$, and $\frac{c-a}{c+a} = \frac{\tan\left[\frac{1}{2}(C-A)\right]}{\tan\left[\frac{1}{2}(C+A)\right]}$.

LAW OF UNIVERSAL GRAVITATION: It states that all objects in the universe attract each other through gravitational force. The size of the force is directly proportional to the product of their masses and inversely proportional to the square of the distance between their centres.

LAWRENCIUM: A short-lived radioactive metallic transuranic element belonging to the actinides with the symbol Lr, atomic number 103, mass number of most stable isotope 262 (half-life of 3.6 hours) and melting point of about 1627 °C (1900 K). It is produced artificially from californium and other elements.

LAWS OF ELECTROMAGNETIC INDUCTION: Formulated by Michael Faraday, English physicist in 1831, they are laws based on experiments, showing the relationship between electric circuit and magnetic field as follows: (i) Whenever there is a change in the magnetic flux surrounding a conductor, an electromotive force (emf) is induced in the conductor. (ii) The strength of the induced electromotive force is proportional to the rate of change of magnetic flux linking the conductor. (iii) If the circuit is complete, the induced current due to the induced electromotive force produces a magnetic effect which

always opposes the change in the magnetic flux inducing it. The laws are applied in electrical machines, industries and medical field, etc; and used in electrical generators, induction cookers, electromagnetic flow meters, musical instruments, etc.

LAWS OF REFLECTION: In physics, Laws of light, which govern the reflection of light-rays off smooth conducting surfaces, such as polished metal or metal-coated glass mirrors. There are two laws of reflection. Law 1 states that the angle at which a ray of light strikes a smooth surface, called the **angle of incidence**, equals the angle at which it is reflected from a smooth surface, called the **angle of reflection**. Law 2 states that the reflected ray, the incident ray and the normal line at the point of incidence all lie in the same plane. The ray of light that strikes a smooth surface is known as the **incident ray**. The ray of light that leaves the smooth surface is known as the **reflected ray**.

LAWS OF REFRACTION (SNELL'S LAW; SNELL-DESCARTES LAW): Laws, which govern the behaviour of light-rays as they propagate across a sharp interface between two transparent dielectric media. They state that: (i) The incidence ray, the refracted ray and the normal at the point of incidence all lie in the same plane. (ii) The ratio of the sine of the angle of incidence in one medium to the sine of the angle of refraction in another medium is constant for the two media. Mathematically, it is given by $n = \sin i / \sin r$, where $\sin i$ is the sine of the angle of incidence, $\sin r$ is the sine of the angle of refraction and n is a constant.

LAWSON CRITERION: Derived by John D. Lawson in 1955, it is the measure of the density of the reacting particles required and the time for which they need to react in order to produce more energy than was used in raising the temperature of deuterium and tritium at the ignition temperature.

LAX: Loose; having parts that are not compact, firm or tense.

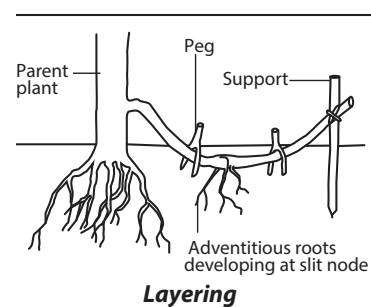
LAXATIVE (PURGATIVE; APERIENTS): Substances such as foods, compounds or drugs taken orally or inserted into the rectum, vagina or urethra (suppository) to cause the bowels to empty easily or any substance which encourages movements of bowels or loosen the stool. It is most often used to treat constipation.

LAYER: 1. A bird, especially a hen that produces eggs regularly. A very successful layer should produce between 19 and 20 dozen eggs in 12 months. 2. A stem of a plant which develops roots when it touches the soil or a plant system that has been fastened partly under the ground in order to grow roots and so become a separate plant. 3. Thickness of material or land over a surface, forming a horizontal division or a flat area of substance under or over another area.

LAYER LATTICE: A crystal structure found in substances such as graphites, micas and clays in which the atoms are chemically bonded in plan layers, with relatively weak forces between atoms in adjacent layers.

LAYER SILICATE: A member of the phyllosilicate group of minerals which dominate the clay fraction of soils. Examples are biotite, montmorillonite and kaolinite.

LAYERING: A method of propagating by inducing plants to regenerate missing parts such as roots from parts that are still attached to the parent plant. For soil layering, a low branch is held firm on the ground with fertile moist soil so that it takes root while maintaining the health and vigour of the parent plant. For air layering, a branch is deeply slit and the wound is covered with a ball of earth or moss and kept moist until roots develop; the branch is then severed and transplanted. It occurs naturally for drooping black raspberry or forsythia stems, whose trailing tips root where they come in contact with the soil.



LAYERS CRAMP: A condition found in pullets after the first few weeks of their laying life. The birds appear weak but the trouble disappears after a few days.

LAYING CAGE (BATTERY FARMING; BATTERY CAGES): A specially built cage for egg-laying hens. The cage is arranged in tiers to allow the birds to stand comfortably. It has generated controversy among advocates for animal welfare and animal rights and industrial egg producers.

LC50: The concentration of a chemical substance required to kill 50% of the test population.

LCD (LIQUID CRYSTAL DISPLAY): A technology used in flat-panel displays that allows displays to be much thinner than cathode ray tube (CRT) technology. It is made by sandwiching a layer of liquid crystal solution between two sheets of a flexible polarising material. The liquid crystals switch pixels on and off to reveal a specific colour. LCD panels do not have a backlight to produce light and require external light such as a cold cathode fluorescent lamp to produce a visible image. When light is projected through the LCD panels, individual pixels are opened or closed to allow light through or block it. The separate colours are then converged and projected on to the screen. The two basic types of LCDs are passive matrix or active matrix display. The active matrix LCD is also known as a thin film transistor (TFT) display. LCDs are used for computer monitors, televisions, instrument panels; smaller LCDs are used in digital cameras, watches, calculators, and mobile telephones, and smartphones. An **LCD projector** is a type of projector based on LCD technology for displaying video, images or computer data on a screen or other flat surface in which generated light is projected onto mirrors which split it up into its 3 primary colours: red, green and blue. The colours then pass through three separate glass prisms. When light is projected through the LCD panels, individual pixels are opened or closed to allow light through or block it. The separate colours are then converged using another prism and projected on to the screen.

LD50: The amount of a chemical or pharmacological or a toxic substance required to kill 50% of a test population or experimental animals. For each LD50, the species and weight of the animal and the route of administration are specified. This test has been criticised by animal protection organisations and by many scientists, but it is still used. They are used in toxicology and in the bioassay of the therapeutic compounds.

LDAP (LIGHTWEIGHT DIRECTORY ACCESS PROTOCOL): In computing, it is a protocol used for accessing information directories. LDAP support is implemented in Web browsers and e-mail programs, which can query an LDAP-compliant directory. LDAP is protocol similar to HTTP and FTP and uses the ldap:// prefix in its URL.

LE CHATELIER'S PRINCIPLE: Formulated by Henri Louis Le Châtelier (1850-1936), French chemist and metallurgist, it describes the response of a system in equilibrium to a change in one of the variables determining the equilibrium. If a system is in equilibrium, any change imposed on the system tends to shift the equilibrium to nullify the effect of the applied change. The principle, which is a consequence of the law of conservation of energy is applied to chemical equilibria. For example, in the generation $2\text{SO}_{2(g)} + \text{O}_{2(g)} \rightleftharpoons 2\text{SO}_{3(g)}$ an increase in pressure on the reaction mixture displaces the equilibrium to the right, since this reduces the total number of molecules present and thus decreases the pressure. The standard enthalpy change for the forward reaction is exothermic. Thus, an increase in temperature displaces the equilibrium to the constant, falling with increasing temperature.

LEACHATE: Liquid that passes through matter and extracts solutes, suspended solids or any other component of the material through which it has passed.

LEACHING: 1. The washing away of water-soluble plant nutrients from the soil of roots usually by rainwater draining quickly through the soil. 2. Applying a small amount of excess irrigation where the water has a high salt content to avoid salts from building up in the soil. 3. A natural process by which water-soluble chemical substances such

as calcium, fertilisers and pesticides are removed from topsoil into lower subsoil by the action of percolating ground water, which is the gravitational movement of water through the soil pores. The leached out chemicals, called leachates can contain environmentally harmful substances, causing pollution to surface and sub-surface water.

LEAD: A chemical element in Group 14 (IVA) of the Periodic Table, with the symbol Pb, atomic number 82, atomic weight 207.19, melting point 327.4 °C (600.55 K) and boiling point 1740 °C (2013 K). It is a heavy metal, bluish in colour, but tarnishes to dull grey. It is pliable, inelastic and easily fusible. Its main ore is the sulfide galena (PbS); other minor sources include anglesite (PbSO₄) cerussite (PbCO₃) and litharge (PbO). The metal is extracted by roasting the ore to give the oxide, followed by reduction with carbon. Silver is also recovered from the ores. Lead forms alloys with many metals and is generally employed in the form of alloys in most applications such as solder, pewter, bearing metals, type metals and fusible alloys. Alloys formed with tin, copper, arsenic, antimony, bismuth, cadmium and sodium are all of industrial importance. Lead has a variety of uses including building construction, lead-plate accumulators, cable covering, pigments, bullets and shot. Chemically, it forms compounds with the +2 and +4 oxidation states, the lead(II) state being the more stable. Lead and its compounds are poisonous.

LEAD CHAMBER PROCESS (CHAMBER PROCESS): A method of making sulfuric acid by the catalytic oxidation of sulfur dioxide with air using a potassium nitrate catalyst in water, now supplanted by the contact process. The process was carried out in lead containers (which was expensive) and only produced dilute acid. It was replaced in 1876 by the contact process.

LEAD EQUIVALENT: A factor that compares any form of shielding against radioactivity to the thickness of lead that gives the same reduction in radiation dose rate as the material in question.

LEAD-ACID ACCUMULATOR (LEAD-ACID CELL): An accumulator in which the electrodes consist of lead and the electrolyte consists of dilute sulfuric acid. It was invented in 1859 by French physicist Gaston Planté. The electrodes are usually cast from a lead alloy containing 7-12% of antimony to give increased hardness and corrosion resistance and a small amount of tin for better casting properties and also calcium or selenium alloy. The electrodes are coated with a paste of lead(II) oxide (PbO) and finely divided lead. After insertion into the electrolyte, a 'forming' current is passed through the cell to convert the PbO on the negative plate into a sponge of finely divided lead. On the positive plate the PbO is converted to lead(IV) oxide (PbO₂). The equation for the overall reaction during discharge is $\text{PbO}_2 + 2\text{H}_2\text{SO}_4 + \text{Pb} \rightarrow 2\text{PbSO}_4 + 2\text{H}_2\text{O}$. The reaction is reversed during charging. Each cell gives an e.m.f of about 2 volts. An example is the 12 volt lead battery commonly used in motor vehicles consisting of six cells of 2 volts each and recharged by the vehicle's alternator or dynamo when the engine is running. The lead-acid battery produces 80-120 kJ per kilogram.

LEAD-FREE PETROL: Petrol that does not contain anti-knocking agent, tetraethyl lead. The use of lead(IV) tetraethyl in petrol is now often restricted or banned because it produces air pollution and poisons catalytic converters.

LEAD(II) CARBONATE: A poisonous white solid, with the chemical formula PbCO₃ and relative density 6.6. It is insoluble in water and occurs naturally as the mineral cerussite, which is isomorphous with aragonite. It may be prepared in the laboratory by the addition of cold ammonium carbonate solution to a cold solution of a lead (II) salt (acetate or nitrate). It decomposes at 315 °C (588 K) to lead (II) oxide and carbon dioxide.

LEAD(II) CARBONATE HYDROXIDE (WHITE LEAD; BASIC LEAD CARBONATE): A white powder with the chemical formula 2PbCO₃·Pb(OH)₂ and relative density 6.14. It is insoluble in water, slightly soluble in aqueous carbonate solutions and decomposes at 400 °C (673 K). It is a mixture of lead carbonate and lead hydroxide. Lead(II) carbonate hydroxide occurs as the mineral hydroxycerussite of variable composition. It was previously manufactured from lead in

processes using spent tanning bark or horse manure, which released carbon dioxide. It is currently made by electrolysis of mixed solutions such as ammonium nitrate, nitric acid, sulfuric acid and acetic acid using lead anodes. For the highest grade product the lead must be exceptionally pure as small amounts of metallic impurity impart grey or pink discolourations. The material was used widely in paints, both for art work and for commerce, but it has the disadvantage of reacting with hydrogen sulfide in industrial atmospheres and producing black lead sulfide. The poisonous nature of lead compound has also contributed to the declining importance of this material.

LEAD(II)ETHANOATE (LEAD(II)ACETATE): A white crystalline solid compound with a sweetish taste and the chemical formula $\text{Pb}(\text{CH}_3\text{COO})_2$. It is soluble in water and glycerin and slightly soluble in ethanol and forms a variety of complexes in solution. It exists as the anhydrous compound trihydrate, $\text{Pb}(\text{CH}_3\text{COO})_2 \cdot 3\text{H}_2\text{O}$ (monoclinic) and as a decahydrate, $\text{Pb}(\text{CH}_3\text{COO})_2 \cdot 10\text{H}_2\text{O}$. The common form is the trihydrate. Lead(II) acetate is used as a mordant in textile printing and dyeing, a drying agent in paints and varnishes and also as a reagent to make other lead compounds and as a fixative for some dyes.

LEAD(II) NITRATE (LEAD NITRATE; PLUMBOUS NITRATE; LEAD DINITRATE): A white, colourless crystal or powder with the molecular formula $\text{Pb}(\text{NO}_3)_2$, molar mass 331.2g/mol, density 4.53 g/cm³ at 20 °C and a melting point of 270 °C (543 K) at which it decomposes. Together with lead ethanoate (lead acetate), they are the only water-soluble compounds of lead. It can be made by two methods: dissolving metallic lead in aqueous nitric acid $\text{Pb} + 2\text{HNO}_3 \rightarrow \text{Pb}(\text{NO}_3)_2 + \text{H}_2\uparrow$; or by dissolving lead(II) oxide in nitric acid: $\text{PbO} + 2\text{HNO}_3 \rightarrow \text{Pb}(\text{NO}_3)_2 + \text{H}_2\text{O}$. When heated, it decomposes readily into oxygen, nitrogen(IV) oxide gas and lead(II) oxide: $2\text{Pb}(\text{NO}_3)_2 \Rightarrow 2\text{PbO} + 4\text{NO}_2 + \text{O}_2$. It is a hazardous substance, which is injurious to health if inhaled or ingested. It is not combustible but it accelerates the burning of combustible materials and is sometimes used in pyrotechnics such as fireworks.

LEAD(II) OXIDE (LEAD MONOXIDE; PLUMBOUS OXIDE): Lead(II) oxide is the International Union of Pure and Applied Chemistry's (IUPAC) name for lead monoxide or litharge. It is a yellow crystalline inorganic compound with the chemical formula PbO. It exists in two polymorphs, red, having a tetragonal crystal structure and yellow, having an orthorhombic crystal structure. Both forms occur naturally as rare minerals. The red form is known as **litharge** and the yellow form is known as **massicot**. It can be prepared by heating the nitrate and is manufactured by heating molten lead in air at approximately 600 °C (873 K). If the temperature used is lower than the melting point of the oxide, the product is massicot; above this, litharge is formed. Variations in the temperature and in the rate of cooling give rise to crystal vacancies and red orange and brown forms of litharge can be produced. The oxide is amphoteric, dissolving in acids to give lead(II) salts and in alkalis to give plumbates. It is used extensively in manufacturing of lead glasses and ceramic glazes as well as in fine dinnerware.

LEAD(II)SULFATE (SULFURIC ACID LEAD SALT; ANGLESITE): A white crystalline solid or powder with the chemical formula PbSO_4 , relative density 6.2 and melting point of 1170 °C (1443 K). It is virtually insoluble in water and soluble in solutions of ammonium salts. It occurs naturally as the mineral anglesite, PbSO_4 . It is toxic by inhalation, ingestion and skin contact and its use has declined. It may be prepared in the laboratory by adding any solution containing sulfate ions to solutions of lead (II) ethanoate. Basic lead(II) sulfate may be made by shaking together lead(II) sulfate and lead(II) hydroxide in water. This material has been used in white paint as it is not so susceptible to discolouration through reaction with hydrogen sulfide.

LEAD(II)SULFIDE: A black crystalline solid with the chemical formula PbS and insoluble in water. It occurs naturally as galena, the main ore of lead. It may be prepared in the laboratory by the reaction of hydrogen sulfide with soluble lead(II) salts. Lead (II)sulfide has been used as an electric rectifier and as detection element materials

in various infrared detectors. As an infrared detector, PbS functions as a photon detector, responding directly to the photons of radiation. It is used to measure radiation either by measuring the tiny photocurrent the photons cause when they hit the PbS material or by measuring the change in the material's electrical resistance that the photons cause.

LEAD(IV) ETHANOATE (LEAD TETRACETATE): A colourless solid with the chemical formula $\text{Pb}(\text{C}_2\text{H}_3\text{O}_2)_4$. It is a lead salt of acetic acid. It decomposes in water and is soluble in pure ethanoic acid. It may be prepared by dissolving dlead(II) lead(IV) oxide in warm ethanoic acid. In solution it behaves essentially as a covalent compound (no measurable conductivity) in contrast to the lead(II) salt, which is a weak electrolyte. It is a strong oxidising agent, a source of acetyloxy groups and a general reagent for the introduction of lead into organo lead compounds.

LEAD(IV)HYDRIDE (PLUMBANE): An extremely unstable gas with the chemical formula PbH_4 , molar mass 211.23 g/mol and boiling point -13 °C (260 K). It is formed by the action of acids on magnesium-lead alloys. It was first reported in 1924, although doubts have since been expressed about the existence of the compound. Derivatives of plumbane include lead tetrafluoride (PbF_4) and tetraethyl lead ($(\text{CH}_3\text{CH}_2)_4\text{Pb}$).

LEAD(IV)OXIDE (LEAD DIOXIDE): Lead(IV) Oxide is the International Union of Pure and Applied Chemistry's (IUPAC) name for lead dioxide with the chemical formula PbO_2 . It is dark brown or black and a poisonous crystalline compound insoluble in water and slightly soluble in concentrated sulfuric and nitric acids. It exists in two crystalline forms: alpha, which has orthorhombic symmetry and occurs naturally as the rare mineral scrutinyite; the beta phases have tetragonal symmetry and occur naturally as rare mineral plattnerite, which may be prepared by the oxidation of lead(II) oxide by heating with alkaline chlorates or nitrates or by anodic oxidation of lead(II) solutions. It is an oxidising agent and readily reverts to the lead(II) oxidation state, by its conversion to Pb_2O_4 and PbO on heating. It reacts with hydrochloric acid to evolve chlorine. It is used to manufacture batteries, explosives, textile and for dyeing, in the manufacture of safety matches and also was widely used until the mid 1970s as an adsorbent for sulfur dioxide in pollution monitoring.

LEAD(IV) TETRAETHYL (TETRAETHYL LEAD): A colourless, extremely poisonous, oily liquid with the chemical formula $\text{Pb}(\text{C}_2\text{H}_5)_4$. It is insoluble in water but soluble in benzene, ethanol, ether and petroleum. It may be prepared by the reaction of hydrogen and ethane with lead but a more convenient laboratory and industrial method is the reaction of a sodium-lead alloy with chloroethane. It is produced industrially by the electrolysis of ethylmagnesium chloride (the Grignard reagent) using a lead anode and slowly running additional chloroethane onto the cathode. Lead tetraethyl is used in fuel for internal-combustion engines (along with 1-2-dibromoethane) to increase the octane number and reduce knocking. The use of lead (IV) tetraethyl in petrol is now often restricted or banned because it produces air pollution and poisons catalytic converters.

LEAD MONOXIDE (LEAD(II) OXIDE; PLUMBOUS OXIDE): Lead(II) oxide is the International Union of Pure and Applied Chemistry's (IUPAC) name for lead monoxide or litharge. It is a yellow crystalline inorganic compound with the chemical formula PbO. It exists in two polymorphs, red, having a tetragonal crystal structure and yellow, having an orthorhombic crystal structure. Both forms occur naturally as rare minerals. The red form is known as **litharge** and the yellow form is known as **massicot**. It can be prepared by heating the nitrate and is manufactured by heating molten lead in air at approximately 600 °C (873 K). If the temperature used is lower than the melting point of the oxide, the product is massicot; above this, litharge is formed. Variations in the temperature and in the rate

of cooling give rise to crystal vacancies and red orange and brown forms of litharge can be produced. The oxide is amphoteric, dissolving in acids to give lead(II) salts and in alkalis to give plumbates. It is used extensively in manufacturing of lead glasses and ceramic glazes as well as in fine dinnerware.

LEADER SEQUENCE: 1. A polynucleotide region between promoter and structural gene, necessary for the correct transcription of DNA into mRNA. 2. Also known as **signal sequence**, it is n-terminal extension of polypeptides which is necessary for the transport of the protein into or through the membrane or for its secretion into the extracellular medium.

LEADING COEFFICIENT: The coefficient of the term of highest degree in a polynomial in a single variable. For example, in the polynomial, $-7x^3 + 6x^2 + 2x - 5$, the leading coefficient is -7.

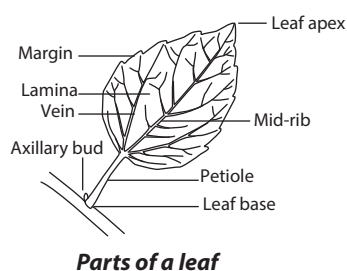
LEADING CURRENT: An alternating current that reaches its maximum value of up to 90° before the voltage that produces it in a cycle. In other words, the waveform of the current leads that of the voltage by a certain angle (called phase angle), as the current rises more quickly than the voltage. This usually happens in inductive circuits due to the presence of capacitors or inductors. In a purely resistive circuit, where there is no capacitor or inductor the current passing through a load rises or decreases at the same time as that of the voltage across it and there is no phase angle between them and are said to be **in phase**.

LEADING DIAGONAL (MAIN DIAGONAL; PRIMARY DIAGONAL; PRINCIPAL DIAGONAL): The elements of a matrix running from the top left corner and proceeding down to right edge of the bottom. The main diagonal of a square matrix is the diagonal which runs from the top left corner to the bottom right corner. The main diagonal of a rectangular matrix is the diagonal which runs from the top left corner and steps down and right, until the right edge or the bottom edge is reached. For example, the matrix in the diagram has ones down its

$$\text{main diagonal: } \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}.$$

LEADING STRAND: The new continuous complementary DNA strand synthesised along the template strand in the mandatory 5' to 3' direction.

LEAF: A flattened (laminar) lateral outgrowth from the node of the stem of a plant or a branch specialised for photosynthesis with a bud in the axil or point of attachment with the stem. Its flatness makes it to expose the chloroplasts to more light and to increase the absorption of carbon dioxide at the expense of water loss. Although most leaves are flat, broad or bladelike, they also may have many other shapes, such as round, oval or feathery. Green leaves derive their colour from a green pigment called chlorophyll. In size, leaves range from only several millimetres long, as in the water plant *Elodea*, to 15 to 18 metres in some palm trees. They also vary in arrangement of veins, type of attachment to the stem and texture. Leaves may be simple or compound (divided into leaflets). The various types include: cotyledons or seed leaves; scale leaves, which lack chlorophyll and develop on rhizomes or protect the inner leaves of a bud; foliage leaves, which are the main organs for photosynthesis and transpiration and consist of a petiole and a blade; and bracts and floral leaves, such as sepals, petals, stamens and carpels, which are specialised



Parts of a leaf

for reproduction. Leaves may be modified for special purposes, for example, the leaf bases of bulbs are swollen with stored food and the reduction of leaves to spines reducing incidence of transpiration and their photosynthetic function is carried out by another organ, such as cladode.

LEAF AREA INDEX (LAI): A dimensionless value, which is the total surface area of the leaves of plants in a given area divided by the area of ground covered by the plants. In an area of dense vegetation such as a forest the LAI will be high (6) and 0 for bare ground.

LEAF AREA RATIO (LAR): The total leaf area of a plant divided by the dry mass of the entire plant, indicating the efficiency with which a plant uses its leaves to produce plant material. Its unit is square centimetres per gram (cm^2/g) or square metres per kilogram (m^2/kg). It is equal to the product of the leaf mass ratio (LMR) and the specific leaf area (SLA).

LEAF BUTTRESS: A lateral prominence on the shoot apex, destined to differentiate into a leaf. It is the earliest stage in the development of a leaf primordium and later forms the leaf base, the remainder of the leaf growing upwards and outwards from the buttress.

LEAF CULTURE: A form of tissue culture in which excised leaves, leaf material or leaf primordia are grown on a sterile growth medium. Mature leaves can be kept healthy under culture conditions for considerable periods. Leaf primordia have been used to study growth and differentiation processes.

LEAF CURL: A plant disease that causes leaves to curl. It is caused by the fungus *Taphrina deformans*, genus *Taphrina* or begomovirus.

LEAF FALL: The shedding of leaves as a result of the formation of a zone of abscission at the base of the petiole. Leaves on dead branches do not fall as no such zone is produced. In deciduous plants, there is a continual shedding of older leaves throughout the growing season but with the onset of the winter or dry season there is a conspicuous shedding of all the remaining leaves. In evergreens, leaves are shed and replaced continually and there is no period when the plant is devoid of foliage. Leaf fall is associated with a drop in the auxin level of the lamina.

LEAF GAP (LACUNA): A parenchymatous area in the stele of many vascular plants associated with and positioned immediately above a leaf trace. Leaf gaps are characteristic of angiosperms but are also present in some gymnosperms and ferns.

LEAF JOINT: A point on the stem of a plant where a new shoot may grow.

LEAF MASS RATIO (LMR): The ratio of leaf dry mass to the dry mass of the entire plant, typically measured in grams per gram or kilograms per kilogram.

LEAF MOULD: 1. Partially decomposed plant leaves or crumbly brown humus typical of forest floors. It is composed of decayed leaves and other plant material mixed with soil and rich in nitrogen, used as a fertiliser. 2. Any of various fungus diseases affecting the leaves of certain plants such as tomatoes.

LEAF PRIMORDIAL: Juvenal leaves, emerging from the shoot apical meristem or a lateral outgrowth from the shoot apical meristem that develops into a leaf.

LEAF ROLL: 1. A viral disease of potatoes transmitted by aphids. The leaves roll up and become dry and this affects crop yield. 2. A serious grapevine disease that is transmitted by grafting or by insects. It affects accumulation of sugar in the grapes.

LEAF SPOT: A symptom of several plant fungal infections, where the leaves show circular brown patches of dead leaf tissue such as groundnut leaf spot caused by *Cercospora arachidicola* and cassava leaf spot caused by *Cercospora henningsii*.

LEAF STRIPE: Viral disease of barley, maize, oat and other monocotyledonous plants in which the leaves show pale longitudinal strips parallel to the veins.

LEAF TRACE: A vascular strand leading from the stele to the leaf.

LEAF VEINS: Vascular tissue in leaves, arranged in a net-like network (reticulate venation) in dicotyledons and running parallel (parallel venation) to the midrib in most monocotyledons or at right angle to the midrib in plants such as banana and plantain.

LEAF-HEIGHT SEED SCHEME (LHS SCHEME): A scheme which describes the ecological strategy of plant species in terms of three parameters: specific leaf area (L); height of the plant at maturity (H) and seed mass (S). It is argued that L reflects whether the species is a good competitor or is more able to tolerate stress and that H and S represent two aspects of coping with disturbance. The strategy of a species is thus depicted by its position in the three-dimensional space bounded by the axes of the three variables.

LEAFLET: A division of a compound leaf. It resembles an entire leaf but it does not arise from the branch. It rather arises on a petiole of the leaf. Examples of plants that possess this kind of compound leaves are the Hemp plant and Acacia.

LEAKAGE: 1. In physics, **earth leakage**, also called **ground leakage** or **leakage current** is the unwanted flow of electrical current from the live wire to the earth wire. It is caused by a conductor being raised to a potential above that of earth, causing some current to flow from that conductor to earth. This may cause voltage to build up and damage the electrical equipment. The current can also leak to metal electrical appliances such as stoves and give electric shock to anybody who handles that equipment. The protective measures taken to guard against this are to earth all metal parts of all appliances; protective devices, called a fuse and a circuit breaker must be installed in all circuits to interrupt the circuit in such an eventuality. 2. **Leakage flux** is the magnetic flux that passes outside useful flux circuits.

LEAP YEAR: A year longer than the normal 365 days, which happens every four years and has an extra day in February. Earth revolves around the Sun once every year or 365.25 days. The extra 0.25 of a day adds up to a day every four years, creating a leap year.

LEAPFROGGING (TECHNOLOGICAL LEAPFROGGING): It refers to the opportunities in developing countries to bypass several stages of technology development, and apply the most advanced presently available technologies in the energy and other economic sectors, through investments in technological development and capacity building.

LEARNED BEHAVIOUR: An acquired change in behaviour during a lifetime through observation or by being taught. This type of behaviour allows an organism to adapt to changes in the environment and modified by previous experiences.

LEARNING: A process of acquiring modifications in existing knowledge, skills, habits or tendencies through experience, practice or exercise, which may alter an animal's response to a particular situation permanently, usually beneficial. On a physiological level, learning involves changes in the connections of neurons in the central nervous system. Categories of learning include: habituation, associative learning, trial-and-error learning, insight learning, latent learning and imprinting.

LEASED LINE: In telecommunications, it is a dedicated digital telephone link between two points such as geographically distant offices within an organisation set up by a telecommunications common carrier. The points do not have a telephone number, as the lines are permanently connected; and are always active. Users are charged a fixed fee; regardless of how much traffic the leased line carry.

LEAST: Of an element in an ordering, uniquely minimal, less than every other member of the set. For example, the set of subsets of a given set ordered by inclusion, has the empty set as least element.

LEAST COMMON MULTIPLE, L.C.M. (LOWEST COMMON MULTIPLE, LCM): The lowest whole number that is divisible without a remainder by all of the members of a particular set of numbers. For example, the L.C.M. for the set {2, 3, 4, 6, 8} is 24.

LEAST INTEGER FUNCTION (CEILING FUNCTION): In mathematics and computer science, the ceiling function of x is the smallest integer greater than or equal to x . For example, ceiling $(x) = \lceil x \rceil$ is the smallest integer not less than x . In the same way that the floor function maps a real number, the smallest following integer: floor $(x) = \lfloor x \rfloor$ is the largest integer not greater than x .

LEAST SQUARES ESTIMATION: A method used for estimating parameters, particularly in regression analysis, by minimising the difference between the observed response and the value predicted by the model. For example, if the expected value of a response variable y is of the form $E(y) = \alpha + \beta x$, where x is an explanatory variable, then least squares estimators of the parameters α and β may be obtained from n pairs of sample values $(x_1, y_1), (x_2, y_2), \dots, (x_n, y_n)$ by minimising S given by $S = \sum_{i=1}^n (y_i - \alpha - \beta x_i)^2$ to give $\alpha = \bar{y} - \beta \bar{x}$ and

$$\beta = \frac{\sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})}{\sum_{i=1}^n (x_i - \bar{x})^2}, \text{ where } \bar{y} \text{ and } \bar{x} \text{ are the mean values of } y \text{ and } x \text{ respectively, given by } \bar{y} = \frac{\sum_{i=1}^n y_i}{n} \text{ and } \bar{x} = \frac{\sum_{i=1}^n x_i}{n}.$$

LEAST SQUARES REGRESSION: When the set of pairs $(x_1, y_1), (x_2, y_2), \dots, (x_n, y_n)$ are plotted on a graph and the points are scattered about a straight line then a linear relationship is said to exist between the variables. The method of least squares fits the "best" line through the points. The method achieves this by minimising the sum of the squared deviations of each point from the "best line". The line of regression of y on x mini f regression has the equation;

$$y - \bar{y} = \frac{S_{xy}}{S_{xx}}(x - \bar{x}), \text{ where } \bar{y} = \frac{\sum y_i}{n}, \bar{x} = \frac{\sum x_i}{n}, S_{xy} = \sum xy - n\bar{x}\bar{y}$$

and $S_{xx} = \sum x^2 - n\bar{x}^2$. The line of regression of x on y minimises $\sum (B_i S_i)^2$ and is used for predicting x values given y values, given by the equation, $x - \bar{x} = \frac{S_{xy}}{S_{yy}}(y - \bar{y})$.

LEAST-SIGNIFICANT BIT (LSB): In computing, it is the extreme right hand digit representing the smallest power or least value in a multi-bit binary number. For example, when 13 is represented by binary 1101, the least significant bit is the 1 at the right. The least significant bit is especially important in transmission of binary numbers.

LEAST SQUARE METHOD (LEAST-SQUARE METHOD): A standard approach to the approximate solution of over determined systems (sets of equations in which there are more equations than unknowns). Least squares mean that the overall solution minimises the sum of the squares of the errors made in solving every single equation. The most important application is in data fitting. The best fit in the least-squares sense minimises the sum of squared residuals, a residual being the difference between an observed value and the fitted value provided by a model.

LEAST UPPER BOUND (SUPREMUM): The unique smallest element of the set of upper bounds for a set that is greater than or equal to all elements of the set, if such an element exists. For example, the sequence $1/2, 2/3, 3/4, \dots$ has as upper bound every real number greater than or equal to 1; it has no maximum, but its supremum is 1.

LEATHER: The skin or hide of an animal tanned to prevent decay and to impart flexibility and toughness or animal skin that has been chemically modified to produce a flexible, strong material that resists decay. Artificial leather is a strong fabric coated with a rubber composition or with a synthetic substance such as pyroxylin. Leather is used in the production of products such as soles of shoes and shoes, machine belting, engine gaskets, bags, jackets and harnesses.

LEBESGUE COVERING: A covering of a set in Euclidean in n -space by countable families of boxes. In other words, a space has Lebesgue

covering dimension n if for every open cover of that space, there is an open cover that refines it such that the refinement has order at most $n + 1$.

LEBESGUE INTEGRAL: The integral of a measurable function, f , over a subset, E , of a measure space with respect to its measure, μ , written as $\int_E f d\mu$. It is defined in terms of upper and lower bounds using the Lebesgue measure of a set.

LEBESGUE INTEGRATION: A generalisation of integration to measurable functions over subsets of a measure space with respect to its measure, often Lebesgue measure.

LEBESGUE MEASURE: An extension of the classical notions of length and area to more complicated sets. It is the standard way of assigning a measure to subsets of n -dimensional Euclidean space. For $n = 1, 2$ or 3 , it coincides with the standard measure of length, area or volume. A unit line segment has Lebesgue measure 1; the Cantor set has Lebesgue measure 0. The Minkowski measure of a bounded, closed set is the same as its Lebesgue measure.

LEBESGUE, HENRI-LÉON (1875-1941): Famous French mathematician who generalised the concept of integration, did fundamental work in real analysis, established fundamental theorems of Borel's measure theory; applied Lebesgue integral to Fourier analysis; and also made major contributions in other areas of mathematics such as topology, potential theory and the Dirichlet problem. Several important mathematical terms have been named after him, including the Lebesgue differentiation theorem, Lebesgue covering, Lebesgue integration, Lebesgue measure, etc.

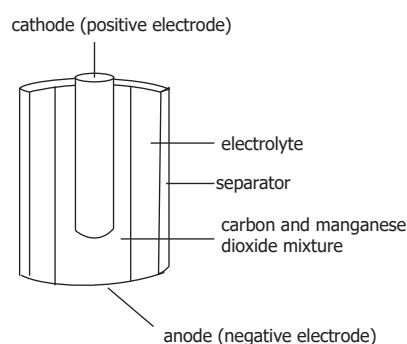
LEBESGUE'S THEOREM: The theorem that if f is a Lebesgue integrable function on the reals, then $\lim_{h \rightarrow 0} \int_0^h |f(x+t) - f(x)| dt$ is zero almost everywhere.

LEBLANC PROCESS: An obsolete process for manufacturing sodium carbonate (soda ash), named after its inventor, Nicolas Leblanc. The raw materials were sodium chloride, sulfuric acid, coke and limestone (calcium carbonate). The process involved two stages: firstly, the sodium chloride was heated with sulfuric acid to give sodium sulfate: $2\text{NaCl(s)} + \text{H}_2\text{SO}_4 \text{ (l)} \rightarrow \text{Na}_2\text{SO}_4 \text{ (s)} + 2\text{HCl(g)}$. The sodium sulfate was then heated with coke and limestone: $\text{Na}_2\text{SO}_4 + 2\text{C} + \text{CaCO}_3 \rightarrow \text{Na}_2\text{CO}_3 + \text{CaS} + 2\text{CO}_2$. Calcium sulfide was a by-product, the sodium carbonate being extracted by crystallisation. The process was the first to be used for producing sodium carbonate synthetically; earlier methods were from wood ash and other vegetable sources. The process gradually has now been replaced by the Solvay process.

LECHATLIERITE: A mineral form of silicon(IV) oxide, with the chemical formula SiO_2 . It forms naturally by melting quartz sand at very high temperature during a lightning strike. It also forms as the result of high pressure shock metamorphism during meteorite impact. It is a common component of a type of glassy ejecta called **tektites**.

LECITHIN: 1. A phosphoglyceride containing the amino alcohol choline esterified to the phosphate group. 2. A lipid or fat, containing nitrogen and phosphorous that forms a vital part of the cell membrane of plants and animal. In animals, higher concentrations are found in nerve tissues and red blood cells. Lecithins also occur in plants and in egg yolks. Commercial lecithin, a wetting and emulsifying agent is a mixture of lecithins and other phospholipids in an edible oil used in animal feeds, baking products and mixes, chocolate, cosmetics and soap, insecticides, paint and plastics.

LECLANCHÉ CELL: A primary voltaic cell consisting of a zinc anode, a carbon (graphite) cathode surrounded by manganese dioxide (acting as a depolariser), and a paste containing ammonium chloride as electrolyte. It was invented and patented in 1866 by Georges Leclanché, a French chemist. Polarisation is prevented by using a mixture of manganese oxide and carbon graphite, held in contact with the anode by means of a porous bag or pot, giving it a relatively high internal resistance; this reacts with the hydrogen produced. An example is the common dry battery with an e.m.f of 1.5 volts that is widely used in torches, radios and calculators. It is not reversible and therefore has a limited operating life.



Anode: Zinc container.

Cathode: Carbon rod surrounded by a mixture of MnO_2 and carbon granules

Electrolyte: Moist paste of ammonium chloride and ZnCl_2 in starch

The Leclanche dry cell

LECTIN: Any of a group of proteins, usually extracted from plants, such as seeds of concanavalin A, that can bind to specific oligosaccharides on the surface of cells, causing the cells to clump together. Lectins are found in seeds of legumes and in other tissues, in which they are thought to act as a toxin. They play a role in biological recognition phenomena involving cells and proteins. They are widely used for diagnostic and experimental purposes. Some viruses use lectins to attach themselves to the cells of the host organism during infection.

LECTOTYPE: A specimen or other element (description, illustration, etc.) subsequently selected from the original material on which the name of a taxon was based. Very often these original elements will be syntypes. A lectotype is only necessary when the original author failed to designate a holotype.

LEECH: An aquatic or terrestrial segmented annelid worm of the class Hirudinea having suckers at both ends which blood-sucking types use to attach to the skin of various animals. Many species are bloodsucking parasites, especially of vertebrates, and others are predators but some others feed on decaying plant materials or various worms and insects. Leeches produce anaesthetic and one species, the *Hirudo medicinalis*, has been used in medicine to extract blood from people without pain allowing for prolonged pain-free extraction. Also, leech releases an anticoagulant, called hirudin, which prevents blood from coagulation. Both anticoagulant and anaesthetic produced by leech are used in medicine for the production of drugs. Leeches are also used in medical procedures to reattach body parts and in reconstructive and plastic surgeries as well as treating osteoarthritis.

LEECH LATTICE: The 24-dimensional integer lattice corresponding to the Steiner triple system $S(5,8,24)$ that contains 759 sets. Let K be the group generated by all octads of the base set $B = \{1, 2, \dots, 24\}$ under the operation of symmetric difference; it has 2^{12} elements, consisting of the octads, their complements, 2576 dodecads, the empty set and B . Let $C \in K$; the lattice then consists of $\bigcup_{m \in C} C(m)$, where $C(m)$ is the

set of all 24-dimensional integer vectors the sum of which is congruent to 0 modulo 4 and such that $x_i \neq 0 \pmod{4}$ if $i \neq C$ and $x_i \neq 2 \pmod{4}$ if $i = C$. It was defined by John Leech in 1967.

LEEK: A hardy winter vegetable, *Allium porrum*, family Alliaceae allied to onion but with a long slender bulb and with a mild onion taste.

LEEUWENHOEK, ANTON VAN (1632-1723): Dutch maker of optical lenses and microscopes, whose microscopes enabled him to give the first accurate description of blood corpuscles, spermatozoa and microbes. His main discoveries include the infusoria (protists in modern zoological classification) in 1674; the bacteria, such as the large Selenomonads from the human mouth) in 1676; the vacuole of the cell; the spermatozoa in 1677; and the banded pattern of muscular fibres in 1682.

LEEWARD: The side opposite from the prevailing wind direction; the side sheltered from the wind by the mountain's elevation. The leeward regions of mountains generally remain dry as compared to the windward.

LEFT: In mathematics, of an operator in a non-commutative theory, acting on the left. I_l is a left identity if $I_l x = x$ for all x ; I_x is a left inverse of x if $I_x = I_l$.

LEFT-CLICK: To press the left hand side button of a mouse used to perform many common computer tasks, such as selecting objects, drag and drop, opening hyperlinks, closing windows, etc. It is the most frequently used click.

LEFT-HAND LIMIT: In mathematics, the one-sided limit of a function, f , defined on an open interval from below or from the left, where x is restricted to values less than a , expressed as $\lim_{x \rightarrow a-} f(x) = f(a-)$.

LEFT-HAND RULE (FLEMING'S RULE): A rule, used to recall the relative directions of motion, magnetic field and current in an electric motor. The fingers of the left hand are placed at right angles to each other around a wire, so that the thumb points in the direction of Motion, the first Finger points in the direction of the Force and the second fingers points in the direction of the Current.

LEFT-HANDED: In mathematics, of a coordinate system, having the orientation corresponding to a left-handed trihedral.

LEFT-HANDED TRIHEDRAL: A configuration of three directed lines that do not lie in the same plane and that have a negative triple product. It is called left-handed because the thumb and first two fingers of the left hand are in this orientation. If the thumb is placed in the positive direction of the first line, then the angle between the other two fingers is less than π . The reverse applies to a right-handed trihedral.

LEFT-TO-RIGHT CONVENTION: Arrangement of the structural formulae of the reactants so that the bonds to be made or broken form a linear array in which the electrons move from left to right.

LEFT-TO-RIGHT SUBTRACTION: A subtraction algorithm that works from the left decimal place to the right in several steps. For example, to solve $94 - 57$, first calculate $94 - 50$ to obtain 44 and then calculate $44 - 7$ to obtain 37. The method is especially suited to mental arithmetic.

LEG: In mathematics, either of the two sides forming the right angle of a right-angled triangle.

LEGACY APPLICATION: In computing, it is an inherited application, usually written for an earlier operating system or hardware platform that may be too expensive for a company or organisation to replace. Legacy applications are usually database systems that run on a large minicomputers or mainframes.

LEGACY SYSTEM: In computing, it is an out-dated system such as programming languages or application software that may still be in use by a company or an organisation.

LEGEND: 1. The letters, numbers, characters and words imprinted on a keyboard's keycaps. 2. In presentation graphics, text that describes the meaning of colours and patterns used in the chart and is linked

to the data being graphically displayed in the plot area of the chart. In other words, it shows, for example, what columns, bars, lines and pie slices represent. In a chart or graph in spreadsheet programs such as Excel, the legend is most often located on the right hand side of the chart or graph and can sometimes be surrounded by a border.

LEGENDRE DIFFERENTIAL EQUATION: The differential equation of the form $(1 - x^2)y'' - 2xy' + p(p + 1)y = 0$. Its solution, when p is a natural number, is the p^{th} Legendre polynomial.

LEGENDRE IDENTITY: The identity that is closely related to the hypergeometric differential equation that relates complete elliptic integrals that for any $0 < k < 1$, $K(k)E[\sqrt{1-k^2}] + E(k)K[\sqrt{1-k^2}] - K(k)K[\sqrt{1-k^2}] = \frac{\pi}{2}$.

LEGENDRE POLYNOMIALS: The solutions to the Legendre differential equation, defined as the collection of orthogonal polynomials, $P_n(x)$, on $[-1, 1]$ and by $P_0 = 1$ and may be expressed

using Rodrigues' formula $P_n(x) = \frac{1}{2^n n!} \frac{d^n}{dx^n} (x^2 - 1)^n$. Each Legendre polynomial $P_n(x)$ is an n^{th} -degree polynomial that solves Legendre differential equation with $p = n$.

LEGENDRE SYMBOL: The symbol (a/p) or (a/p) , defined for prime p and a relatively prime to p that is a multiplicative function with values $+1$ and -1 ; if a is a quadratic residue modulo p , the value is 1 and -1 on a non-quadratic residue.

LEGENDRE, ADRIEN-MARIE (1752-1833): French mathematician, who made many important contributions in differential equations, calculus, function theory, number theory and elliptic integrals. He also conjectured the prime number theorem and the law of quadratic reciprocity. He published a textbook on geometry, *Éléments de géométrie*. Mathematical terms such as Legendre differential equation, Legendre identity, Legendre polynomials and Legendre transformation are named after him.

LEGGETT'S THEOREM: A result in quantum mechanics that is a generalisation of Bell's theorem put forward in 2003 by the British-born American physicist, Anthony Leggett (1938–). It states that nonlocal hidden variable theories make predictions that contradict the predictions of quantum mechanics. It was subsequently found that experiments agree with the predictions of quantum mechanics rather than of nonlocal hidden variables theories.

LEGHAEMOGLOBIN (LEGHEMOGLOBIN; LEGOGLOBIN): Plant haemoglobin found in the root nodules of legumes which functions as an oxygen-carrying pigment in symbiotic nitrogen fixation.

LEGHORN: An excellent laying breed of hen. It is a hardy bird coloured black, brown and white.

LEGIONNAIRES' DISEASE: A severe type of pneumonia caused by the bacterium *Legionella pneumophila*, which usually breeds in warm water (at temperatures of between 25 and 45 °C), such as in plumbing, shower heads and water-storage tanks. It is not airborne and not transmitted from person to person, but usually through inhaling the bacteria through infected minute water droplets. It was first identified when it broke out at a convention of the American Legion in Philadelphia in 1976. Symptoms include headache, muscle pain, chills, fever, cough (which may be dry or producing mucus and sometimes blood), shortness of breath, chest pain, etc.

LEGUMES (PULSES, LEGUMINOUS CROPS): Plants belonging to the family Leguminosae or Fabaceae, which are grown primarily for their edible seeds. They are cultivated for their seeds, which are used for human and animal consumption or for oil production for industrial use. Since they are legumes these plants can fix atmospheric nitrogen for their own needs or soil enrichment, hence minimizing the cost of fertilizer input. They are a major source of protein and are mainly the various types of beans, peas, lentils, lupins, peanuts, etc.

LEGUMINOSAE: A family of plants including peas and beans that produces seed in a pod. The family is now known as **Fabaceae**.

LEGUMINOUS CROPS (LEGUMINOUS PLANTS): Crops, plants, fruit or seed belonging to the family Fabaceae or Leguminosae, grown

basically for human consumption, livestock forage, silage and as a green manure for enhancing soil fertility. The fruits of this plant are called legumes and examples include beans, clover, peas, lentils, soy, peanut and alfalfa.

LEHMANN SYSTEM: A system of pig breeding, where bulk food such as potato and fodder beet are fed after a basic ration.

LEHMANN-SCHEFFÉ THEOREM: Named after Erich Leo Lehmann and Henry Scheffé, in statistics, it is a theorem, which states that any estimator which is unbiased for a given unknown quantity and that depends on the data only through a complete, sufficient statistic is the unique best unbiased estimator of that quantity. It can be defined mathematically that if $T(x)$ is a complete sufficient statistic and $S(x)$ is an unbiased estimate of θ , then the conditional expectation of $S(x)$ given $T(x)$ is a uniform minimum variance unbiased estimate for θ .

LEIBNIZ THEOREM: A formula for finding the n th derivative of a product of two functions, named after the German mathematician, philosopher, historian and physicist Gottfried Wilhelm Leibniz (1646-1716). The n th derivative with respect to x of a function $f(x) = u(x)$

$v(x)$ written $D^n(uv) = \frac{d^n(uv)}{dx^n}$ is equal to the series $uD^n v + {}_nC_1 DuD^{n-1}v + {}_nC_2 D^2uD^{n-2}v + \dots + {}_nC_{n-1} D^{n-1}uDv + vD^nu$, where ${}_nC_r = \frac{n!}{(n-r)!r!}$. The formula holds for all positive integer values of n .

LEIBNIZ, GOTTFRIED WILHELM (1646-1716): German philosopher, logician, mathematician, scientist and also with a PhD in law, who made significant contributions in several fields including mathematics, physics, logic and philosophy. He is credited with developing the differential and integral calculus independently of Newton, perfected Pascal's calculating machine and laid the foundations of dynamics. He described the binary system and laid the foundations of topology. He postulated a perfect universal language of thought, in which logical relations would be transparent.

LEIBNIZ'S LAW: 1. In logic, the principle of the identity of indiscernibles, stating that two expressions satisfy exactly the same predicates if and only if they both refer to the same individual. It can be expressed as $(\forall F)(Fa \rightarrow Fb) \equiv a = b$, the universal quantifier ranges over predicates and so is second-order. 2. The weaker principle of the indiscernibility of identicals, that if $a = b$ then whatever is true of a is true of b .

LEISHMAN'S STAIN (LEISHMAN STAIN): A neutral stain for blood smears, named after its inventor, Scottish pathologist William Boog Leishman. It consists of a mixture of eosin (an acid stain) and methylene blue (a basic stain) in alcohol and is usually diluted and buffered before use. It stains the different components of blood in a range of shades between red and blue.

LEISHMANIASIS: Any of several tropical and subtropical parasitic diseases mainly affecting the skin (cutaneous leishmaniasis) or the internal organs of the body (visceral leishmaniasis or kala-azar). It is caused by microscopic protozoans of the genus leishmania and transmitted by the bite of sandflies. It occurs mainly in the Mediterranean region, Africa, Asia and Central/South America. It was first identified by Lieutenant General Sir William Leishman (1865-1926), British Professor of Pathology at the Army Medical School.

LEK: An area of ground divided into territories, where males in various birds species, for example, the black grouse and peafowl and in some mammals assemble before and during the breeding season, on a daily basis for purposes of sexual display and mating. The most dominant males known as **alpha-males** occupy the smallest and central territories where they are most likely to attract and copulate with visiting females. The peripheral and larger territories are occupied by the beta-male, gamma-male, etc or subordinate males. The lek

territories contain no resources of value to the breeding female, such as food or nesting materials.

LEMMA: 1. One of two bracts that protects the floret of grasses or the outer bract which encloses the flower of grass. The lemma is situated beneath the second bract, known as the **palea** which it surrounds. 2. A mathematical theorem, which is proved purposely for use in the proof of other theorems.

LEMNISCATE (LEMNISCATE OF BERNOULLI): A closed plane curve with a characteristic shape consisting of two symmetrical loops meeting at a node. In the (x, y) plane, it can be expressed by the equation $(x^2 + y^2)^2 = a^2(x^2 - y^2)$, where a is the greatest distance from the curve to the origin.

LEMON: A small tree (*Citrus limon*) of the family Rutaceae (orange family) and its edible fruit. The tree is about 5 metres tall with thorny branches and purple-edged white blossoms, often cultivated in mild, equable climate. The fruits are yellow or, in some cases, green with an oval shape with a thick fragrant rind and sour juicy flesh. The lemon is grown primarily for its acid flavour. It is rich in ascorbic acid or vitamin C. It is used in beverages, soft drinks and other culinary activities. The principal by-products are citric acid from the juice and lemon oil from the peel.

LEMPER-ZIF-WELSH COMPRESSION (LZW COMPRESSION):

Data compression system used to reduce the size of data files by using a lossless look-up algorithm and thus reducing the time needed to transmit them. It was developed in 1977 by Abraham Lempel (1936-), an Israeli computer scientist, Jacob Ziv (1931) an Israeli electrical engineer and Terry Welch, American computer scientist; and named after them. LZW compression is used in the graphics interchange format (GIF), the tagged image file format (TIFF), portable document file (PDF), etc.

LENGTH: 1. A measure of the extent of a line segment that is zero only when its endpoints coincide and is otherwise positive. For an object, it is the longest dimension of an object. The length of a line segment is unaffected by changing the orientation of the line. In a Cartesian coordinate system, the length of the straight line segment

AB, where A is the point $A(a_1, b_1)$ and $B(a_2, b_2)$, is $\sqrt{(a_1 - a_2)^2 + (b_1 - b_2)^2}$.

2. A basic quantity in the system of quantities upon which SI unit is based. The SI unit for length is the metre, with the symbol m. The metre is defined as the length of the path travelled by light in vacuum during a time interval of $1/299\,792\,458$ of a second.

LENGTH CONTRACTION: An effect of relativity which is the shortening of the measured length of an object moving relative to the observer's frame. If the length of a moving object is measured, its length L is shorter than the proper length, L_0 if the object were stationary.

LENGTH MODEL: A linear representation (model) of fractions as line segments, where the value of each fraction equals its distance from zero on a number line diagram or length of the segment/rod when using Cuisenaire rods (a type of manipulative).

LENGTH-PRESERVING TRANSFORMATION: A linear transformation, A , such that $\|Ax\| = \|x\|$ for all vectors x in a normed vector space.

LENNARD-JONES POTENTIAL (L-J POTENTIAL; 6-12 POTENTIAL): A mathematical approximation that illustrates the energy of interaction between two non-bonding atoms or molecules based on their distance of separation, given by $v = (A/r^{12}) - (B/r^6)$,

where r is the distance between the centres of the molecules and A and B are constants. The equation takes into account the difference between attractive forces (dipole-dipole, dipole-induced dipole and London interactions) and repulsive forces. It was proposed by Sir John Edward Lennard-Jones, a British mathematician who was a professor of theoretical physics at Bristol University and then of theoretical science at Cambridge University.

LENS: A piece of transparent piece of ground and polished or moulded material such as plastic, glass or other transparent material with two surfaces that is used to form an image of an object by converging or diverging rays of light from the object. Because of the curvature of its surface, different rays of light are refracted through different angles. Lenses are used singly or in groups in instruments such as cameras, projectors, microscopes, telescopes, binoculars, opera glasses and eyeglasses. Lenses are classified by the curvature of the two optical surfaces. For example, if both surfaces are convex, is biconvex or double convex. Some other types are shown in the diagram.

LENSE-THIRING EFFECT (ROTATIONAL FRAME-DRAGGING):

An effect predicted to occur in general relativity theory by J. Lense and Hans Thirring in 1918 in which a compact rotating body causes the space near it to rotate in the same direction. Under the Lense-Thirring effect, the frame of reference in which a clock ticks the fastest is one which is revolving around the object as viewed by a distant observer. This also means that light travelling in the direction of rotation of the object will move past the massive object faster than light moving against the rotation, as seen by a distant observer.

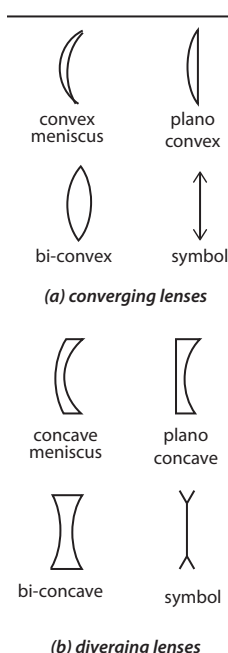
LENTIC: Describing a freshwater ecosystem such as a lake or pond in which there is not a continuous flow of water.

LENTICEL: A pore of loosely arranged spherical cork cells developed in the phellem (cork layer) of woody stems, roots and other parts of vascular plants when the epidermis is replaced by bark. It allows for gaseous exchange. **Lenticular** means having the shape of a lentil or lens.

LENTIGINOUS: 1. A small, coloured or pigmented spot on the skin with a clearly defined age, surrounded by a normal appearing skin. 2. Covered with bran-like scales or freckles.

LENTIL: A plant (*Lens culinaris*) of the legume family that originates from southwest Asia. It has flat pods containing lens-shaped, edible seeds. The seeds are very high in proteins.

LENZ'S LAW: The law, discovered by the German scientist H. F. E. Lenz in 1834, which states that the direction of an induced electromotive force (emf) tends to oppose its cause or that an induced electric current always flows in such a direction that it opposes the charge producing it. For example, in an electric motor, the armature is turning in a magnetic field, thus inducing an electromotive force in it. Lenz's law requires that this electromotive force, called **back electromotive force** or **counter electromotive force**, opposes the motion of the armature and also the original electromotive force, causing the motor to rotate. It is a most important statement that describes electromagnetic induction.



Types of lenses and symbols used

LEPANTHIUM: A petal with a nectary.

LEPIDOPTERA: An order of the class Insecta (insects) containing over 150,000 species comprising the butterflies and moths, found mainly in tropical regions. Adults possess two pairs of membranous wings, often brightly coloured. The wings, body and legs are covered with minute scales. Adult mouthparts are modified to form a long proboscis for sucking nectar and juices. Examples are moths, which have larger bodies, are nocturnal and rest with their wings in various positions; and butterflies, which are typically small-bodied, active during day light and rest with wings folded vertically. The larvae of moths and butterflies are called **caterpillars** and undergo metamorphoses via pupae (chrysalises) to the adult forms.

LEPIDOTE: 1. In biology, covered with scurfy scales or scaly leaves. Tiny scales cover the undersides of the leaves. Lepidotes have smaller leaves and are usually low growing. An example is found in rhododendron species with tiny scales on the undersurface of the leaves. 2. An extinct genus of neopterygian ray-finned fish. They inhabited fresh water lakes and shallow seas and were about thirty centimetres long. Their bodies were covered with thick, enamelled scales. Fossils have been found in marine sediments in France, England and Germany.

LEPROSY: A chronic infectious tropical disease caused by the bacterium *Mycobacterium leprae*, which affects the skin, peripheral nerves and mucous membranes of the nose, throat and eyes. The skin becomes rough and thick with small round hard whitish mark and the flesh and nerves are slowly destroyed. In severe cases, it can cause disfigurement. It is transmitted through close personal contact and has an incubation period of 1-30 years.

LEPTIN: A protein hormone, comprising 167 amino acids in humans, secreted by adipose tissue that acts to regulate long-term appetite. It acts on leptin receptors in the hypothalamus and inhibits expression of neuropeptide Y, thereby countering the appetite-stimulating effects of the latter and inhibiting food intake. It also promotes the synthesis of the appetite suppressant melanocyte-stimulating hormone. Deficiency of leptin or its receptors leads to severe obesity.

LEPTOID: An elongated food-conducting cell, similar to a sieve cell of seedless vascular plants that transports photosynthetic materials in certain mosses. Mature leptoids have inclined cell walls with small pores and lose their nuclei but retain their cytoplasm and have prominent connection with adjacent cells.

LEPTOKURTIC: Of a statistical distribution, having points along the X-axis clustered, resulting in a higher peak or higher kurtosis than the curvature found in a normal distribution.

LEPTON: Any of a class of elementary particles, alongside quarks and gauge bosons that consist of the electron, muon, neutrino and tau particles. For each lepton, there is an equivalent antiparticle. Like quarks, leptons are fermions (spin- $1/2$ particles) and are subject to the electromagnetic force, the gravitational force and weak interaction, but unlike quarks, leptons do not participate in the strong interaction.

LEPTON NUMBER: A conserved quantum number, which is equal to the number of leptons minus the number of antileptons in a system.

LEPTOQUARK: In physics, any hypothetical particle that would decay into a lepton and a quark.

LEPTOSPIROSIS: An infectious disease of humans and domestic animals, especially cattle, which affects the kidneys and liver. It is characterised by fever, jaundice, muscle pains, headaches, hepatitis, albuminuria and multiple small haemorrhages in the conjunctiva or skin. It is caused by spiral-shaped bacteria spirochetes of the genus *Leptospira*.

LEPTOSPORANGIUM: A type of sporangium that is formed from a single epidermal cell. It is typical of ferns of the order Filicales (the

leptosporangiate ferns). The wall of a leptosporangium is usually only one cell thick and does not contain tissue derived from the archesporium.

LEPTOTENE: The beginning of the first prophase of meiosis, when the chromatids appear as thin threads having well-defined chromomeres.

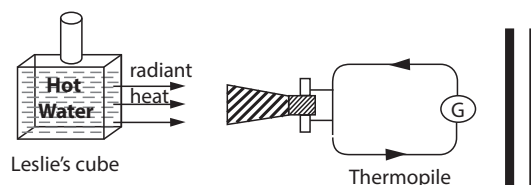
LESION: Any pathological or traumatic discontinuity of tissue or loss of function of a part, especially an area of skin that is infected or broken or a physical change in a body part that is the result of injury or illness. Lesion includes wounds, sores, ulcers, tumours, cataracts and any other tissue damage such as occurs in tuberculosis.

LESLIE MATRIX MODEL: A model often applied in demography and animal population studies in which the vector of the number of individuals of each age at time t , N_t , is related to the initial number of individuals of each age, N_0 , by the equation $N_t = M^t N_0$ where M is known as the population projection matrix given by,

$$M = \begin{bmatrix} B_0 & B_1 & \dots & B_v \\ P_0 & \dots & 0 & 0 \\ 0 & \dots & P_{v-1} & P_v \end{bmatrix}, \text{ where } B_v \text{ equals the number of females}$$

born to females of age x in one unit of time that survive to the next unit of time P_x equals the proportion of females of age x at time t that survive to time $t+1$ and v is the greatest age attained.

LESLIE'S CUBE (LESLIE CUBE; LESLIE'S CANISTER): A metal box in the shape of a cube in which each of the four vertical sides have different surface finishes. It is used in the measurement or demonstration of the variations in energy radiating from different surfaces. When hot water is placed in the cube, the thermal detector shows much greater emission from the side with varnish than from any of the other three. The device was first devised in 1804 by John Leslie (1766–1832), a Scottish mathematician and physicist.



Leslie's Cube

LESS THAN: Approximate measure of the difference between numbers with the symbol $<$. The symbol for "less than or equal to" is $\leq$.

LET-DOWN: The release of milk from the mammary gland activated usually in response to sucking or crying by an infant, in which milk previously secreted into the alveoli of the breasts, is released into the ducts that lead to the nipple.

LETHAL: Causing death; sufficient or designed to cause death.

LETHAL ALLELE (LETHAL GENE): Any alteration in the nucleotide sequence of a gene that eventually results in the death of an organism if expressed in the phenotype. Most lethal genes are recessive. Sickle-cell anaemia results from a recessive lethal gene that causes the production of abnormal and inefficient haemoglobin.

LETHAL CONCENTRATION: Concentration of a potentially toxic substance in an environmental medium that causes death following a certain period of exposure. It is denoted by LC.

LETHAL DOSE (LD): Amount of a substance or physical agent (radiation) that causes death when taken into the body by a single absorption.

LETHAL SYNTHESIS (SUICIDE METABOLISM): Metabolic formation of a highly toxic compound from one that is relatively non-toxic (bioactivation), often leading to death of affected cells.

LETTER-BOX PRINCIPLE (DIRICHLET'S PRINCIPLE; DRAWER PRINCIPLE; PIGEON-HOLE PRINCIPLE): It states that if n items are put into m containers, with $n > m$, then at least one container must contain more than one item. It was introduced in 1834 by German mathematician, Peter Gustav Lejeune Dirichlet (1805-1859).

LETTUCE: A member of *Lactuca*, family *compositae*. Its flowers are borne in paniced heads and produce feathered fruits called achenes. It is a crisp-leaved annual plant used raw in salads. It comes in a variety of forms and leaf mixtures.

LEUCINE (2-AMINO-4-METHYLPENTANOIC ACID): A colourless crystalline essential amino acid, a constituent of many proteins with the chemical formula $C_6H_9CH(NH_2)COOH$. It is not synthesised in animals; hence it must be ingested, usually as a component of proteins. It is synthesised in plants and microorganisms in several steps starting from pyruvic acid.

LEUCINE ZIPPER (LEUCINE SCISSORS): A common three-dimensional molecular motif originally discovered in DNA-binding proteins. They are also found in other proteins, in which a set of four or five consecutive leucine residues is repeated every seven amino acids in the primary sequence. Leucine zippers are found in both eukaryotic and prokaryotic regulatory proteins, but are mainly a feature of eukaryotes. In a helical configuration this produces a line of leucines on one side of the helix. With two such helices alongside each other, the arrays of the leucines can interdigitate like a zip fastener, thus forming a stable link between the two helices.

LEUCO BASES: Colourless compounds formed by reduction of triphenylmethane dyes. Generally, they are amino or hydroxy derivatives of triphenylmethane.

LEUCOCYTES (WHITE BLOOD CELLS): Any of several types of cells with nucleus found in blood and lymph of most vertebrates that help defend the body from infection. They are formed in lymph nodes and red bone marrow and are capable of amoeboid movement. There are two major types: those without granules in the cytoplasm, such as lymphocytes and monocytes or agranulocyte; and those with granular cytoplasm (granulocytes), such as basophils and neutrophils.

LEUCOMALACHITE GREEN TEST: A presumptive test for blood. The reagent is the dye leucomalachite green dissolved in water along with sodium perborate ($NaBO_3$). A blue-green colour indicates a positive result.

LEUCOPLAST: A colourless plastid found in cells of plant tissues not normally exposed to light, lack pigments and are therefore located in roots and non-photosynthetic tissues of plants. They are organelles used for storing food. Examples are amyloplasts which store starch and elaioplast which stores oil.

LEUKAEMIA: Any one of a group of cancers in which the bone marrow produces abnormal white blood cells. Some types affect mostly children and others adults. It may be caused by exposure to radiation, smoking and children with Down's Syndrome are at greater

risk. Signs and symptoms include anaemia, fever or chills, weakness and fatigue, frequent infections, weight loss, swollen lymph nodes, enlarged liver or spleen, bleeding or bruising, recurrent nosebleeds tiny red spots in the skin, etc.

LEUKOTRIENES (LTs): A class of compounds derived from arachidonic acid in a variety of tissues, including the lungs and leucocytes as part of the inflammation response. There are at least six different types of leukotrienes, all of which play an active role in neutrophils and eosinophils and to sites of infection, increase the permeability of small blood vessels and cause vasoconstriction. Large amounts are released during an allergic reaction or asthmatic attack when their effects on the smooth muscle can make it very difficult to breathe.

LEEVE (DIKE; DYKE; LEVÉE): Naturally or artificially formed raised bank along the side of a river channel to regulate water levels.

LEVEL: 1. Logarithm of the ratio of a quantity and a reference value of that quantity. 2. An instrument used to determine heights or for establishing a horizontal plane. It is used in surveying, building, construction and archaeology. It consists of a small, sealed glass tube containing liquid and an air bubble. The tube is fixed horizontally in a block or frame with a smooth lower surface. When the bubble is in the middle of the glass tube, the device is on a level surface. 3. Concentration of analytes in an analytical sample.

LEVEL CURVE (LEVEL SURFACE): The points of constant value for a given scalar function; $f(x, y) = c$ typically gives rise to a level curve or contour line, while $f(x, y, z) = c$ typically gives rise to a level surface.

LEVEL OF SCIENTIFIC UNDERSTANDING (LOSU): This is an index on a 4-step scale (High, Medium, Low, and Very Low) designed to characterise the degree of scientific understanding of the radiative forcing agents that affect climate change.

LEVEL OF SIGNIFICANCE: The level of probability at which the null hypothesis is rejected, if there is not enough evidence to support it.

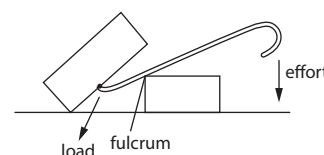
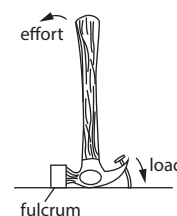
LEVEL SET: For a given real-valued function, any lower level set of the form $L(r) = \{x : f(x) \leq r\}$ or $L(r) = \{x : f(x) < r\}$. The upper level set is of the form $U(r) = \{x : f(x) \geq r\}$ or $L(r) = \{x : f(x) > r\}$.

LEVELLING EFFECT: The tendency of a solvent to make all Brønsted acids whose acidity exceeds a certain value appear equally acidic. It is due to the complete transfer of a protophilic solvent of a proton from a dissolved acid stronger than the conjugate acid of the solvent. The only acid present to any significant extent in all such solutions is the hydroxonium ion. For example, the solvent water has a levelling effect on the acidities of perchloric acid (HClO_4), hydrochloric acid (HCl) and hydrogen iodide (HI). Aqueous solutions of these acids at the same (moderately low) concentrations have the same acidities. A corresponding levelling effect applies to strong bases protogenic solvents.

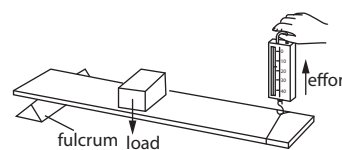
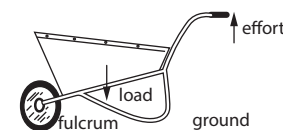
LEVELS OF ORGANISATION: In multicellular organisms, sequence of structures from the least complex to the most complex which coordinate and work together for proper functioning. The human body has five biological levels of organisation from the simplest to most complex as follows: cells, tissues, organs, organ systems, and organisms.

LEVER: A simple machine made of a rigid bar and a turning point or

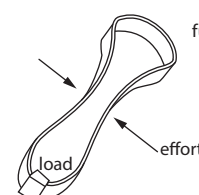
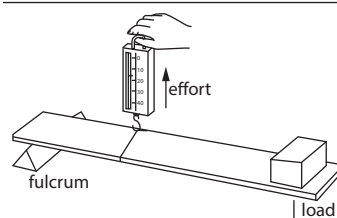
consisting of a rigid bar pivoted about a fulcrum, used to lift heavy weights with the least amount of effort. It amplifies an input force to provide a greater output force, which provides leverage. The ratio of the output force to the input force is the ideal mechanical advantage of the lever. Applications of the lever range from the simple nutcracker and paper punch to complex multiple-lever systems found in scales and in testing machines used in the study of properties of materials. There are three types of levers that are classified according to their relative position of the load, effort and pivot. For first class levers, the pivot is situated between the load and the effort. For second class levers, the load is found between the pivot and the effort. For third class levers, the effort is situated between the load and the pivot. Examples are shown in the diagrams.



First class levers



Second class levers



Third class levers
Levers

LEVER RULE: A rule enabling the relative amounts of two phases a and b , which are in equilibrium, to be found by a construction in a phase diagram. For example, a can be gas and b can be liquid, the distance la and lb along the horizontal tie line of the phase diagram are measured. The lever rule states that $na la = nb lb$, where na is the amount of phase a and nb is the amount of phase b . The rule takes its name from the similar form of the rule, $ma lb = mb lb$, relating the moments of two masses ma and mb about a pivot in a lever.

LEVITATION: The process by which an object is suspended by a physical force against gravity, in a stable position without solid physical contact, usually done using electric or magnetic fields. Other methods include aerodynamic, acoustic, electrostatic, gas film and optical. There are many technological applications of levitation, including levitation melting, in which a metal or other substance being melted is suspended in an electric or magnetic field and does not come into contact with the material of any container.

LEVODOPA (L-DOPA): The levorotatory form of dopa, normally produced by the body and converted by an enzyme to dopamine in the brain. It is used to treat Parkinson's disease.

LEVOROTATORY: A term describing an optically active compound that causes the plane of a polarised light to rotate in an anti-clockwise direction or a term designating a chemical compound that rotates the plane of plane-polarised light to the left or anticlockwise for someone facing the oncoming radiation.

LEWIS ACID: A molecular entity (and other corresponding chemical species) that is an electron-pair acceptor and therefore able to react with a Lewis base to form a Lewis adduct, by sharing the electron pair furnished by the Lewis base. An example of of Lewis acid base reaction is $\text{BF}_3 + \text{NH}_3 \rightarrow [\text{H}_3\text{N} \rightarrow \text{BF}_3]$. In this reaction, BF_3 is the Lewis acid, NH_3 the Lewis base and $[\text{H}_3\text{N} \rightarrow \text{BF}_3]$ the adduct.

LEWIS ACID-BASE: The third of three structural definitions of acids and bases, which defines an acid as a substance capable of accepting a pair of electrons and a base as a substance capable of donating a pair of electrons to a covalent bond. It was proposed by American chemist, Gilbert Newton Lewis.

LEWIS ACIDITY: The thermodynamic tendency of a substrate to act as a Lewis acid. Comparative measures of this property are provided by the equilibrium constants for Lewis adduct formation of a series of Lewis acids with a common reference Lewis base.

LEWIS ADDUCT: The adduct formed between a Lewis acid and a Lewis base.

LEWIS BASE: A molecular or an ionic entity (and the corresponding chemical species) able to donate a pair of electrons and thus capable of coordination to a Lewis acid, thereby producing a Lewis adduct. Examples are ammonia (NH_3), hydroxide (OH^-), water (H_2O) and Fluoride (F^-).

LEWIS BASICITY: The thermodynamic tendency of a substance to act as a Lewis base. Comparative measures of this property are provided by the equilibrium constants for Lewis adduct formation for a series of Lewis bases with a common reference Lewis acid.

LEWIS OCTET RULE: A classical rule for describing the electronic configuration of atoms in molecules: the maximum number of electron pairs that can be accommodated in the valence shell of a first-row element is four. For the second and subsequent row elements there are many exceptions to this rule.

LEWIS STRUCTURE (LEWIS DOT DIAGRAMS; ELECTRON DOT DIAGRAMS; ELECTRON DOT STRUCTURES): A representation of covalent bonding using dots and lines or dot pairs to represent the valence electrons on different atoms. Shared electron pairs are shown either as lines or as pairs of dots between two atoms and lone pairs are shown as pairs of dots on individual atoms.

LEWIS, GILBERT NEWTON (1875-1946): US physical chemist who taught at Harvard University, at the Massachusetts Institute of Technology (1907-12) and spent most of his career at Berkeley, California. His ideas on chemical bonding were extremely influential and he introduced the idea of a stable octet of electrons and of a covalent bond being a shared pair of electrons. He formulated the Lewis theory of acids and bases and with Harold C. Urey, discovered heavy water in 1932.

LEXICAL ORDER (LEXICOGRAPHIC ORDER): The arrangement of a set of items in accordance with a recursive algorithm, such as the entries in a dictionary, whose order depends on their first letter unless those are identical, in which case it is the second that decides and so on. The **reverse-lexical order** orders by last letters, which is particularly useful for ordering finite words constructed from an infinite alphabet, as in an infinite-dimensional vector space. When polynomials are ordered in this way, a polynomial is greater than zero if its highest order term has a positive coefficient and the order is a total ordering.

LEY: A field on which grass or clover is grown for pasture for a few years called **short ley** or for up to twenty years called **long ley**.

LEYDEN JAR: An early form of capacitor consisting of a narrow-necked glass jar with a layer of conductive metal foil on the outside

and a similar layer on the inside. It was invented at the University of Leyden in Holland in about 1745. Contact to the inner foil is by means of a loose chain hanging inside the jar.

LIANA (LIANE): A long-stemmed woody climbing plant that grows from ground level to the canopy of trees, with individual stems as long as 70 metres. Lianas abound in tropical forests and is one of the essential structural differences between the tropical and temperate forests. They form bridges alongside forest canopies which serve as a path for arboreal animals across the forest. The plants begin their growth in deep shade but by climbing up and over the top of very tall trees they benefit from full sunlight at maturity. They may bind the trees together so if one dies it is held in position until it decays. Examples are Monkey ladder (*Entada Gigas*) and Water vine (*Cissus hypoglauca*).

LIAPUNOV/LYAPUNOV CONVEXITY THEOREM: It states that the range of a (non-atomic) vector measure is closed and convex. In other words, given a finite number of signed finite measures $\{\mu_1, \dots, \mu_n\}$, the associated vector measure μ with values in n -space defined by $\mu(E) = \{\mu_1(E), \dots, \mu_n(E)\}$ for all E in the measure algebra M , has a compact range for each measurable set E . That is, $R_\mu(E) = \{\mu(F) : F, E \in M, F \subseteq E\}$ is compact and is convex if each measure is non-atomic. It is named after Alexey Andreevich Lyapunov (1911–1973), Soviet mathematician.

LIAPUNOV/LYAPUNOV FUNCTION: A function V constructed in order to establish that a point, for example, zero, is stable for an autonomous differential equation system $y' = f(y)$ with f continuous. In one form, one requires that locally the function V has continuous partial derivatives, is strictly positive except at zero and has the derivative along a trajectory $\frac{dV(y(t))}{dt}$ negative for all solutions. The

conclusion is that zero is a Liapunov-stable point of the system, called **Liapunov's second** or **direct method**. If, in addition, the derivative along a trajectory is strictly negative away from the origin, then the origin is actually **asymptotically stable**.

LIBIDO: The emotional and psychic energy in people's psychological makeup that is related to the basic human instincts, especially the sex drives, a concept originated by Sigmund Freud. He believed that psychiatric illnesses were the result of misdirecting or suppressing the libido.

LIBRARY PROGRAM (LIBRARY): Any of a collection of regularly used software routines, held in a computer-backing store that a computer can interpret and execute. In MS-Windows environments, library files have a .DLL extension.

LIBRATION: The phenomenon that enables 59% of the Moon's surface to be observed from Earth over a 30-year period, in spite of its synchronous rotation. Physical libration arises from slight variation in the rotation of the Moon on its axis, caused by minor distortions in its physical shape. Geometric librations are apparent oscillations arising from the fact that the Moon is observed from slightly different directions at different times. The geometric libration in longitude results from the non uniform orbital motion of the moon. The geometric libration in latitude arises because the Moon's axis of rotation is not perpendicular to its orbital plane, enabling more of the lunar Polar Regions to be observed.

LICHEN: A symbiotic association between a fungus and an alga which develops into unique morphological form quite distinct from each symbiont. The combination becomes an identifiable new plant with a name. In lichens 98% of the plant consists of Ascomycota and the rest are mainly Basidiomycota. Lichens are able to survive extreme heat, cold and drought, but unable to withstand air pollution and are especially sensitive to sulfur dioxide and fluorine. There are three morphological types: **crutose** which are flat, closely-oppressed patches on the substratum; **foliose**, which resembles leaves loosely attached to the substratum; and **frutitos** which produces cylindrical

branches. Crutose lichens play a role in soil formation. Cladonia species and other lichens serve as food for humans and reindeers. Other uses include as dyes in the textile industry, components of perfume, cosmetics and lotions.

LIE DETECTOR: An instrument intended to detect an involuntary physiological response that all persons exhibit when lying. It records graphically certain body activities such as thoracic and abdominal respiration, blood pressure, pulse rate and galvanic skin responses. Marked changes in these activities when a person answers questions may indicate that the person is lying.

LIE GROUP: A topological group whose topological space is a finite-dimensional (or locally Euclidean) manifold. In other words, a topological group that can be given an analytic structure such that the group operation and inversion are analytic. Thus, the coordinates of the product of two elements are analytic functions of the coordinates of the elements. It is named after Sophus Lie 1842-1899), Norwegian mathematician.

LIE PRODUCT (LIE COMMUTATOR): The binary operation that sends elements a and b to the Lie bracket $[a, b]$ which is defined as $ab - ba$.

LIEBERMANN TEST: A presumptive test sometimes used for cocaine and morphine. The Liebermann reagent is a solution of potassium nitrite (KNO_2) in sulfuric acid. With morphine a black colour is produced; cocaine gives a yellow colour.

LIEBERMANN'S REACTION: A method of testing for phenols. A small sample of the substance and a crystal of sodium nitrite are dissolved in warm sulfuric acid. The solution is then poured into excess aqueous alkali. The formation of a blue green colour indicates the presence of a phenol.

LIEBIG, JUSTUS VON (1803-73): German organic chemist, who made many important contributions to the early systematisation of organic chemistry and to biochemistry, chemical education and agricultural chemistry. Liebig was the first to recognise that the two different chemical compounds can have the same formula. He also developed a method of analysing organic compounds by burning them to produce carbon dioxide and water; analysed many tissues and body fluids and showed that plants use carbon dioxide, water and ammonia.

LIEBIG'S CONDENSER: Named after the German chemist Justus Baron von Liebig, although he did not invent it, it is an apparatus used to convert vapour into liquid or a condenser used for distillation. It consists of a glass tube surrounded by a jacket through which cooling water is run.

LIEBMANN METHOD (GAUSS-SEIDEL ITERATION; METHOD OF SUCCESSIVE DISPLACEMENTS): An iterative method similar to the Jacobi iteration, which is used to solve a linear matrix system of equations, $\mathbf{Ax} = \mathbf{b}$ by repeatedly solving for $\mathbf{x}_{\text{NEW}}$ from

$$\mathbf{x}_{\text{NEW}(i)} = \left(\frac{1}{a_{ii}} \right) \left[\mathbf{b}_i - \sum_{j < i} a_{ij} \mathbf{x}_{\text{OLD}(j)} - \sum_{j > i} a_{ij} \mathbf{x}_{\text{NEW}(j)} \right]. \text{ This assumes that}$$

$a_{ii} \neq 0$ and will not always converge even then.

LEIBNIZ'S ALTERNATING SERIES TEST (ALTERNATING SERIES TEST): A method used to prove that an alternating series with terms that decrease in absolute value is a convergent series; and to establish conditional convergence of an alternating series one need only to verify that the terms decrease monotonically to zero in absolute value. In this case the error after adding n terms is always less than the magnitude of the next term. For example, $1 - \frac{1}{3} + \frac{1}{5} - \frac{1}{7} + \frac{1}{9} - \dots$

is convergent, since $|a_n| = \left| \frac{(-1)^n}{2n+1} \right| = \frac{1}{2n+1}$ decreases monotonically to zero as n tends to infinity; its sum is $\pi/4$.

LIEBNIZ'S SERIES (GREGORY'S SERIES): The Maclaurin series for arctan: $\arctan x = x - \frac{x^3}{3} + \frac{x^5}{5} - \frac{x^7}{7} + \dots$ It is the special case of $x = 1$ sums to $\pi/4$.

LIFE: The ability to generate energy or process materials for energy and tissue building, grow, reproduce and respond to such stimuli as light, heat and sound. All known life-forms possess either DNA or RNA. Viruses possess DNA and RNA, but cannot reproduce without a host cell and do not metabolise nutrients.

LIFE CYCLE: The sequence of developmental stages through which all different generations of a species succeed each other through means of reproduction, whether asexual or sexual. For example, human life cycle includes egg, embryo, foetus, baby, child, adolescent, adult, old age and death. The life cycle of a flowering plant begins when the seed germinates. It progresses through the growth of roots, stems and leaves; formation of flower buds; pollination and fertilisation; and seed and fruit development. The life cycle ends with senescence or old age and death.

LIFE EXPECTANCY: The average number of years which an individual can expect to live in a given society. It depends on the nutrition, disease control, environmental contaminant, war, stress and living standards in general. The various phases for most multicellular organisms are: **embryonic phase**, starting with the first division of the fertilised egg; a **juvenile phase**, in which the organism grows to sexual maturity; and an **adult phase**, in which the organism reaches a peak of sexual productivity and begins a decline of physiological activities that ends in death. In 2013, Monaco had the highest life expectancy with 89.63 years; and Chad had the lowest life expectancy with just 49.07 years.

LIFE FORM: The body form that characterises a kind of organism (as a species) at maturity. This forms the basis of which a species may be described as a plant, animal, bacterium, protist or fungus. In describing an area of vegetation much information is conveyed by naming the type or types of life form that predominate. The Raunkiaer system of classification is often used for this purpose.

LIFE SCIENCES: The scientific study of living organisms, which comprising bacteriology (the study of bacteria), bioinformatics (the development and application of computational methods in biology), biophysics (the use of physics to study biological systems), biostatistics (the application of statistics to biology), botany (the study of plants), cell biology (the study of life at the cellular level), entomology (the study of insects), ethology (the study of animal behaviour), microbiology (the study of microscopic organisms), molecular biology (the study of life at the molecular level), mycology (the study of fungi), parasitology (the study of parasites), physiology (the study of how living organisms function), soil science (the study of soils), taxonomy (the classification of living organisms), zoology (the study of animals), etc.

LIFE SPAN: Length of time that embraces all the events of an organism's life, from conception to death.

LIFE TABLE: A table of data of summarising the probable years of survivorship of a defined population of subjects. Survivors to age x are denoted by the symbol l_x and the expectation of life at age x is denoted by the symbol e_x .

LIFE-CYCLE ANALYSIS: A systematic assessment of the environmental impact of all the stages of a product's life, including resources used, manufacturing, packaging, distribution, use and disposal or recycling. This 'cradle-to-grave' approach can expose inconsistencies in many so-called 'eco-friendly' labels, applied to products such as soap powders, which may be biodegradable but which are perhaps contained in non-recyclable containers.

LIFE-HISTORY PATTERN: A group of traits, such as size and number of offspring, length of maturation, age at first reproduction and the number of times reproduction occurs, that affect reproduction, survival and the rate of population growth.

LIFETIME: 1. The period of time during which an individual lives. 2. The average time spent by an elementary particle or radioactive nucleus in a certain state before decay occurs or before there is a transition from an atom in an excited state to a lower energy state.

LIFO (LAST IN, FIRST OUT): A method of processing data in which the last items entered are the first to be removed. It is sometimes used by computers when extracting data from an array or data buffer. The opposite **FIFO (FIRST IN, FIRST OUT)**, describes items removed in the order they have been entered.

LIFSHITZ-SLYOZOV LAW: The relation $L \propto t^{1/2}$, between a length-scale L , characteristic of a system undergoing spinodal decomposition and the time t , for a long time after the system has been quenched. The Lifshitz-Slyozov law applies below the spinodal curve to a system in which the order parameter is conserved. This law is in accord with light-scattering experiments on binary liquid mixtures. The Lifshitz-Slyozov law was put forward by the Soviet physicists M. Lifshitz and V.V. Slyozov.

LIFT: 1. In aerodynamics, the force that acts at a right angle to the direction of motion through the air and directly opposes the weight of a flying object such as an airplane and holds it in the air. Lift is created by differences in air pressure. It is a vector quantity, having both a magnitude and a direction. 2. The force produced by the action of air flow (aerofoil) that enables birds to fly and also controls their movements in flight.

LIFTING CONDENSATION LEVEL: The height in metres above the ground surface at which a rising parcel of moist air first becomes saturated.

LIFTING UNIT: A pair of wheel or triangular-shaped shear, used in harvesting roots and passing them through the elevator.

LIGAMENT: A strong flexible connective tissue that supports internal organs and holds bones together properly in joints such as elbow and knee joints. White ligament is made up of sturdy, inelastic collagen fibres; yellow ligament is rich in tough elastic fibres, which allow more movement.

LIGAND: 1. An atom, ion or molecule that donates a pair of electrons to a metal atom or ion in forming a coordinating complex. Molecules that function as ligands are acting as Lewis bases. 2. A molecule that binds to a protein with a high degree of specificity. Examples are the substrate of an enzyme and a hormone binding to a cell receptor.

LIGAND FIELD SPLITTING (CRYSTAL FIELD SPLITTING): The difference in energy between d-orbitals of ligands in a crystal field, that explains the difference in colour between two similar metal-ligand complexes. Its number is denoted by the capital Greek letter Δ . It tends to increase with oxidation number and increases down a group on the periodic table.

LIGAND FIELD THEORY (LFT): An extension of crystal-field theory describing the properties of compounds of transition-metal ions or rare-earth ions in which covalent bonding between the surrounding molecules and the transition-metal ions is taken into account. This may involve using valence-bond theory or molecular-orbital theory. As with crystal-field theory, ligand-field theory indicates that energy levels of the transition-metal ions are split by the surrounding ligand, as determined by group theory. The theory has been very successful in explaining the optical, spectroscopic and magnetic properties of the compounds of transition metal and rare-earth ions.

LIGAND TO LIGAND CHARGE TRANSFER (LLCT) TRANSITION: An electronic transition of a metal complex that corresponds to excitation populating an electronic state in which considerable electron transfer between two ligands has occurred.

LIGAND TO METAL CHARGE TRANSFER (LMCT) TRANSITION: An electronic transition in a metal complex that corresponds to excitation populating an electronic state in which considerable electron transfer from a ligand to a metal centre has occurred.

LIGAND-GATED ION CHANNEL (LGICs): A type of ion channel that opens when a signal molecule binds to an extracellular receptor

region of the channel protein. This changes the conformation of the channel protein and hence opens the channel. Many neurotransmitters act as ligands for ion channels. For example, at neuromuscular junctions, nicotinic acetylcholine receptors form a cation channel that opens when two molecules of acetylcholine bind to the channel's extracellular region. The resultant influx of sodium ions causes depolarisation of the muscle cell membrane. Other types of ligand-gated ion channels are found at synapses and in the brain.

LIGASE (SYNTHETASE): A class of enzymes that catalyse the formation of covalent bonds using the energy released by the cleavages of adenosine triphosphate (ATP). Ligases are important in the synthesis and repair of many biological molecules, including DNA and are used in genetic engineering to insert foreign DNA into cloning; and also in recombination techniques to seal the joints between DNA sequences. For example, in DNA, it joins 3'-OH to 5' phosphates.

LIGATE: 1. To tie up or constrict with a ligature. 2. To coordinate to a central atom as a ligand. 3. To join together (as DNA or protein chains) by a chemical process.

LIGHT: An electromagnetic radiation having a wavelength between 400 (extreme violet) and 760 nanometres (extreme red), which can be detected by the eyes of humans and animals. Light from the sun provides the energy to fuel the photosynthetic fixation of carbon dioxide by green plants and is thus the basis of life on earth.

LIGHT AMPLIFICATION BY STIMULATED EMISSION OF RADIATION (LASER): A light amplifier, which works on the same principle as the maser using the principle of amplification of electromagnetic waves by stimulated emission of radiation to produce monochromatic coherent radiation in the infrared, visible and ultraviolet regions of the electromagnetic spectrum. Laser light is very pure in colour, can be extremely intense and can be directed with great accuracy. Lasers are used in many modern technological devices including bar code readers, compact disc (CD) players and laser printers.

LIGHT AMPLIFIER (IMAGE INTENSIFIER): Any electronic device such as photocell relay-light source combination which, when actuated by a light image or incident light, reproduces a similar image of enhanced brightness and which is capable of operating at very low light levels without introducing spurious brightness variations (noise) into the reproduced image.

LIGHT CLOCK: A beam of light reflected between two parallel mirrors, used to measure time.

LIGHT COMPENSATION POINT: The point on a light curve where the rate of cellular respiration is exactly equal to the rate of photosynthesis. This depends on factors such as plant type, changes in temperature and other environmental factors. Shady plants tend to have lower light compensation points than sun plants because they have lower respiration rates and can absorb light more efficiently.

LIGHT CURVE: A graph showing how the brightness of an astronomical object such as a star varies with time; plotted with brightness on the y -axis and time on the x -axis. Analysis light curves of variable stars gives information about the physical processes causing the variation.

LIGHT DETECTION AND RANGING (LIDAR): A real time remote sensing technique for the determination of aerosol and trace gas concentrations in air by measurement of scattered laser radiation.

LIGHT ENERGY: A type of kinetic energy that is carried by waves in the visible region of the electromagnetic (EM) spectrum. Light is a form of electromagnetic radiation, and hot objects, such as light bulbs, the sun and lasers, emit it. Light energy can be converted into other forms of energy. Light energy is used by plants in photosynthesis and by extension, provides food for most organisms, provides vision, colour, maintains the temperature, provides energy from the sun for solar

power, etc. All types of electromagnetic radiation travel at a constant speed, the speed of light. The speed of light is about 300,000 km/s.

LIGHT INTENSITY: Brightness of light, which is obtained by dividing either a power or a luminous flux by a solid angle, a planar area or a combination of the two. There are several intensities of light, which include **radiant intensity**, which is energy emitted per unit time per unit solid angle about the direction considered, measured in watts per steradian (W/sr); **irradiance or intensity** is the power of electromagnetic radiation per unit area incident on a surface, measured in watts per square metre (W/m²); **radiance**, commonly called **intensity** in astronomy and astrophysics is radiant energy emitted per unit time in a specified direction by a unit area of an emitting surface, measured in watts per steradian per square metre (W·sr⁻¹·m⁻²); **luminous intensity** is the luminous flux density per solid angle as measured in a given direction relative to the emitting source, measured in lumens per steradian (lm/sr) or candela (cd); **luminance** is the amount of light that passes through or is emitted from a particular area and falls within a given solid angle, measured in candela per square metre (cd/m²); **illuminance** is the light incident on a surface measured in lumens per square metre (lm/m²) or lux (lx); and **luminous emittance or luminous exitance** is light emitted from a source, also measured in lumens per square metre (lm/m²) or lux (lx).

LIGHT MICROSCOPE (OPTICAL MICROSCOPE): An optical instrument that contains one or more lenses and is used in the laboratory to enlarge objects that are too small to be examined in detail by the naked eye. Its maximum resolution (the capacity to observe fine detail clearly), is about 0.3 µm (i.e. it can distinguish points only 0.3 µm apart) as compared to about 80 µm for the average human eye.

LIGHT PEN: A computer input device, used in association with a special software that enables an operator to write or draw on screens.

LIGHT POLARISATION (LIGHT POLARIZATION): A process or state in which rays of light exhibit different properties in different directions, especially the state in which all the vibration takes place in one plane. A light beam is linearly polarised if the end-point of the electric vector moves in a straight line when viewed along the direction of propagation of the beam. If it moves along a circle the beam is **circularly polarised**; and if it moves along an ellipse the beam is **elliptically polarised**. Circular polarisation is said to be right-handed if the direction of rotation is clockwise when viewed against the direction of propagation; and left-handed if the sense of the rotation is opposite. When the position of the endpoint of the electric vector is viewed at a given time t as a function of distance along x , it forms a left-handed helix if the light polarisation is left-handed and a right-handed helix if it is right-handed.

LIGHT REACTION (LIGHT-DEPENDENT REACTION): The first stage in photosynthesis that requires light energy that is captured by light-absorbing pigments such as chlorophyll to be converted into chemical energy in the form of adenosine triphosphate (ATP) and NADPH that will be used in synthesis of glucose in the dark stage. The light reactions take place in the lumen of the thylakoid membrane in the chloroplast. The light-dependent reaction takes place on the thylakoid membranes of the chloroplast. It can be cyclic or non-cyclic. In cyclic reaction, when the electron is displaced from the photosystem, the electron is passed down the electron acceptor molecules and returns to photosystem I, from where it was emitted. In the non-cyclic reaction, the photons are captured in the light-harvesting antenna complexes of photosystem II by chlorophyll and other accessory pigments.

LIGHT SCATTERING BY LIGHT: A process predicted to occur in quantum electrodynamics (QED) that does not occur in classical electrodynamics. Light scattering by light does not occur in

classical physics because Maxwell's equations are linear, whereas electrodynamics becomes nonlinear when quantum effects are taken into account. The process has never been conclusively observed but powerful lasers may enable it to be observed in the future. Light scattering by light was first predicted in the 1930s, when calculations of the process were performed by the German physicist Werner Heisenberg and others. Complete calculations of light scattering by light were not performed until the renormalisation of QED was developed in the late 1940s.

LIGHT SECOND: A unit of length that is equal to the distance travelled by a beam of light in a vacuum in one second. It is equal to 2.99792458×10^8 metres.

LIGHT SUSSEX: A dual-purpose breed of poultry, white with black stripes on the feathers of the neck and black feathers on the wings.

LIGHT TIME: The time taken for light, travelling at 299,793 km/s, to reach Earth from a distant object. A correction for the effect of light time has to be made when calculating rotation periods of planets, eclipses and transits. The observed times of maxima and minima of variable stars also need a light-time correction depending upon the position of Earth in its orbit at the time of observation, as periods of variation are stated for the Earth at mean distance from the star.

LIGHT YEAR (LIGHT-YEAR): In astronomy, it is the distance travelled by a beam of light in a vacuum in one year. Its symbol is ly. It is equal to 9.465×10^{12} km or approximately 9.46 trillion kilometres. The light-year is often used to measure distances to stars and other distances on a galactic scale.

LIGHT-DEPENDENT REACTION (LIGHT REACTION): The first stage in photosynthesis that requires light energy that is captured by light-absorbing pigments such as chlorophyll to be converted into chemical energy in the form of adenosine triphosphate (ATP) and NADPH that will be used in synthesis of glucose in the dark stage. The light reactions take place in the lumen of the thylakoid membrane in the chloroplast. The light-dependent reaction takes place on the thylakoid membranes of the chloroplast. It can be cyclic or non-cyclic. In cyclic reaction, when the electron is displaced from the photosystem, the electron is passed down the electron acceptor molecules and returns to photosystem I, from where it was emitted. In the non-cyclic reaction, the photons are captured in the light-harvesting antenna complexes of photosystem II by chlorophyll and other accessory pigments.

LIGHT-DEPENDENT RESISTOR, LDR (PHOTORESISTOR):

A high resistance electronic component, whose resistance varies according to the intensity of light it is exposed to, usually ranging as much as several megohms in the dark and as low as a few hundred ohms when exposed to light. For example, when exposed to a light level of 1000 lux, the resistance is just 400 ohms; but when exposed to a light level of 10 lux, the resistance rises to 10.43 megohms (10,430,000 ohms). It is made of a high resistance semiconductor that absorbs photons and based on the quantity and frequency of the absorbed photons the semiconductor material gives bound electrons enough energy to jump into the conduction band. The resulting free electrons conduct electricity resulting in lowering the resistance of the photoresistor. LDRs are classified as intrinsic or extrinsic based on the materials used. Intrinsic photo resistors are semiconductors that are not doped, consisting of pure semiconductor materials such as silicon or germanium. Extrinsic photo resistors are made of doped semiconductor materials. LDRs are often used as light sensor devices such as camera light meters, clock radios, alarm clock, burglar alarm circuits, in street light, etc.

LIGHT-EMITTING DIODE (LED): A semiconductor device that emits light when a suitable voltage is applied to it. LED consists of a P-type semiconductor and N-type semiconductor that are placed together, forming a P-N junction. The "P" represents "positive" and are holes (known as the anode), and the "N" represents "negative"

(cathode) and are electrons. Therefore, the P region of the semiconductor is positively charged with holes, while the N region is negatively charged with electrons. When a suitable voltage is applied to the leads of the P-N region, current flows in the forward direction and blocks the current in the reverse direction enabling electrons to flow from the N-region and recombine with the holes in the P-region. This creates energy in the form of visible light, a process called electroluminescence. LEDs emit various colours of light depending on the type of material used for the semiconductor. For example, for blue, white, green and UV colours, the material used is Indium gallium nitride (InGaN); and for red, yellow, and orange, the material is Aluminum gallium indium phosphide (AlGaInP or AlInGaP). LEDs are very energy efficient and have a long lifespan and are used in many applications such as indicator lights, displays, flashlights, traffic light signals, lights in homes and industries, street lights, motor vehicle lighting, etc. LEDs called infrared-emitting diode (IRED) emit infrared (IR) energy that is not visible to the eye but can be used as an infrared sensor in remote controls or other electronic devices.

LIGHT-INDEPENDENT REACTIONS (DARK REACTIONS): The series of biochemical carbon-fixing reactions of the second stage of photosynthesis, where energy stored in adenosine triphosphate (ATP) and NADPH by the light-dependent reactions is used to reduce carbon from carbon dioxide to simple sugars. Because light is not required for these reactions, it is also known as the dark reaction stage of photosynthesis.

LIGHTHOUSE: A structure built on or near the shore, which exhibits a light of distinctive characteristics to serve as an aid to navigation.

LIGHTNING: An atmospheric discharge of electricity between two-charged rain clouds or between a cloud and the Earth caused by a build-up of electrical charges, often accompanied by thunder which typically occurs during thunderstorms and sometimes during volcanic eruptions or dust storms.

LIGHTNING CONDUCTOR (LIGHTNING ROD): A sturdy electrical conductor, running from a high point such as a tall building to ground. It protects a structure from lightning strikes by providing an easier path for the current to flow to the earth than through the building.

LIGHTWEIGHT DIRECTORY ACCESS PROTOCOL (LDAP): A protocol used for accessing information directories. LDAP support is implemented in Web browsers and e-mail programs, which can query an LDAP-compliant directory. LDAP is protocol similar to HTTP and FTP and uses the ldap:// prefix in its URL.

LIGNANS: A type of polyphenolic compound that occurs naturally in some plants. It has antioxidant and oestrogenic activities. Examples of lignans are pinoresinol, podophyllotoxin and steganacin. Sources include flax, sesame, sunflower and pumpkin seeds. Lignan compounds are also found in many vegetables, fruits, tea and wine. One lignan compound, secoisolariciresinol diglycoside (SDG), is a precursor to a variety of phytoestrogens with antioxidant properties.

LIGNEOUS: Woody; having the appearance of wood.

LIGNICOLOUS: A term used to describe organisms that live on or in wood, such as bracket fungi.

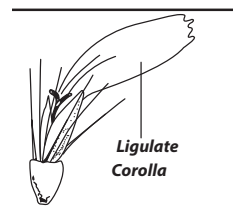
LIGNIFY: The deposition of lignin in the walls of plant cells such as fibre, tracheid and vessel elements.

LIGNIN: A complex organic compound deposited in the walls of plant cells giving them strength. Lignin is a mixture of substances of high carbon content but distinct from carbohydrates. Lignification makes the walls woody and therefore rigid.

LIGNITE (BROWN COAL): A type of coal that is brown and fibrous or visible plant remains with a relatively low carbon content (25-35%), a high moisture content (about 66%) and an ash content ranging from 6% to 19%. It is used in generating electricity in some countries.

LIGROIN: A flammable mixture of hydrocarbons that boils at 20 °C - 135 °C, obtained from petroleum by distillation and used as a solvent.

LIGULATE: Strap or tongue shaped such as the ray florets of plants of the daisy family. Ligulate flowers superficially resemble the ray flowers of radiate heads in having a corolla that is tubular at the base and prolonged on the outer side into a flat, strap-shaped ligule.



LIGULE: 1. A membranous or hairy structure or outgrowth located on the inside of the leaves of certain plants. Many grasses have a ligule at the base of the leaf blade. 2. A strap-shaped extension from the corolla tube in certain florets of a capitulum. They are termed **ligulate florets**.

LIKE TERMS: Terms of an algebraic expression that differ only in their coefficients. Examples are $4xy^2$, $6xy^2$, $-5xy^2$ and $1/2(xy^2)$. They can be combined into like terms by adding or subtracting only their coefficients and then combining the result with the variables. For $4xy^2$ and $6xy^2$, the sum is given by $4xy^2 + 6xy^2 = 10xy^2$.

LIKELIHOOD: In statistics, the probability of a given sample being randomly drawn, regarded as a function of the parameters of the population. Likelihood functions play a key role in statistical inference, especially methods of estimating a parameter from a set of statistics. A **likelihood ratio** is any ratio of likelihoods.

LIKELIHOOD RATIO TEST (LIKELIHOOD-RATIO TEST): The statistical test in which the null hypothesis is rejected for small values of $\lambda = p_0/p_1$, where each p_1 is the maximum of the probabilities $P[X | \theta]$, where θ ranges over the possibilities permitted by the respective hypotheses and X is a vector of observations. This is an invaluable hypothesis testing technique, used to compare the goodness of fit of two models, the null model and the alternative model.

LILIACEAE: A large monocotyledonous family, commonly known as the lily family, containing about 3500 species in about 250 genera. Many of the Liliaceae possess swollen underground perennating organs, such as bulbs, corms and rhizomes. The only important food plant in the family is the onion (*Allium cepa*). Minor food plants include garlic, leek and asparagus.

LILIIDAE: One of four subclasses or superorders of Monocotyledones comprising 17 families including Liliaceae, Alliaceae, Amaryllidaceae, Iridaceae, orchidaceae and Trilliaceae. They are mainly herbaceous plants often with relatively large flowers.

LIM SUP (LIMIT SUPERIOR; UPPER LIMIT): 1. The limit as n tends to infinity of the suprema of the subsequences of the elements beyond the n^{th} of a given real sequence, expressed as $\lim_{n \rightarrow \infty} \sup a_n = \lim_{n \rightarrow \infty} [\sup \{a_m : m \geq n\}]$. This produces the largest cluster point of the sequence and may be positive infinity. The sequence has a limit if and only if the limit superior and the limit inferior are identical and in that case the limit is their common value. 2. The set of points that are infinitely many members of a given sequence $\{A_n\}$, given

as $\lim_{x \rightarrow \infty} A_n = \bigcap_{m=1}^{\infty} \left\{ \bigcup_{n \geq m} A_n \right\}$. The sets are said to have a limit if the limit

superior and limit inferior, agree and the limit is then this common set.

LIMACON OF PASCAL: The locus of a point, P, on a line at a fixed distance from its intersection, X, with a fixed circle as it revolves around a point, C, on the circle. Its polar equation is as follows:

$r = 2a \cos \theta + b$, then it becomes a cardioid, if $b = a$ then it becomes a trisectrix. Its Cartesian equation is given by:

$(x^2 + y^2 - 2ax)^2 = b^2(x^2 + y^2)$. If $b \geq 2a$ then its area is $(2a^2 + k^2)\pi$. If $b = a$ then the area of the inner loop is given by $a^2(\pi - 3\sqrt{3}/2)$ and the area between the loops is given by $a^2(\pi + 3\sqrt{3})$. It was discovered by Étienne Pascal (1588-1651), French lawyer and amateur mathematician.

LIMB: 1. An appendage of a vertebrate such as the leg or arm of a mammal or the wing of a bird used for locomotion, such as walking, running, flying or climbing. 2. The upper part of a sepal, petal or leaf, which is expanded or the widened upper section of a gamopetalous corolla.

LIMBATE: A leaf which has a distinct border or edging, with the edges having a different colour from the rest.

LIMBIC SYSTEM (PALEOMAMMALIAN BRAIN): A group of brain structures involved in the expression and control of mood and instincts playing a major role in long term memory. The limbic system includes the hippocampus, amygdala, anterior thalamic nuclei and limbic cortex.

LIME: 1. Also called **quicklime**, **caustic lime**, or **calcium oxide**, it is a white powdery substance with the chemical formula CaO , density 3.35 g/cm^3 , melting point 2572°C (2845 K) and boiling point 2850°C (3123 K). It is made by heating limestone above 900°C (1173 K) in a kiln. It is used in making mortar and cement, porcelain and glass, in purifying sugar, in preparing bleaching powder, calcium carbide and calcium cyanamide, in water softeners and to neutralise acid in soil. 2. Any of several species and hybrids of spiny trees and shrubs in the rue family (*Rutaceae*), widely grown in tropical and subtropical areas for their edible acidic fruits. It is a small, evergreen tree, about 5 m tall with short, stiff, sharp spines. The leaves are small and heart-shaped, the yellowish fragrant flowers come in clusters. The fruit, also called lime is round or globose to ovoid in shape, like a small orange (3-6 centimetres in diameter), and green in colour. The juice has a sour and acidic taste. Limes contain vitamin C, used to flavour foods and beverages; oil is extracted from the fruit, used for some health reasons, such as to treat severe diarrhoea or dysentery, to kill germs on the skin, etc.; also used to prepare some cosmetics. Citrate of lime and citric acid are also prepared from the fruit. In some traditions the leaves are used as mosquito repellants.

LIME CONCRETION: An aggregate of precipitated calcium carbonate or other material cemented by precipitated calcium carbonate.

LIME KILN: Oven used to make quick lime (calcium oxide) by heating limestone, CaCO_3 above 900°C (1173 K) in the absence of air to produce lime.

LIME POTENTIAL: The negative logarithm of the ratio of the hydrogen ion activity to the square root of the sum of the activities of calcium and magnesium in the soil solution.

LIMESTONE: A common sedimentary rock that is composed largely of carbonates of calcium and magnesium. Clacite and aragonite are the chief minerals; dolomite is also present in the dolomitic limestone. There are many varieties of limestones but most are deposited in shallow water organic limestones (chalk) formed from the calcareous skeletons of organisms. Precipitated limestones include oolite, which is composed of ooliths (spherical bodies formed by the precipitation of carbonate around a nucleus); and clastic limestones which are derived from fragments of pre-existing calcareous rocks. Limestones and dolomites are used commercially as building materials and as a source for industrial and agricultural lime. Also, limestones and dolomites are important reservoirs for oil and gas and are the hosts for important mineral deposits, such as lead, zinc, silver and fluorite.

LIMESTONE PAVEMENT: Bare blocks of rock, separated from each other by fissures or grikes, called clints, formed by the weathering of limestone. It is found on limestone plateaus.

LIMEWATER: A saturated solution of calcium hydroxide $[\text{Ca}(\text{OH})_2]$ in water. When carbon dioxide gas is bubbled through limewater, a 'milky' precipitate of calcium carbonate is formed:

$\text{Ca}(\text{OH})_{2(aq)} + \text{CO}_{2(g)} \rightarrow \text{CaCO}_{3(s)} + \text{H}_2\text{O(l)}$. If the carbon dioxide continues to be bubbled through, the calcium carbonate eventually dissolves to form a clear solution of calcium hydrogen carbonate:

$\text{CaCO}_{3(s)} + \text{H}_2\text{O(l)} + \text{CO}_{2(g)} \rightarrow \text{Ca}(\text{HCO}_3)_{2(aq)}$. It is therefore, used to test for the presence of carbon dioxide gas. If cold limewater is used the original calcium carbonate precipitated has a calcite structure. Hot limewater yields an aragonite structure.

LIMING: The application of lime (calcium hydroxide) to soils to increase levels of calcium and to decrease acidity. It enables bacteria on the roots of legumes such as alfalfa and clover to secure essential nitrogen from the air, increasing soil fertility. It also increases the available phosphorus in soils.

LIME REQUIREMENT: The amount of agricultural limestone, or the equivalent of another liming material, required per hectare to a soil depth of 15 cm to raise the pH of the soil to a specific value under field conditions.

LIMIT: 1. The value that a function approaches as the independent variable approaches a specified value or infinity or one of the two given values specifying the range over which a definite integral is evaluated. The limit of a function $f(x)$ as x approaches a is L , written as $\lim_{x \rightarrow a} f(x) = L$, if, for every real number $\varepsilon > 0$, there exists a real number $\delta > 0$ such that $|f(x) - L| < \varepsilon$ and $0 < |x - a| < \delta$. 2. The limit of reading a measurement is equal to the smallest graduation on the scale of an instrument.

LIMIT CYCLE: A closed trajectory in phase space having the property that at least one other trajectory spirals into it either as time approaches infinity or as time approaches negative infinity. Such behaviour is exhibited in some nonlinear systems. In the case where all the neighbouring trajectories approach the limit-cycle as time approaches infinity, it is called a **stable** or **attractive limit-cycle** (**ω -limit cycle**). If instead all neighbouring trajectories approach it as time approaches negative infinity, it is an **unstable** or **non-attractive limit-cycle** (**α -limit cycle**).

LIMIT INFERIOR (LIM INF; LOWER LIMIT): 1. The limit as n tends to infinity of the infima of the subsequences of the elements beyond the n^{th} of a given real sequence, expressed as:

$\liminf_{n \rightarrow \infty} a_n = \lim_{n \rightarrow \infty} [\inf \{a_m : m \geq n\}]$. This produces the smallest cluster point of the sequence and may be negative infinity. The sequence has a limit if and only if the limit inferior and limit superior are identical and in that case the limit is their common value. 2. The set of points that are in all but finitely many members of a given sequence of sets

$\{A_n\}$, expressed as $\liminf_{n \rightarrow \infty} A_n = \bigcup_{m=1}^{\infty} \bigcap_{n \geq m} A_n$. The sets are said to have

a limit if the limit inferior and limit superior agree and the limit is then this common set. For example, the limit of a nested decreasing sequence of sets is the intersection of the collection.

LIMIT LOAD: The maximum load calculated to be experienced by the structure under specified conditions of operation.

LIMIT OF A POLYNOMIAL FUNCTION: The limit of a polynomial function, $f(x)$, as x approaches b is $\lim_{x \rightarrow b} f(x) = f(b)$.

LIMIT OF A SEQUENCE: The limit of a sequence is the value which the terms of the sequence approach as n approaches infinity.

LIMIT ORDINAL: A non-zero ordinal with no immediate predecessor. For example, the ordinal of the natural number ω is a limit ordinal.

LIMIT POINT (ACCUMULATION POINT; CLUSTER POINT): In mathematics, a point where neighbourhoods intersect with the set other than at the point itself.

LIMIT PRESSURE: The maximum operating pressure including the effect of system environment, such as vehicle acceleration, etc. For hydraulic and pneumatic equipment, limit pressure excludes the effect of surge.

LIMIT SUPERIOR (LIM SUP; UPPER LIMIT): 1. the limit as n tends to infinity of the suprema of the subsequences of the elements beyond the n^{th} of a given real sequence, expressed as

$\limsup_{n \rightarrow \infty} a_n = \lim_{n \rightarrow \infty} [\sup \{a_m : m \geq n\}]$. This produces the largest cluster point of the sequence and may be positive infinity. The sequence has a limit if and only if the limit superior and the limit inferior are identical and in that case the limit is their common value. 2. The set of points that are in infinitely many members of a given sequence $\{A_n\}$, given

as $\lim_{x \rightarrow \infty} A_n = \bigcap_{m=1}^{\infty} \left\{ \bigcup_{n \geq m} A_n \right\}$ The sets are said to have a limit if the limit

superior and limit inferior, agree and the limit is then this common set.

LIMIT TEST: Acute toxicity test in which, if no ill-effects occur at a pre-selected maximum dose, no further testing at greater exposure levels is required.

LIMITING ADSORPTION CURRENT: The potential-independent value that is approached by an adsorption current as the rate of reduction or oxidation of the electroactive substance is increased by varying the applied potential. The terms adsorption current and limiting adsorption current should not be applied to faradaic currents that have been increased or decreased by adding a non-electroactive surfactant to a solution containing an electroactive substance, nor to apparent waves resulting from the effect of adsorption or desorption on double-layer currents.

LIMITING CATALYTIC CURRENT: The potential-independent value that is approached by a catalytic current as the rate of the charge transfer process is increased by varying the applied potential.

LIMITING CONDITION OF OPERATION: Range of physical and operational parameters in which the method meets given values of performance characteristics with 95% probability.

LIMITING CURRENT: The limiting value of a faradaic current that is approached as the rate of the charge-transfer process is increased by varying the potential. It is independent of the applied potential over a finite range and is usually evaluated by subtracting the appropriate residual current from the measured total current. A limiting current may have the character of an adsorption, catalytic, diffusion or kinetic current and may include a migration current.

LIMITING DIFFUSION CURRENT: The potential-independent value that is approached by a diffusion current as the rate of the charge transfer process is increased by varying the applied potential.

LIMITING FACTOR: Any factor such as availability of food, predation pressure or availability of shelter that affects the rate of a metabolic reaction or any factor that controls a process such as organism growth or species population, size or distribution. In desert ecosystem, low rainfall and high temperature are some of the factors limiting colonisation. Photosynthesis is affected by many factors such as light, temperature and carbon dioxide concentration. But on a sunny day, carbon dioxide concentration is the limiting factor since light and temperature will be at optimum.

LIMITING FRICTION (STARTING FRICTION): The frictional force that resists the initiation of sliding motion of one body over the other with which it is in contact or frictional force that just balances a moving force applied to a solid body resting on a solid surface when the body fails to move. If the moving force exceeds the limiting friction, the body will begin to move.

LIMITING KINETIC CURRENT: The potential-independent value that is approached by a kinetic current as the rate of the charge transfer process is increased by varying the applied potential.

LIMITING MAGNITUDE: The faintest detectable magnitude that can be observed or detected by an instrument or telescope. Limiting magnitude depends not only on the instrument, but also on the observing technique and, crucially, upon the seeing conditions.

LIMITING MIGRATION CURRENT: The limiting value of a migration current, which is approached as the rate of the charge-transfer process is increased by varying the applied potential.

LIMITING REACTANT (LIMITING REAGENT): The reactant in a chemical reaction that determines the amount of product that is produced in a chemical reaction. The reaction will stop when all of the limiting reactant is consumed. For example, given 1 mol of hydrogen and 1 mol of oxygen in the reaction: $2\text{H}_2 + \text{O}_2 \rightarrow 2\text{H}_2\text{O}$, the limiting reactant would be hydrogen because the reaction uses up hydrogen twice as fast as oxygen.

LIMITS OF INTEGRATION: The end points over which a definite integral is to be evaluated. It is given by

$\int_a^b f(x) dx = [F(x)]_a^b = F(b) - F(a)$. The lower limit is a and b the upper.

LIMNO LAYER: In organic soil, a layer at least 5 cm thick composed of marl, diatomaceous earth, or coprogenous earth (sedimentary peat).

LIMNOLOGY: The study of lakes and open fresh water bodies including their biological, physical and chemical aspects. It was established by Alphonse Forel (1841-1912).

LIMONITE (BOG IRON ORE): An ore of iron consisting of a mixture of hydrated iron(III) oxide-hydroxides in varying composition, with the generic chemical formula $\text{FeO}(\text{OH}) \cdot n\text{H}_2\text{O}$. Amorphous goethite and haematite are the most important constituents, together with colloidal silica, clays and manganese oxides. Limonite is formed by direct precipitation from marine or fresh water in shallow seas, lagoons and bogs and by oxidation of iron-rich minerals. It is used as an ore of iron and as a pigment.

LIMPING TRIANGLE: A right-angled triangle whose two shorter sides (sides other than the hypotenuse) differ in length by one unit. An example is the 20-21-29 triangle ($20^2 + 21^2 = 29^2$).

LINCH PIN (LINCHPIN; LYNCH PIN; LYNCHPIN): A pin used to lock an implement onto the three point linkages at the rear of a tractor.

LINCOLN RED: A breed of cattle from the short horn which is deep red in colour. It was originally dual-purpose but used now mainly to cross dairy cattle to produce beef calves.

LINDANE (GAMMA-HEXACHLOROCYCLOHEXANE, (γ-HCH); GAMMAXENE; GAMMALLIN): A white, poisonous, crystalline organochlorine pesticide, with the chemical formula $\text{C}_6\text{H}_6\text{Cl}_6$ that biodegrades very slowly. It is a persistent organic pollutant and has been banned for all agricultural use in Europe.

LINDE PROCESS: A cyclic process for liquefying gas (air), in which compressed gas is cooled by Joule-Thomson expansion through a valve to a pressure of about 40 atmospheres (4 megapascals). It further cools the incoming gas in a heat exchanger and is compressed for the next cycle. It is normally used in the extraction of liquid oxygen and nitrogen.

LINDEMANN THEOREM: In mathematics, it is the result that if $\alpha_1, \dots, \alpha_n$ are algebraic and distinct, while $\beta_1, \dots, \beta_n$ are algebraic and not all zero, then $\beta_1 \exp(\alpha_1) + \dots + \beta_n \exp(\alpha_n) \neq 0$. It is named after the German mathematician, Carl Louis Ferdinand von Lindemann (1852-1939).

LINDEMANN-HINSELWOOD MECHANISM (LINDEMANN MECHANISM): A schematic reaction mechanism, which breaks down a stepwise reaction into two or more elementary steps, then gives a rate constant for each elementary step. It was put forward by the British physicist Frederick Lindemann (1886-1957) in 1921 and examined in more detail by the British chemist Sir Cyril Hinshelwood (1897-1967) in 1927. In the process, a molecule of A becomes excited by colliding with another molecule of A and that having been excited there is a possibility that it undergoes unimolecular decay. If the process of unimolecular decay is sufficiently slow, the reaction has a first-order rate law, in agreement with experiment. If the concentration of A is reduced, the reaction kinetics become second order. This change from first to second order agrees with the experiment qualitatively, does not do so quantitatively. The mechanism fails quantitatively because the molecule has to be excited in a specific way for a reaction to take place. The Rice-Ramsperger-Kassel (RRK) and Rice-Ramsperger-Kassel-Marcus (RRKM) theories improve on this deficiency of the Lindemann-Hinshelwood mechanism linear combination of atomic orbitals.

LINDELÖF SPACE: In mathematics, a topological space in which every open cover has a countable subcover. Lindelöf's theorem states that every second countable space is a Lindelöf. It is named after Ernst Leonard Lindelöf (1870-1946), Finnish mathematician.

LINE: 1. A long thin mark made by a pen, pencil, etc. 2. In Cartesian geometry, a line is a straight one-dimensional geometrical figure without curves, has no thickness and extends in both directions infinitely. If it has an end it is called a line segment, of which the segment between any given points is the shortest path between them. In two dimensions, the line joining (x_1, y_1) and (x_2, y_2) has the equation

$$y = mx + c, \text{ where the slope or gradient } m = \frac{y_2 - y_1}{x_2 - x_1} \text{ or } \left(\frac{y_2 - y_1}{x_2 - x_1} \right) \text{ and}$$

c is the intercept. 3. An undefined element of an axiomatic geometry (along with point). A line is often viewed as a distinguished set of points, satisfying given incidence relations. 4. A one-dimensional subspace of a vector space or an algebraic geometry; the affine hull of two distinct points. 5. A circuit between a customer terminal and the control office. 6. The portion of a transmission system including the transmission media and associated repeaters between terminal locations.

LINE AT INFINITY: In mathematics, the set of improper points added to the Euclidean plane to form the real augmented Euclidean geometry for the plane; the set of ideal points at which parallel lines are held to meet.

LINE BLANKETING: The decrease in intensity of a star's spectrum due to many closely spaced, unresolved absorption lines. Line blanketing is particularly noticeable in the case of cool stars, the atmospheres of which contain many different types of atoms and molecules that tend to absorb at shorter (bluer) wavelengths and reemit in the red and infrared.

LINE BREEDING: The deliberate mating of closely related individuals in order to retain desired characteristics of a common ancestor.

LINE BROADENING: A widening of the absorption and emission lines in a spectrum due to any of several factors. These include Doppler broadening, caused by movements within the emitting gas; pressure broadening or the Stark effect, due to collisions between atoms and molecules; and the Zeeman effect, due to a strong magnetic field.

LINE DEFECT: An imperfection in the crystal structure of a metal or other solid resulting from an absence of an atom or atoms in one or more layers of a crystal.

LINE DISPLACEMENT: The change in a spectral line's wavelength due to a Doppler shift. Approaching sources have spectral lines displaced toward the blue end of the spectrum, receding ones toward the red.

LINE FORMULA: A two-dimensional representation of molecular entities in which atoms are shown joined by lines representing single or multiple bonds, without any indication or implication concerning the spatial direction of bonds.

LINE GRAPH: A graph in which a single line is drawn through the points plotted on a graph.

LINE INTEGRAL (CURVILINEAR INTEGRAL): An integral in which the function to be integrated is evaluated along a curve, which may be of the first kind or second kind. The line integral of the second kind is a linear combination of line integrals of the first kind. An ordinary definite integral is defined over a line segment, whereas a line integral may use a more general path, such as a parabola or a circle. Line integrals are used extensively in the theory of functions of a complex variable. They are also used in a vector field. One interpretation of the line integral of a vector field is the amount of work that a force field does on a particle as it moves along a curve. A **linear integral of the first kind** is denoted by $\int_C \phi(x) dx$, of a scalar field, $\phi(x)$, along

a curve C in n -dimensional Euclidean space, with respect to i^{th} position variable; if C can be represented as $x(t)$, in terms of a parameter, t in the interval $[a, b]$, then it is equal to $\int_a^b \phi(x(t)) x_i(t) dt$.

LINE NOTATION: A typographical notation system using ASCII characters, most often used for writing the structure of a chemical compound as a string of letters, numbers and symbols. Examples of line notation are Wiswesser line notation (WLN), simplified molecular input line (SMILES), line notation (SLN), ROSDAL and InChI.

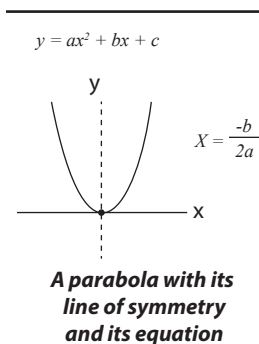
LINE OF BEST FIT: Lines drawn on a line graph which best denotes a trend in the graph.

LINE OF FLOW (STREAMLINE): Paths followed by a particle whose velocity field is the given vector field. The vectors in a vector field are tangent to the flow lines.

LINE OF FORCE: In physics, an imaginary line such as magnetic, gravitational or electrical field that represents the direction of force at any point such that the tangent at any point is the direction of the field vector at that point.

LINE OF SYMMETRY (AXIS OF SYMMETRY; LINE OF REFLECTION; REFLECTION SYMMETRY): A line through a plane

or a 2-dimensional (2-D) shape so that each side of this line is a mirror image of each other. In other words, it is a line that divides a figure into two symmetrical parts in such a way that the figure on one side is the mirror image of the figure on the other side. For example, a quadratic curve and the letter **W** each has one axis of symmetry; the equilateral triangle has three lines of symmetry; and a circle has an infinite lines of symmetry of infinitely many lines of symmetry. The six axes of symmetry of a regular hexagon, for example, are all the bisectors of its angles and sides.



The equation $y = ax^2 + bx + c = a \left[\left(x + \frac{b}{2a} \right)^2 + \frac{4ac - b^2}{4a^2} \right]$ is that of a parabola. Its line of symmetry passes through the vertex with its equation $x = -\frac{b}{2a}$.

LINE PLOT: A statistical graph that shows frequency of data along a horizontal number line. Check marks, Xs, or other symbols above a labelled line represent the frequency of each value.

LINE PRINTER: A fast computer output device which prints a line of characters at a time, typically operating at 200 to 3000 lines per minute. The disadvantages of line printers are that they cannot print graphics, the print quality is low and they are very noisy.

LINE PRINTER TERMINAL (LPT): An identification method for parallel port connection to a printer or other device on a personal computer designated as LPT1, LPT2 or LPT3 used by IBM compatible computers. This is commonly required when installing a printer on an IBM compatible computer. The majority of computers utilises LPT1 and do not have an option for another LPT port unless additional ports are added to the computer.

LINE SEARCH METHOD: In mathematics, one of a number of techniques for determining the minimum or maximum of a function on a line or line segment.

LINE SEGMENT: The part of a line such as the sides of a triangle or square between two given distinct points on that line including the two points.

LINE SPECTRUM: 1. A spectrum made up of a series of distinct lines, occurring as a result of absorption or emission by atoms of specific wavelengths. 2. A spectrum produced by a gas selectively absorbing light emitted by another source or a gas emitting light that consists of a series of distinct parallel lines.

LINE WIDTH: Width of the power spectral density of emitted electric field in terms of frequency, wave number or wavelength. Alternatively, it is the measure of the width of the band of frequencies of radiation emitted or absorbed in an atomic or molecular transition, given by the difference between the upper and lower frequencies at which the intensity of radiation reaches half its maximum value. It is usually defined as full or half width of the line at half of the maximum intensity, FWHM or HWHM, respectively.

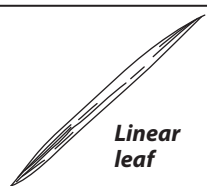
LINE-OF-CENTRES MODEL: A form of the collision theory of chemical reactions in which the assumption is made that reaction can only occur if on collision the component of energy along the line of centres of the masses of the colliding species is greater than a particular threshold energy.

LINE-OF-SIGHT COMMUNICATION (LINE OF SIGHT COMMUNICATION; LINE-OF-SIGHT PROPAGATION): A type of electromagnetic propagation in which there is a direct and no obstruction between the path of the transmitting device and the receiving or reflecting device. Examples are found in microwave, FM radio and satellite transmissions. Repeaters can be used to regenerate the weakened signals and extend the range of communication.

LINE-OF-SIGHT VELOCITY (RADIAL VELOCITY): Velocity at which a celestial body moves towards or withdraws from an observer. It is usually given in relation to the Sun to avoid complication arising from the Earth's orbital motion. It is normally calculated from the Doppler Effect on the body's spectrum. A red shift indicates a receding body which shows a positive velocity. A blue shift indicates an approaching body and a negative velocity.

LINE-SHAPE ANALYSIS: Determination of rate constants for a chemical exchange from the shapes of spectroscopic lines of dynamic processes. The method is most often used in nuclear magnetic resonance spectroscopy.

LINEAR: 1. Relating to a straight line or capable of being represented by a straight line. 2. Variables that change proportionally and can be represented on a graph as a straight line. 3. In the first degree relative to a mathematical variable; having no variable raised to any power consisting only of constant multiples of the variables. For example, a **linear combination** of x , y and z has the form $ax + by + cz$, where a , b and c are numerical coefficients; a linear differential equation is a linear combination of derivatives of order less than some finite n . 4. **Linear leaf** is a leaf hat is long and narrow. 5. An output that varies directly with the input. 6. Relying for its visual effect on clearly defined lines rather than on colour.



LINEAR ABSORPTION COEFFICIENT: The fraction of a beam of radiation absorbed per unit thickness of the absorber.

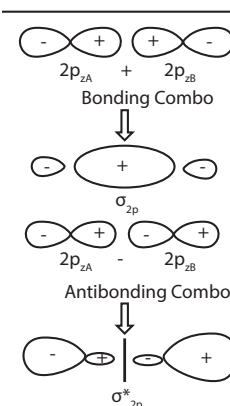
LINEAR ACCELERATOR (LINAC; TRAVELLING-WAVE ACCELERATOR; LINEAR PARTICLE ACCELERATOR): A type of particle accelerator that imparts a series of relatively small increases in energy to subatomic particles such as protons and electrons, as they pass through a sequence of alternating electric fields set up in a linear structure. The small accelerations add together to give the particles a greater energy than could be achieved by the voltage used in one section alone. It consists of electrodes and gaps arranged in a straight line. It is proportioned so that when electrical potentials are varied with the proper amplitude and frequency, particles passing through the waveguide receive successive increments of energy and are therefore accelerated. It is used in radiotherapy, in physics

research and in the production of radionuclides. In radiotherapy, lines of accelerated x-ray radiation are shaped to target a tumour or surgery site, limiting radiation exposure to healthy tissue while killing cancer cells. A linear accelerator, for example, is used during whole breast radiation, accelerated partial breast radiation and Intensity-Modulated Radiation Therapy (IMRT).

LINEAR ALGEBRA: The study of vectors and vector spaces. Linear algebra began as the study of vectors in two- and three-dimensions but has now been extended to a generalised n -space. It has a major role in modern mathematics and used widely in both abstract algebra and functional analysis and finds a concrete representation in the form of analytical geometry.

LINEAR BUS TOPOLOGY: A type of physical topology with a single cable to which all network nodes are directly connected. The cable has terminators at each end to prevent the loss of signal.

LINEAR COMBINATION: 1. A sum of the respective products of the elements of some set with constant coefficients. A linear combination of two vectors p and q is a vector of the form $pa + qb$, where a and b are scalars. 2. The **linear combination of atomic orbitals (LCAO)** is a molecular orbital formed by the linear combination of atomic orbitals. The LCAO approximation arises because with an electron that is very close to a nucleus, the potential energy of the electron is dominated by the interaction between the electron and the nucleus. The LCAO approximation shows the increase in electron density associated with chemical bonding. Basically, combination of a number of atomic orbitals forms a number of molecular orbital (conservation of orbitals).



Linear Combination of Atomic Orbitals (LCAO)

LINEAR CONGRUENCE: A congruence of the form $ax \equiv b \pmod{m}$ or $f(x) \equiv 0 \pmod{m}$, where f is a Linear function of several variables.

LINEAR CONVERGENCE: Convergence of a sequence subject to the condition that $\limsup_{n \rightarrow \infty} \frac{|x_{n+1} - x|}{|x_n - x|} = L$ is strictly between 0 and 1

so that when the limit exists the error is effectively reduced by some fixed proportion. If the convergence is worse or better than linear it is said to be sub-linear or super-linear respectively.

LINEAR DIFFERENTIAL EQUATION: Differential equations having differential equation solutions which can be added together to form other solutions. The dependent variables and all derivatives appearing in the equation are raised to the first power. There are two types: an ordinary differential equation and partial differential equation. A first order linear ordinary differential equation has the form $y' + a(r)y = b(r)$, which is homogeneous if $b(r) = 0$.

LINEAR ECCENTRICITY (ECCENTRICITY): Often denoted e , it is a property that identifies a conic section (parabola, hyperbola or ellipse), which is equal to the distance from the curve to a fixed point (focus) divided by the distance to the fixed line (directrix) or the ratio of the distance of any point on a conic section from a focus to its distance from the corresponding directrix.

LINEAR ELECTRON ACCELERATOR: An evacuated metal tube in which electrons pass through a series of small gaps, usually in the form of cavity resonators in the high frequency range. It is so arranged and spaced that, at a specific excitation frequency, the stream of electrons on passing through successive gaps gains additional energy from the electric field in each gap.

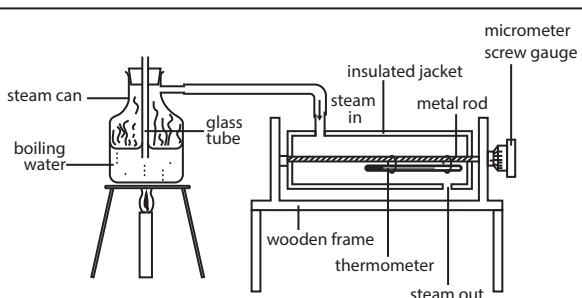
LINEAR ENERGY TRANSFER, LET (LINEAR STOPPING POWER; STOPPING POWER): The energy transferred per unit path length by a moving high-energy charged particle such as an electron or

proton to the atoms and molecules along its path. It is a measure of the ability of matter to reduce the kinetic energy of a particle passing through it, which is the energy loss of a particle per unit distance or the energy lost by a charged particle passing through a substance per unit length of path. It is proportional to the square of the charge on the particle and increases as the velocity of the particle decreases. It is of particular importance when the particles pass through living tissues as it modifies the effect of a specific dose of radiation.

LINEAR EQUATION: An algebraic equation or relationship between two variables that gives a straight line when plotted on a graph or an algebraic equation in which each term is either a constant or the product of a constant and (the first power of) a single variable. It is of the form $y = mx + c$, where m is the gradient of the line and c is the intercept of the line on the y axis. There is a linear equation in one variable, for example, $3x - 5y + 4 = 0$.

LINEAR ESTIMATE: In statistics, an estimate that is a linear combination of observations.

LINEAR EXPANSIVITY (LINEAR EXPANSITY): The fractional change in length of a substance per degree rise in temperature. Most materials expand when heated as long as there is no phase change. The change in length is proportional to its original length, L_0 and the change in temperature. Linear thermal expansion of an object c can be expressed as $\Delta L = L_0 \alpha (t_1 - t_0)$, where ΔL is the change in length in metres, L_0 the initial length in metres, α the linear expansion coefficient in metre per metre/ $^{\circ}\text{K}$ or $^{\circ}\text{C}$ ($\text{m}/\text{m}^{\circ}\text{K}$ or $^{\circ}\text{C}$), t_0 the initial temperature and t_1 the final temperature. Note that the amount of expansion depends upon the material; each material has its own unique coefficient of linear expansion. For example, the coefficient of linear expansion for steel is 13×10^{-6} ($\text{m}/\text{m}^{\circ}\text{C}$). To find the linear expansivity of a material using the apparatus shown in the figure, we must measure the expansion of a known length for a known temperature rise. The original length of a rod of the material is measured with a metre rule. It is placed in the insulated jacket with one end against the stop. The micrometer screw gauge is adjusted until it touches the other end. The reading is taken, as is the temperature of the rod. The micrometer is unscrewed and steam passed through the jacket for several minutes. It is screwed up again and the new reading taken. The procedure is repeated to ensure that there is no change in temperature, showing that the rod is at the steam temperature. The difference in the micrometer readings equals the expansion of the rod. The linear expansivity can then be calculated.



Measuring linear expansivity of a metal

LINEAR EXPRESSION: A mathematical expression such as $4a + b + 3c - 7$, which has no variable with exponents above 1.

LINEAR FACTOR: A factor that includes only the first power of an unknown. For example, in the expression $f(x) = (x-2)(x^2 + 5x + 9)$, the factor $(x-2)$ is a linear factor, but the factor $x^2 + 5x + 9$ is a quadratic factor.

LINEAR FREE-ENERGY RELATION (LINEAR GIBBS ENERGY RELATIONS): A linear correlation between the logarithm of a rate constant or equilibrium constant for one series of reactions and the logarithm of the rate constant or equilibrium constant for a related series of reactions. Typical examples of such relations are the Brønsted relation and the Hammett equation.

LINEAR FUNCTION: A polynomial of degree one that has the form $f: x \rightarrow ax + b$, where a and b are constants. For example, $f: x \rightarrow 2x + 3$ is a linear function. Its graph is a straight line.

LINEAR GRAPH: A graph of a linear function, where all plotted points lie on a straight line.

LINEAR MACROMOLECULE: A macromolecule, whose structure essentially comprises the multiple repetitions in linear sequence of units derived, actually or conceptually, from molecules of low relative molecular mass.

LINEAR MAPPING: A mapping, θ , between vector spaces such that $\theta(x+y) = \theta(x) + \theta(y)$ and $\theta(\lambda x) = \lambda\theta(x)$, where λ is a field element. A linear mapping is a homomorphism of vector spaces and if the domain and range are finite-dimensional and may be represented by a matrix.

LINEAR MODEL: A model in which the expected value of a random variable is expressed as a linear function of the parameters in the model.

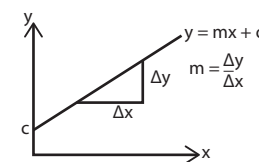
LINEAR MOLECULES: Molecules in which the atoms are arranged so that the bond angle between each is 180° . Linear molecules have only two rotational degrees of freedom.

LINEAR MOMENTUM (MOMENTUM): The product of the mass and velocity of an object ($p = mv$). Its SI unit is $\text{kg}\cdot\text{m}/\text{s}$. Newton's second law of motion states that the rate of change of momentum is proportional to the force acting on the particle. Like velocity, momentum is a vector quantity, possessing a direction as its velocity component, as well as a magnitude. Momentum is a conserved quantity (law of conservation of linear momentum), meaning that if a closed system is not affected by external forces, its total momentum cannot change.

LINEAR MOTOR (LINEAR INDUCTION MOTOR): A form of induction motor in which the stator and armature are linear and parallel, rather than cylindrical and coaxial and produces a linear force along its length instead of a torque (rotation). It was first built by Charles Wheatstone (1845). It consists of a series of electromagnets, with current being passed through each along the line in turn, so that a moving component (usually of aluminium) is drawn along the line. The two major categories are low-acceleration and high-acceleration linear motors. Low-acceleration linear motors are suitable for maglev trains (Magnetic levitation) and other ground-based transportation applications. High-acceleration linear motors are normally quite short and are designed to accelerate an object up to a very high speed and then release it, like roller coasters. In some experimental trains the magnetic force between the primary winding in the vehicle and the secondary winding on the ground support the vehicle on an air cushion thus eliminating track friction. However, because of the high cost of the installation and the low efficiency, the device has not yet found commercial application. They are usually used for studies of hypervelocity collisions, as weapons or as mass drivers for spacecraft propulsion.

LINEAR OPERATOR: An operator between vector spaces that preserves addition and scalar multiplication, often written $L(X,Y)$, where X and Y are the vector spaces. Finite-dimensional linear operators are identifiable with matrices.

LINEAR POLARISATION: Polarisation of an electromagnetic wave on a single plane, either vertical or horizontal. In vertical polarisation, the electric field vector oscillates from top to bottom as the wave travels and its electric field vector of electromagnetic wave is perpendicular to the earth; caused by having antennas vertical to the earth. Horizontally polarised signals oscillate from left to right and their electromagnetic wave frequencies are parallel to and touch the earth. This is generated by having antennas vertical to the earth. The position of electric and magnetic field vectors are very important in radio wave communication. The direction of electric



A diagram showing a line with its intercepts (c) and gradient (m) on the cartesian plane

Linear Graph

field vector specifies polarisation of the antenna. Linear polariser is an optical device that allows the transmission of radiation of which the electric vector is restricted to one plane, resulting in linearly polarised radiation.

LINEAR PROGRAMMING: A mathematical programming using linear inequalities for determining maximum and minimum values. In linear programming the mathematical model under study is expressed by using linear relations. A linear programming problem appears in the form of a set of linear equations (or inequalities), called **constraints** and a linear function that states the objective, which is called the **objective function**.

LINEAR PULSE AMPLIFIER: A pulse amplifier which, within the limits of its normal operating characteristics, delivers an output pulse of amplitude proportional to that of the input pulse.

LINEAR RANGE: Concentration range over which the intensity of the signal obtained is directly proportional to the concentration of the species producing the signal.

LINEAR REGRESSION: A term usually used for simple linear model involving a response variable y , that is a continuous variable, x , related by the equation $E(y) = \alpha + \beta x$, where E denotes the expected value.

LINEAR SCALE: In mathematics, a scale whose divisions are proportional to the quantities they represent, for example, a ruler.

LINEAR SEQUENCE (ARITHMETIC PROGRESSION): A number pattern which increases or decreases by the same amount known as the **common difference** each time. It is generated by writing down the terms in a definite order, for example, increasing or decreasing by the same amount, such as 4, 8, 12, 16 with an increase of 4. The terms are connected by the formula $U_n = a + (n - 1)d$, where a is the first term, d is the common difference and n is the number of terms.

LINEAR SOLVATION ENERGY RELATIONSHIPS: Equations involving the application of solvent parameters in linear or multiple (linear) regression expressing the solvent effect on the rate or equilibrium constant of a reaction. Its symbol is e .

LINEAR SPAN: In mathematics, the smallest sub-space containing a given set in a vector space; the set of all linear combinations of the vectors in the given set.

LINEAR SYSTEM: Any system whose change of values of its variables can be represented as a series of points suggesting a straight line on a coordinate. More generally, a linear system is one in which small changes result in small effects and large changes in large effects. In a linear system, the components are isolated and non-interactive. Real linear systems are rare in nature since living organisms and their components are not isolated and do interact.

LINEAR THERMAL EXPANSION COEFFICIENT (COEFFICIENT OF EXPANSION; EXPANSIVITY): The amount of expansion (or contraction) per unit length of a material resulting from one degree change in temperature. Its symbol is α and its S.I unit is K^{-1} or $^{\circ}C^{-1}$. Mathematically,

it is expressed as $\alpha = \frac{\text{Increase in length}}{\text{Original length} \times \text{temperature change}}$ or $\alpha = \frac{l_2 - l_1}{l_1(\theta_2 - \theta_1)}$, where l_1 and l_2 are initial and final lengths respectively, θ_2 and θ_1 are final and initial temperatures respectively.

respectively, θ_2 and θ_1 are final and initial temperatures respectively.

LINEAR TOPOLOGICAL SPACE (TOPOLOGICAL VECTOR SPACE): A vector space, such as a normed space that constitutes a topological space over a topological field in which addition and scalar multiplication are continuous functions, where the domains of these functions are endowed with product topologies.

LINEAR TRANSFORMATION: A transformation in which linear equations are used to transform a point from one plane into another plane. For example, the linear transformation described by $\begin{pmatrix} x \\ y \end{pmatrix} \rightarrow \begin{pmatrix} ax+b \\ cx+d \end{pmatrix}$ can be used to transform a point $p(x,y)$ in the xy -plane to $p'(x',y')$ in the same xy -plane.

LINEAR TREND: A relationship between two variables in which the value of one changes at a constant rate as the other increases. Linear trend at a constant rate and variables are connected by a relation which is in the form of a straight line.

LINEAR VIDEO EDITING: An obsolete analogue video editing system in which scenes are copied from one video tape to another, using two tape VCRs, in the order required. Its disadvantage is that it is not possible to insert or delete scenes from the new tape without re-copying all the subsequent scenes.

LINEARLY ACCESSIBLE SET (IMPOSSIBLE SET; NIKODYM SET): A subset of the unit square with unit area, such that for every point in the set there is a line that intersects the set only in this point.

LINEARLY DEPENDENT: Of elements, such that there is a linear combination of the given elements with not all coefficients zero that equals zero. For example, a set of vectors is said to be linearly dependent if one of the vectors in the set can be defined as a linear combination of the other vectors.

LINEARLY INDEPENDENT: A set of vectors a , b and c is linearly independent if it is impossible to find three scalars m , n and p (not all zero) such that $ma + nb + pc = 0$. Two vectors clearly are non linearly independent if they are multiples of each other, for example, if $a = (3,5,4)$ and $b = (15,25,20)$, then $5a - b = 0$.

LINEATE: Marked with lines.

LINEIC: Attribute to a physical quantity obtained by division by length. The lineic mass of a thread is the mass of the thread divided by its length.

LINEN: A yarn and fabric manufactured from the fibres of the flax plant. The stems of the flax plant must first be softened by soaking in water (retting).

LINER: 1. The rubber inner tube of a teat cup in a milking machine. 2. A passenger ship or airplane operated by a shipping or airline company. 3. A protective sleeve for a component.

LINES OF FORCE: Imaginary lines in fields of force that enable the direction and strength of the force to be visualised. The tangents at any point give the direction of the field at that point and whose number through unit area perpendicular to the field represents the intensity of the field.

LINES PER INCH (LPI): A measure of the resolution of images printed in halftones. Because halftone images are printed as a series of dots, the higher the LPI number, the denser the dots, resulting in a finer resolution. Newspapers are typically printed in a resolution of 85 lpi, while magazines may use 133 lpi or higher. Because the naked eye can distinguish halftone dots up to about 120 lpi, the dots in a newspaper print may appear to the eyes more than that of magazines.

LINGUIFORM (LINGULATE): A part which has the shape of a tongue.

LINING: A material or a layer used to cover, insulate or protect the inner or unexposed surface of a tube.

LINK: 1. A unit of length used in surveying, equal to 20.12 cm and one hundredth of a chain. 2. In computer terms, it describes the following: (i) In communications, a circuit or transmission path, including all equipment between a sender and a receiver. (ii) An address that points to a Web page or other file (image, video, PDF, etc.) on a Web server. Links reside on Web pages, in e-mail messages and word processing documents as well as any other document type that supports hypertext and URL addressing. (iii) In data management, a pointer embedded within a record that refers to data or the location of data in another record.

LINK ACCESS PROCEDURE FOR MODEMS (LAPM): An error control protocol that uses cyclic redundancy checking (CRC) and retransmission of corrupted data (ARQ) to ensure data reliability. It is defined in ITU-T recommendations V.42.

LINKAGE (GENE LINKAGE): The tendency for two different genes on the same chromosome to remain together during the separation of homologous chromosomes at meiosis or the association between two or more genes that tends to be inherited together because they are on the same chromosome. Linkage can be broken by crossing over or by a chromosome mutation, when sections of chromosomes are exchanged and new combinations of genes are produced.

LINKAGE DISEQUILIBRIUM: A condition in linkage mapping of genes, in which a certain combination of alleles at two or more marker loci occurs more frequently than expected. This happens when there is a relatively low level of recombination in the chromosomal region containing the loci, so that the alleles tend to be inherited as a set or haplotype. This is very useful in mapping disease related genes because a marker such as a single nucleotide polymorphism (SNP) can be used to tag an entire haplotype block containing a particular set of alleles. This makes it easier to scan the genome for markers that are associated with a disease and is the rationale underlying the International HapMap Project.

LINKAGE ISOMERS: Isomers in which a particular ligand bonds to a metal ion through different donor atoms. Examples are violet-coloured $[(\text{NH}_3)_5\text{Co-SCN}]^{2+}$ and orange-coloured $[(\text{NH}_3)_5\text{Co-NCS}]^{2+}$.

LINKAGE MAP: A plan showing the relative positions of genes along the length of the chromosomes of an organism. It is constructed by making crosses and observing whether certain characteristics tend to be inherited together. The proportion of offspring that shows recombination of alleles concerned thus reflects their spacing and is used as a unit of length in mapping chromosome. The information obtained from such a classical linkage map can be combined with a restriction map, which is a linkage map of sites cleaved by restriction enzymes. It provides a huge number of potential marker sites for genes of interest. Linkage maps provide valuable frameworks for constructing detailed physical maps giving the base sequence of the chromosomal DNA.

LINKED GENES: Genes that are located on the same chromosome.

LINKEDIN: 1. A social networking website designed for business professionals. It was founded on December 14, 2002, and launched on May 5, 2003. It allows sharing of work-related information with other users and keeping an online list of professional contacts. It is the most popular social media site for professional networking. By August 2018, it had 106,000,000 monthly users. 2. An object such as a graphic, video clip and spreadsheet that is inserted into a document but still exists in the source file. When information is linked, the files remain separate, but any updated information in one file is automatically updated in all the files that contain the same information.

LINNAEUS, CAROLUS (LINNE VON, CARL) — 1707-1778: Swedish physician and botanist, often called the "Father of Taxonomy", who set up the binomial system of naming species or binomial nomenclature. It consists of the name of the genus to which it belongs, followed by the specific name, which is a descriptive phrase name, for example, *Linum usitatissimum*.

LINOLEIC ACID: A liquid polyunsaturated fatty acid with two double bonds and the chemical formula $\text{C}_{18}\text{H}_{32}\text{O}_2$. It occurs in many grains, seeds, fats and oil such as groundnut and soya bean. Chemically, it is a carboxylic acid with an 18-carbon chain and two *cis* double bonds. The first double bond is located at the sixth carbon from the omega end. It is an essential fatty acid that humans and other animals must ingest for good health because the body requires them for various biological processes, but cannot synthesise them from other food components. It is also used in paints.

LINOLENIC ACID: A liquid polyunsaturated fatty acid, with the chemical formula $\text{C}_{18}\text{H}_{30}\text{O}_2$ with three double bonds in its structure. It is found in certain plant oils such as linseed, soya bean and in algae. It is one of the essential fatty acids required in the diet because it cannot be synthesised in the body.

LINSEED: The seed of the flax plant, *Linum usitatissimum* used for producing linseed oil. The seeds are borne in five double-celled carpels, which are part of the pistil. Linseed oil contains a mixture of glycerides of fatty acids, including linoleic acid and linolenic acid. It is a drying oil, used in oil paints, varnishes and linoleum.

LINT: 1. The mass of soft fibres surrounding the seeds of unginned cotton. 2. A soft absorbent material made from linen or cotton, used in dressing wounds.

LINUX: A freely-distributable open source operating system, developed by Linus Torvalds and others that runs on a number of hardware platforms including the Intel and Alpha, PCs and Macintoshes. Linux has become an extremely popular alternative to proprietary operating systems. Web hosting companies often install Linux on their Web servers because Linux-based servers are cheaper to set up and maintain than Windows-based servers.

LION: A large yellowish-brown carnivorous mammal (*Panthera leo*) of the cat family found mainly in Africa. Male lions weigh between 150 and 250 kg and females weigh between 120 and 182 kg. Lions have massive shoulders and strong forelimbs; long, sharp claws; and short, powerful jaws. The male has a mane, which is a thick growth of hair over its head and shoulders.

LIIOUVILLE FUNCTION: Denoted by $\lambda(n)$, it is the number theoretic function $\lambda(n) = (-1)^{e(n)}$, defined for an integer n where $e(n)$ is the number of prime factors of n , counting repetition. It is named after Joseph Liouville.

LIIOUVILLE NUMBERS: An uncountable set of numbers that are transcendental by virtue of being too well approximated by rationals. It is a number x such that, for each integer n , there is a rational number

$$p/q \text{ (with } q > 1) \text{ with } 0 < \left| x - \frac{p}{q} \right| < \frac{1}{q^n}.$$

The infinite Liouville series $\sum x^{n!}$

produces a Liouville number for all rational x in $]0,1[$. Liouville numbers are transcendental numbers, but not all transcendental numbers are Liouville numbers.

LIIOUVILLE'S THEOREM: 1. The theorem that a bounded entire function is constant. This leads to Morera's theorem and to an analytic proof of the fundamental theorem of algebra. 2. A classical theorem that postulates that the number of states accessible to a system is proportional to the accessible volume of phase space.

LIP: 1. Either of the two edges of the mouth where the skin is delicate and usually redder than the surrounding skin. 2. In botany, also known as **labellum**, it is any of the two variously shaped projections or segments of an irregular, two-lipped corolla or calyx; a labium. Lip cell is the line of cells between which the sporangium dehisces.

LIPASE: Enzyme responsible for breaking down fats into fatty acids and glycerol (glycerine) in the small intestine. It is secreted by the pancreas.

LIPID: A group of naturally occurring molecules including fats, waxes, sterols, fat-soluble vitamins such as vitamins A, D, E and K, monoglycerides, diglycerides, phospholipids and others. Lipid should not be confused with fat. Fat is actually a subgroup of lipids called triglycerides of fatty acids. The biological functions of lipids include energy storage, acting as structural components of cell membranes and participating in important signalling molecules. Lipids are broadly classified into complex lipids and simple lipids. **Complex lipids** are esters of long chain fatty acids and includes the glycerides (fats and oils of plants and animals, glycolipids, phospholipids and waxes). **Simple lipids** contain no fatty acids and contain the steroids and terpenes. Phospholipids and sterols are major components of cell membrane; waxes provide vital waterproofing for body surfaces. Terpenes include vitamin A, E and K and phytol, occurring in essential

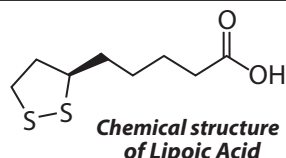
oils such as menthol and camphor. Steroids include the adrenal hormones, sex hormones and bile acids. Lipids can combine with proteins to form lipoproteins.

LIPID BILAYER: 1. The arrangement of lipid molecules in plasma membrane, which takes the form of a double sheet. The cell membrane of almost all living organisms and many viruses are made of a lipid bilayer, as are the membranes surrounding the cell nucleus and other sub-cellular structures. Each molecule comprises a hydrophilic head and a hydrophobic tail. In the lipid bilayer, the molecules are aligned so that their hydrophilic heads face outwards forming the outer and inner surfaces of the membrane, while the hydrophobic tails face inwards away from the external aqueous environment. 2. Model for the structure of the cell membrane based on the hydrophobic interaction between phospholipids. Bilayers are particularly impermeable to ions, which allow cells to regulate salt concentrations and pH by pumping ions across their membranes using proteins called ion pumps.

LIPOCHROME: A pigment that is soluble in fat to impart colour to fatty materials. An example is carotene.

LIPOFUSCIN: Any of a class of fatty pigments formed by the solution of a pigment in fat. They take the form of golden granular deposits derived from lipid components of membranous organelles and commonly occur with advancing age or vitamin E deficiency.

LIPOIC ACID [(R)-5-(1,2-DITHIOLAN-3-YL)PENTANOIC ACID; ALPHA LIPOIC ACID; THIOCTIC ACID; 6,8-DITHIOOCTANOIC ACID]: An organo sulfur compound derived from octanoic acid with the molecular formula $C_8H_{14}O_2S_2$. It is part of the vitamin B complex and plays a role in carbohydrate metabolism. It is one of the coenzymes involved in the decarboxylation of pyruvate by the enzyme pyruvate dehydrogenase. This reaction has to take place before carbohydrate can enter the Krebs cycle during aerobic respiration. Good sources of lipoic acid include liver and yeast.



LIPOLYSIS: The breakdown or hydrolysis of storage lipids in living organisms. Metabolically it is the breakdown of triglycerides into free fatty acids within cells. This process takes place when the body needs energy during starvation from the stored lipids in the form of triglycerides. During the process the enzyme lipase converts triglycerides into glycerol and the component fatty acids. These are transported to tissues and oxidised to provide energy. The hormones that induce lipolysis are epinephrine, norepinephrine, glucagon, growth hormone and cortisol. When fats are broken down for energy the process is known as **beta oxidation**. Ketones are produced and are found in large quantities in ketosis.

LIPOPHILICITY: The ability of a chemical compound to dissolve in fats, oils, lipids and non-polar solvents such as hexane and toluene. The term **lipophilic** describes or refers to such chemical compounds.

LIPOPHOBICITY: In chemistry, it is the inability of chemical substances to dissolve in fats and oils, lipids, but soluble in water and other hydrophilic substances such as methanol and ethanol. **Lipophobic** describes or refers to such substances.

LIPOLYSACCHARIDE, LPS (LIPOGLUCANS): A unique glycolipid found in Gram-negative bacteria or a complex of lipid and polysaccharide that forms the outer layer of some bacteria, protecting the membrane from certain kinds of chemical attack. It acts as endotoxins eliciting strong immune responses in animals.

LIPOPROTEIN: A group of compounds occurring in blood and lymph consisting of lipid combined with a protein. It is the form in which lipids are transported in blood and lymph. They are the main structural materials of the membrane of cell organelles. They may be soluble such as those in egg yolk and blood plasma or insoluble like

those in cell membranes in water and water solutions. Cholesterol is transported in the blood stream mainly in the form of low-density lipoproteins (LDLs) and is removed by means of LDL receptors in cell membranes. LDL-bound cholesterol is primarily responsible for deposits in arteries causing arteriosclerosis, which can lead to coronary heart disease, angina pectoris, myocardial infarction or stroke. Plasma levels of LDL can be reduced by dieting and exercise and or by drugs. Very low-density lipoproteins (VLDLs) are formed in the liver and are precursors of LDLs. High-density lipoproteins (HDLs) are the smallest of all lipoproteins. They transport cholesterol from the tissues to the liver for breakdown and excretion. HDL does not form such deposits and may actually retard or reduce their buildup.

LIPOSOME: A microscopic spherical membrane-enclosed vesicle or sac about 20 -30 nm in diameter, whose walls are a double layer of phospholipids, used to carry relatively toxic substances such as drugs, vaccines and enzymes to specific cells or organs of the body, where they can exert maximum effects. For example, when liposomes containing the drug methotrexate, used in the treatment of cancer are injected into the patient's blood, the cancerous organ is heated to a temperature higher than the body temperature, so that when the liposome passes through its blood vessels, the membrane melts and the drug is released. It is made artificially in the laboratory by the addition of an aqueous solution to phospholipid gel. Liposomes can also be used as vectors in the transfection of cells with foreign DNA or RNA. The study of the behaviour of liposome membranes is used in research into membrane function, particularly to observe membranes during anaesthesia with respect to permeability changes.

LIPOTROPIN: Either of the two peptide hormones, adrenocorticotropin (ACTH) and β -lipotropin (β -LPH) produced in the anterior pituitary gland that triggers the mobilisation of fat deposit and the transfer of lipid components to the bloodstream. Lipotropin is formed by cleavage of the precursor pro-opiomelanocortin and is itself cleaved to form lipotropin and other active peptides, including endorphins and melanocyte stimulating hormone.

LIPSCHITZ FUNCTION: Named after Rudolf Otto Sigismund Lipschitz (1832-1903), a German mathematician, it is a function between normed spaces with the property that the distance between function values is bounded by a constant multiple of the distance between the arguments. If the function satisfies the Lipschitz condition that $\|f(x) - f(y)\| \leq k\|x - y\|$ for all x and y in a set A or at a point x_0 , then f is k -Lipschitz on A or at x_0 .

LIQUATION: A technique for separation of mixtures of solids by heating to a temperature at which lower melting components liquefy.

LIQUEFACTION: 1. The conversion of a soft deposit such as clay to a jelly-like state by severe shaking. 2. The process of converting a gas to a liquid, normally associated with low temperature and high pressure, usually achieved by one of four methods or by a combination of two of them: (i) by vapour compression, provided that the substance is below its critical temperature; (ii) by refrigeration at constant pressure, typically by cooling it with a colder fluid in a countercurrent heat exchanger; (iii) by making it perform work adiabatically against the atmosphere in a reversible cycle; and (iv) by the joule-Thomson effect. Large quantities of liquefied gases are now used commercially, especially Liquefied Petroleum Gas (LPG) and Liquefied Natural Gas (LNG). Liquefied petroleum gases, principally propane and butane, are stored as a liquid under pressure and used as an engine fuel and has the advantage of causing very little cylinder head deposits. 3. Also called **spontaneous liquefaction**, it is the sudden large decrease of the shearing resistance of a cohesion-less soil. It is caused by a collapse of the structure by shock or other strain and is associated with a sudden, temporary increase of the interstitial water pressure. It involves a temporary transformation of the material into a fluid mass.

LIQUEFIED NATURAL GAS (LNG): Natural gas predominantly methane, CH_4 that has been converted temporarily to liquid form for ease of storage or transport. It is similar to LPG but cannot be liquefied simply by pressure because it needs a low critical temperature before it will liquefy. It is odourless, colourless, non-toxic and non-corrosive; and takes up about 1/600th the volume of natural gas in the gaseous state.

LIQUEFIED PETROLEUM GAS (LPG): Various petroleum gases, comprising mainly of propane and butane, produced commercially from petroleum and stored under pressure to keep it in a liquid state. It is used as an engine fuel and for domestic uses such as for cooking and heating. It has the advantage of causing very little cylinder-head deposits. It burns with little air pollution and little solid residue, does not dilute lubricants and it has a high octane rating.

LIQUID: A phase of matter intermediate between that of a crystalline solid and a gas. Liquids are distinguished from crystalline solids in their capacity to flow under the action of extremely small shear stresses and to conform to the shape of a confining vessel. It may alternatively be defined as a substance in a condition in which it flows. Its shape changes, taking that of the container but its volume remains the same. Liquids differ from gases in possessing a free surface and in lacking the capacity to expand without limit. Liquids have bundles of ordered atoms, molecules or ions, which move about in relation to each other, enabling liquids to have almost fixed volumes, which adopt the shape of their containers. Liquids possess important transport properties, such as their capacity to transmit heat (thermal conductivity), to transfer momentum under shear stresses (viscosity) and to attain a state of homogeneous composition when mixed with other miscible liquids (diffusion).

LIQUID AIR: A pale blue, intensely cold liquid, obtained by the compression and cooling of air. Fractional distillation is used to separate oxygen, nitrogen and noble gases from liquid air compression. After filtering the air, a portion of the air stream is decompressed by a centrifugal expander. As the air expands, it loses kinetic energy, which lowers its temperature. The air expands and cools until it liquefies at a very low temperature, called cryogenic temperature, about -190°C (83 K). It is usually kept in a vacuum insulated flask and used as a source of oxygen, nitrogen and inert gases and as a refrigerant.

LIQUID CHROMATOGRAPHY (LC): A separation technique in which the mobile phase is a liquid. The stationary phase can be either a liquid or solid. Liquid chromatography can be carried out either in a column or on a plane. Present-day liquid chromatography generally utilising very small particles and a relatively high inlet pressure is often characterised by the term high-performance (or high-pressure) liquid chromatography and the acronym HPLC.

LIQUID CRYSTAL (LCs): A state of matter that have properties between those of a conventional liquid and those of a solid crystal or a substance that flows like a liquid but whose molecules can be arranged in a particular way. There are different types of liquid crystals. **Nematic crystals** have long molecules all aligned in the same direction, but otherwise randomly arranged. **Cholesteric** and **smectic liquid crystals** also have aligned molecules, which are arranged in distinct layers. In cholesteric crystals, the axes of the molecules are parallel to the plane of the layers. In smectic crystals they are perpendicular. An example of liquid crystals is electronic displays. They are also found in proteins and cell membranes, in solutions of soap and various related detergents, as well as tobacco mosaic virus.

LIQUID CRYSTAL POLYMER (LCPs): A polymer with a liquid-crystal structure, which is extremely unreactive and inert and highly resistant to fire. Liquid-crystal polymers contain long rigid chains and combine strength with lightness. They are, however, difficult to produce commercially. Liquid crystallinity in polymers may occur either by dissolving a polymer in a solvent (lyotropic liquid-crystal polymers) or by heating a polymer above its glass or melting transition point (thermotropic liquid-crystal polymers).

LIQUID EXCIMER LASER: A liquid laser in which excimer species in liquids are used as the active medium.

LIQUID FERTILISER: A simple solution of the normal raw materials of solid fertilisers, as opposed to pressurised solutions such as aqueous ammonia.

LIQUID HYDROGEN (LH_2): Hydrogen in its liquid state, used as a cryogenic rocket fuel, in which hydrogen gas turns to liquid under standard atmospheric pressure at -262.9°C . Its molecular weight is 2.016, with a density of 0.071 g/ml. When oxidised by liquid oxygen, liquid hydrogen delivers about 40% more thrust per unit mass than other liquid fuels, such as kerosene.

LIQUID ION EXCHANGE: A term used to describe a liquid-liquid extraction process that involves a transfer of ionic species from the extractant to the aqueous phase in exchange for ions from the aqueous phase. The term does not imply anything concerning the nature of the bonding in the extracted complex. The term solvent ion exchange is not recommended.

LIQUID ION LASER: A liquid laser in which rare earth ions in solution are used as the laser active medium. An example is krypton difluoride, (KrF) in liquid krypton (Kr) laser.

LIQUID JUNCTION: Any junction between two electrolyte solutions of different composition. Across such a junction there arises a potential difference, called the liquid junction potential. In the operational pH cell the junction is between the test or pH standard, solution and the filling solution or the bridge solution of the reference electrode.

LIQUID LASER: A laser in which a liquid solution is the active species.

LIQUID MANURE: Manure consisting of dung and urine in a liquid form.

LIQUID MEMBRANE: Liquid phase existing either in supported or unsupported form that serves as a membrane barrier between two phases.

LIQUID OXYGEN (O_2 ; LOX): Oxygen in its liquid state, used as the oxidiser in many liquid-propellant rocket engines. Oxygen gas turns to liquid under standard atmospheric pressure at -183°C . Its molecular weight is 32, with a density of 1.141 g/ml.

LIQUID PROPELLANT: A propellant used in a liquid-propellant rocket engine. The three main categories of liquid propellants may be petroleum-based, hypergolic or cryogenic. Additionally, liquid propellants may be classed as bipropellants (in which a liquid fuel and a liquid oxidiser are stored separately) or monopropellants.

LIQUID-AIR CYCLE ENGINE: An advanced engine cycle that uses liquid hydrogen fuel to condense air entering an inlet. Liquid oxygen is separated from the condensed air and pumped into the combustion chamber, where it is burned with the hydrogen to produce thrust.

LIQUID-CRYSTAL DENDRIMER: A highly branched oligomer or polymer of dendritic structure containing mesogenic groups that can display mesophase behaviour.

LIQUID-CRYSTAL DISPLAY (LCD): A thin, flat electronic visual display that uses the light modulating properties of liquid crystals (LCs). The image on an LCD screen is created by sandwiching an electrically reactive substance between two electrodes. This colour of this substance can be changed by increasing or reducing the electrical current. Since LCD screens are based on the principle of blocking light (rather than emitting it), they use up much less power than standard CRT (Cathode-Ray Tube) monitors. LCDs are used in laptop computer screens, video players, gaming devices, clocks, watches, calculators and telephones.

LIQUID-CRYSTALLINE PHASE: A phase occurring over a definite temperature range within the liquid-crystalline state.

LIQUID-CRYSTALLINE POLYMER: Polymeric material that, under suitable conditions of temperature, pressure and concentration, exists as a liquid-crystalline mesophase. A liquid-crystalline polymer can exhibit one or more liquid state(s) with one- or two-dimensional,

long-range orientational order over certain ranges of temperatures either in the melt (thermotropic liquid-crystalline polymer) or in solution (lyotropic liquid-crystalline polymer).

LIQUID-DROP MODEL (DROP MODEL): A model of the atomic nuclei in which the nuclei are regarded as being analogous to the molecules in a liquid, the interactions between which maintain the droplet shape by surface tension. The model is used to study binding energies, fission, collective motion, decay and reactions.

LIQUID-FUELED ROCKET: A rocket that derives its thrust from one or more liquid-propellant rocket engines.

LIQUID-LIQUID DISTRIBUTION/PARTITION/EXTRACTION: The process of transferring a dissolved substance from one liquid phase to another (immiscible or partially miscible) liquid phase in contact with it. The term **liquid-liquid extraction** may be used in place of liquid-liquid distribution when the emphasis is on the analyte(s) being distributed or extracted.

LIQUID-PHASE LOADING: A term used in partition chromatography to express the relative amount of the liquid stationary phase in the column packing. It is equal to the mass fraction (%) of liquid stationary phase in the total packing (liquid stationary phase plus support).

LIQUID-CRYSTAL PROJECTOR (LCD): A thin, flat electronic visual display that uses the light modulating properties of liquid crystals (LCs). The image on an LCD screen is created by sandwiching an electrically reactive substance between two electrodes. This colour of this substance can be changed by increasing or reducing the electrical current. Since LCD screens are based on the principle of blocking light (rather than emitting it), they use up much less power than standard CRT (Cathode-Ray Tube) monitors. LCDs are used in laptop computer screens, video players, gaming devices, clocks, watches, calculators and telephones.

LIQUID-PROPELLANT ROCKET ENGINE: A rocket propulsion system that uses a liquid propellant, combining a liquid fuel such as kerosene or liquid hydrogen in a thrust chamber with a liquid oxidiser such as liquid oxygen or fuming nitric acid. Very efficient and controllable, such engines are used extensively in spaceflight. Unlike solid-propellant rocket motors, they can be shut off by remote command, simply by closing off their fuel line. In some cases, the thrust can also be varied over a certain range.

LIQUIDISE (LIQUIDIZE): To crush, especially fruit or vegetable into a liquid-like form. A device used for crushing fruit or vegetables into a liquid-like form is called a **liquidiser**.

LIQUIDUS: A line on a phase diagram above which a substance is liquid or the maximum temperature at which crystals can co-exist with the melt in thermodynamic equilibrium. Above the liquidus temperature the material is homogeneous and below the liquidus temperature more and more crystals are formed. Below the liquidus temperature homogeneous glasses can be obtained through sufficiently fast cooling (through kinetic inhibition of the crystallisation process). The liquidus temperature is important in the glass industry because crystallisation can cause severe problems during the glass melting and forming processes and it also may lead to product failure.

LIQUOR: 1. A concentrated alcoholic liquid such as whisky made by distillation rather than by fermentation. 2. Also called **pot liquor**, it is a rich broth resulting from the prolonged cooking of meat or vegetables, especially greens. 3. An aqueous solution of a nonvolatile substance. 4. A solution, emulsion or suspension for industrial use.

LIQUORICE: Perennial herb (*Glycyrrhiza glabra*) of the pea family and the flavouring, confection and medicine made from its roots. It is 42 times sweeter than table sugar. It is native to southern Europe and cultivated around the Mediterranean and in parts of the U.S. It grows to 1 metre in height and bears graceful compound leaves, blue-violet flower clusters and flat, flexible seedpods 7-10 cm long.

LISP (LIST PROCESSING): In computing, it is a high-level programming language designed for manipulating lists of data items and usually used for artificial intelligence (AI) programming.

LISSAJOUS FIGURE (BOWDITCH CURVES): Closed plane curves

formed by the combination of two or more simple periodic sinusoidal motion at right angles to each other or a curve in one plane traced by a point moving under the influence of two independent harmonic motions. In the common case the harmonic motions are simple, perpendicular to each other and have a simple frequency ratio. They can be displayed by applying sinusoidal alternating potentials to the X- and Y-inputs of a cathode-ray oscilloscope. The simplest Lissajous figures are circles or ellipses, but they can also take the form of lemniscates and other, more complex shapes.

LISTER, JOSEPH (1827 – 1912): An English surgeon, who developed antiseptic surgery. He showed that the inflammation or sepsis of wounds during surgery was due to bacteria which gained entrance either from the air or from contamination by dirt or even from the surgeon's instruments or his skin. He, therefore, introduced the practice of washing all his instruments in carbolic acid (phenol) and dressing the wound with material soaked in the same substance, thus killing the bacteria. The air of the operating theatre was likewise disinfected by keeping up a continuous spray of carbolic acid during an operation. He transformed surgery from a hazardous affair to a safe predictable procedure, initiating the trend toward modern aseptic techniques in medicine.

LISTERIA: A genus of rod-shaped aerobic motile Gram-positive bacteria containing seven species, named after the English pioneer of sterile surgery, Joseph Lister. Only one species, *Listeria monocytogenes*, causes listeriosis disease. It is resistant to physical and chemical treatment and occurs as contaminant in certain foods, in faeces, etc. It can take various forms, depending on the site of infection: localisation in the central nervous system causes meningoencephalitis, while urine infection can result in abortion or congenital handicap in the foetus.

LISTERIOSIS: A foodborne bacteria disease, caused by the bacterium *Listeria monocytogenes*, found in soil, water and animal faeces. The disease may sometimes affect humans through eating undercooked animal products such as meat and unpasteurised milk products and through eating contaminated raw vegetables. The disease can cause miscarriage, stillbirth and premature delivery in pregnant women. It can spread to the nervous system and may be very serious in people with weakened immune systems such as babies and elderly people. It is named after English surgeon Joseph Lister (1827-1912), the founder of antiseptic surgery.

LISTSERV (LIST SERVER): A small program that automatically sends messages to multiple e-mail addresses on a mailing list. When a user subscribes to a mailing list, the listserv automatically adds the address and distributes future e-mail messages to that address along with all the others on the list. When a user unsubscribes, the listserv removes the address.

LITERAL: 1. In mathematics, using letters to denote constants, coefficients, etc., in an expression or equation, as contrasted with a numerical one. For example, in $ax^2 + b$, x is a variable and a and b are literal constants. 2. In logic, an expression consisting of a single atomic symbol or its negation, especially a constant as contrasted with a variable. A positive literal is just an atom and a negative literal is the negation of an atom.

LITHIC: 1. Relating to or consisting of stone. 2. Relating to clastic rocks, either sedimentary or volcanic, containing a large proportion of debris from previously formed rocks. 3. Relating to stony concretions, or calculi, formed within the body, especially in the bladder.

LITHIC LAYER: Bedrock under the control section of a soil. In organic soils, it is the bedrock occurring within a depth of between 10 cm and 160 cm from the surface.

LITHIFICATION: The process such as compaction or cementation by which sedimentary materials are converted into solid stone or rock.

LITHIUM: An alkali metal, which is a soft silvery white metal that belongs to the alkali metal group of chemical elements, with the chemical symbol Li. It is found in spodumene ($\text{LiAlSi}_2\text{O}_6$), petalite ($\text{LiAlSi}_4\text{O}_{10}$), the mica lepidolite and certain brines. It is the lightest metal known. It reacts with oxygen and water; on heating it also

reacts with nitrogen and hydrogen. Its chemistry differs somewhat from that of the other Group 1 elements because of the small size of the Li^+ ion. It is usually extracted by treatment with sulfuric acid to give the sulfate, which is converted to the chloride. This is mixed with a small amount of potassium chloride, melted and electrolysed. The metal's stable isotopes are lithium -6 and lithium -7. Lithium -5 and lithium -8 are short-lived radioisotopes. The metal is used to remove oxygen in metallurgy and as a constituent of some aluminium and magnesium alloys. It is also used in batteries and is a potential source for fusion research. Lithium compounds are used for medical treatment of bipolar disorders.

LITHIUM ALUMINIUM HYDRIDE [LITHIUM TETRAHYDROALUMINATE (III)]: A powerful reducing agent, with the chemical formula LiAlH_4 , density 0.917 g/cm^3 (solid), melting point 150°C (423 K) and decomposes. It is used in organic synthesis, especially for the reduction of esters, carboxylic acids and amides.

LITHIUM BATTERY: A type of voltaic cell containing lithium or lithium compounds. The most commonly used has a metallic lithium anode and a manganese dioxide (MnO_2) cathode. The electrolyte is a solution of lithium salts in an organic solvent. Batteries of this type have an output of about 3 volts. They are more expensive than alkaline batteries but last longer. They are also produced in a flat disk form for use in digital watches and other small portable devices.

LITHIUM CARBONATE: A white solid composed of lithium, carbon and oxygen with the chemical formula Li_2CO_3 . It is found in nature as the rare mineral zabuyelite. It is less stable and decomposes above 1310°C (1583 K). It is produced commercially by treating the ore with sulfuric acid at 2500°C (2773 K) and leaching the product to give a solution of lithium sulfate. The carbonate is then obtained by precipitation with sodium carbonate solution. Lithium carbonate is used very widely including in the prevention and treatment of manic depressive disorders, and industrially in ceramic glazes. It forms low-melting fluxes with silica and other materials, used in ovenware, cement, in tile adhesives, in the manufacture of most lithium-ion battery cathodes, etc. It is also used to lower the melting point of alumina in the extraction of aluminium by electrolysis. Its hydride complex $\text{Li}[\text{AlH}_4]$ is employed in reducing organic compound such as aldehydes to alcohol.

LITHIUM HYDRIDE: A white solid with the chemical formula LiH , which decomposes at about 8590°C (8863 K). It is produced by direct combination of the elements at temperature above 500°C (773 K). The bonding in lithium hydride is believed to be largely ionic, i.e. $\text{Li}^+ \text{H}^-$ (as supported by the fact that hydrogen is released from the anode on electrolysis of the molten salt). The compound reacts violently and exothermically with water to yield hydrogen and lithium hydroxide. It is used as a reducing agent to yield hydrogen and lithium hydroxide and to prepare other hydrides. It is the source of hydrogen used for military activities and filling weather balloon. The 2H isotopic compound, lithium deuteride, is particularly valuable for deuterating a range of organic compounds. Lithium hydride has also been used as a shielding material for thermal neutrons. Its hydride complex ($\text{Li}[\text{AlH}_4]$) is employed in reducing organic compounds such as aldehyde to alcohol.

LITHIUM HYDROGEN CARBONATE: A compound with the chemical formula LiHCO_3 , formed by the reaction of carbon dioxide with aqueous lithium carbonate and known only in solution. It has found medicinal uses similar to those of lithium carbonate and is sometimes included in proprietary mineral waters.

LITHIUM HYDROXIDE (LITHINE): A white crystalline solid, with the chemical formula LiOH , density 1.46 g/cm^3 (anhydrous), 1.51 g/cm^3 (monohydrate) and melting point 462°C (735 K). It decomposes at a boiling point of 924°C (1197 K). It is soluble in water, slightly soluble in ethanol and insoluble in ether. It is available commercially in anhydrous (tetragonal) form or as the monohydrate (monoclinic). The compound is made by reacting lime with lithium salts or lithium ores. Lithium hydroxide is basic but has a closer resemblance to group 2 hydroxides than to the other group 2 hydroxides (an example of the

first member of a periodic group having atypical properties). Lithium hydroxide is used to remove carbon dioxide for purification of gases and air, as a heat transfer medium, as a storage-battery electrolyte, as a catalyst for polymerisation, in ceramics, manufacturing other lithium compounds and esterification especially for lithium stearate.

LITHIUM OXIDE (LITHIA; LITHIUM MONOXIDE): A white crystalline compound with the chemical formula Li_2O , which can be obtained from a number of lithium ores. Its main uses are in lubricating greases, ceramics, glass and refractories and as a flux in brazing and welding.

LITHIUM SULFATE: A white or colourless crystalline material, with the chemical formula Li_2SO_4 , density 2.221 g/cm^3 (anhydrous), 2.06 g/cm^3 (monohydrate), melting point 859°C (1132 K) and boiling point 1377°C (1650 K). It is soluble in water and insoluble in ethanol. It forms a monohydrate (monoclinic) and an anhydrous form, which exists in Cl^- (cubic) forms. The compound is prepared by the reaction of the hydroxide or isomorphous with other group 1 sulfates and does not form alums. It is used to treat bipolar disorder and also used in ultrasound-type non-destructive testing because they are very efficient sound generators.

LITHIUM TETRAHYDROALUMINATE(III) (LITHIUM ALUMINIUM HYDRIDE; LAH): A white or light grey powder with the chemical formula LiAlH_4 , density 0.917 g/cm^3 (solid), melting point 150°C (423 K) and decomposes. It is prepared by the reaction of excess lithium hydride with aluminium chloride. The compound is soluble in ethoxyethane (diethylether) and reacts violently with water to release hydrogen. It is widely used as a powerful reducing agent in organic chemistry. It should always be treated as a serious fire risk in storage. It is used to reduce carbonyl compounds to alcohol.

LITHIUM-ION BATTERY (LI-ION BATTERY, LIB): A type of rechargeable cell in which the anode is carbon and the cathode is a metal oxide such as cobalt(IV) oxide or CoO_2 that provides more energy for a longer time than a standard battery. The electrolyte is a lithium salt such as borate (LiBO_4) or chlorate (LiClO_4) in an organic solvent. Lithium ions move from the negative electrode to the positive electrode during discharge and back when charging. They are one of the most popular types of rechargeable batteries for portable electronics, with one of the best energy densities, no memory effect and a slow loss of charge when not in use. They are used in consumer electronics and for military, electric vehicle and aerospace applications.

LITHOCYST: A cell that contains a cystolith. Cystolith is a stalk-like crystal of calcium carbonate formed inside an epidermal cell by crystal growth on to an ingrowth from the cell wall.

LITHOMETEOR: A particle of dry substance in the atmosphere, as contrasted to a hydrometeor.

LITHOPHYTE: 1. A plant that grows on rock outcrops or on rocky or stony ground. An example is the fern, *Asplenium ruta-muraria* (wall-rue). 2. An organism partly composed of siliceous or calcareous material, for example, the stoneworts (Charophyta).

LITHOSEQUENCE: A group of related soils that differ from each other by the differences in the parent rock.

LITHOSPHERE: The Earth's solid crust and upper region of the mantle that behaves elastically on time scales of thousands of years or greater. It is fragmented into tectonic plates, which move over the semi fluid underlying asthenosphere. It is 8-70 km thick and covers the inner molten parts. It is underlined by the **asthenosphere**, the weaker, hotter and deeper part of the upper mantle.

LITHOTRIPTICS: Agents or substances containing properties that cause dissolution of stones in the kidneys and lungs. They include herbs such as barberry, chaparral, cornsilk, couchgrass and wild carrot.

LITMUS: A water-soluble organic dye extracted from certain lichens, which are mashed, treated with potassium carbonate and ammonia and allowed to ferment. It turns red under acidic conditions and blue under alkaline conditions. The colour change occurring over the pH range 4.5-8.3 (at 25°C) is not suitable for titrations because of the wide range over which the colour changes. It is used as a rough indicator of acidity or alkalinity, both in solution and as litmus paper (absorbent paper soaked in litmus solution). The active component of litmus (the part sensitive to acids or bases) is called **erythrolitmin**.

LITOTORAL DRIFT (LONG SHORE DRIFT, LSD; LONGSHORE TRANSPORT): The movement of sediments along a coast at an angle to the shoreline, which is dependent on prevailing wind direction, swash and backwash, occurring in or within close proximity to the surf zone.

LITRE: A unit of volume in the metric system regarded as a special name for the cubic decimetre. Its symbol is *l*. 1 litre is equal to 1 000 cm³.

LITTER: 1. A group of young animals such as kitten or piglets born at the same time to one mother. 2. A pile of straw used as an animal's bed. 3. Dead organic matter in the soil that has not yet decomposed. It consists of fallen leaves and other plant remains, animal excrement, etc. After decomposition by decomposers, litter becomes humus.

LITTLE ICE AGE: The period from the 14th to the 19th century, marked by expansion of mountain glaciers and cooling of global temperatures, especially in the Alps, Scandinavia, Iceland and Alaska. This occurred between warmer periods.

LITTORAL: 1. A term used to describe organisms that live in the littoral zone. 2. A term used to describe the area of a lake or pond that extends from the edge to the lower limit of rooted aquatic plants. There may be three distinct zones of plants: emergent plants at the water's edge; plants with floating leaves in deeper water; and finally completely submerged plants. Certain mosses, notably Fontinalis, can grow at lower light intensities than higher plants and may form meadows on the lake bed. 3. A term used to describe the seashore between low and high tide marks that is exposed alternately to air and water. It may be muddy, sandy or rocky. The size of the zone varies with the slope of the coast and the height of the tides. Typical plants are brown algae of the order Fucales.

LITUUS: In mathematics, it is a trumpet-shaped spiral that is the locus of points in which the angle is inversely proportional to the square of the radius with a polar equation of $r^2 = a/\theta$. It is asymptotic to the x -axis and winds round the origin without ever reaching it.

LIUSTEMIK'S THEOREM: In mathematics, the infinite-dimensional inverse function theorem that establishes that if a mapping, T , between Banach spaces that is continuously differentiable in the sense of the Frechet differential, has a surjective derivative at a point a , then there is a neighbourhood U of $T(a)$ and a constant $K > 0$ such that, for each element $y \in U$, there is a solution to $y = T(x)$ with $\|x - a\| \leq K\|y - T(a)\|$.

LIVE: In mathematics, of a measure, to be concentrated on (some set). In other words, if there is a set A such that for all measurable E , $\mu(E) = \mu(A \cap E)$, then μ is said to live on A .

LIVE TIME: For a measurement, the time during which a radiation measuring assembly is capable of processing events occurring in the radiation detector. It equals the clock time minus the integrated resolving or dead time (to be distinguished from 'life time').

LIVE WIRE: In electricity, wire that maintains a high potential difference compared with the neutral wire which is at or near the earth potential. Electrons in the live wire carry electrical energy from the power station to an electrical appliance. In a single phase system, the insulation of a live wire is coloured brown or red and this is the wire that is usually fused. In a three phase system, the insulation of the live wires are coloured red, yellow and blue respectively.

LIVER: A large visceral organ found in all vertebrates that occupies much of the upper part of the abdomen. Structurally, the liver is made up of lobules, each of which is a roughly hexagonal structure consisting largely of hepatocytes arranged around a central vein. It has many regulatory and storage functions, such as the removal of simple sugars from the general circulation after ingestion of food and storing them as glycogen; converting noncarbohydrate metabolites such as amino acids and fats into glucose to maintain blood sugar levels; detoxification of poisonous by-products; release and destruction of heat; and converting ammonia to urea or uric acid in a process known as **deamination**, which is excreted via the kidneys. The liver is the sole source of such necessary constituents of the blood as fibrinogen, serum albumin and cholinesterase. In the embryonic

stage of most vertebrates the liver serves as the major manufacturing site of erythrocytes, a process known as **erythropoiesis**. The liver receives the products of digestion dissolved in the blood via the hepatic portal vein.

LIVER FLUKE: A parasitic worm that infests the liver of mammals, including humans, feeding on blood. It belongs to the phylum Platyhelminthes and its scientific name is *Fasciola hepatica*. Adult flukes produce eggs which are passed into the intestine.

LIVERWORT: A non-vascular plant of the division Bryophyta that is related to mosses and found growing in damp places. They do not have flowers, stems, proper roots or leaves and look like fleshy leaves growing flat on the ground.

LIVERY: A stable which keeps a horse for the owner and usually feeds, grooms and exercise the horse.

LIVESTOCK: Animals reared for food or animals kept on a farm such as cattle, chicken or sheep.

LIVING FOSSIL: Any living organism very closely resembling fossil relatives in most anatomical details and whose closest relatives are extinct. Characteristics, which truly define a living fossil are: truly close anatomical similarity to an ancient fossil species (generally at least 100,000,000 years old); and living members of the group are represented by only a single or at best a few species, which are often found in a very limited geographic area. Examples are horseshoe crabs, ginkgo trees and coelacanth fish, a primitive fish that was common in the Devonian era, the first specimen of which was discovered in 1938.

LIVING POLYMERISATION: A chain polymerisation from which chain transfer and chain termination are absent. In many cases, the rate of chain initiation is fast compared with the rate of chain propagation, so that the number of kinetic-chain carriers is essentially constant throughout the polymerisation.

LIXIVIATION: The process of separation of mixtures by dissolving soluble constituents in water. For example, to extract silver from ores of other metals by lixiviation method, the silver is dissolved in a solution of a salt, usually sodium cyanide, after which metallic silver is precipitated by bringing the solution in contact with metallic zinc or aluminium.

LIZARD: Any of about 4,450 species of reptiles constituting the suborder Larcetilia (Sauria) of the reptilian order Squamata. They range in size from 3 cm for geckos to 3 m for Komodo dragons. Lizards have scales, paired male copulatory organs and a flexible skull. They have crests on the head, back or tail; spines; brightly coloured throat fans; and throat frills. Most species feed on insects and rodents, but some, such as the iguana, eat plants.

LLOYD'S MIRROR: An optical arrangement for producing interference fringes. A slit is illuminated by monochromatic light and placed close to a plane mirror. Interference occurs between direct light from the slit and light reflected from the mirror. It was first used by Humphrey Lloyd (1800-81) of the Royal Irish Academy of Science in 1834.

LOAD: 1. The object that a force works against or which a machine does work on, expressed as follows:

Load = Effort $\times$ Mechanical Advantage (M. A.),

i.e., $\frac{\text{Load}}{\text{Effort}} \Rightarrow \text{Load} = \text{Effort} \times \text{M.A.}$. 2. To put a disk or tape in a drive

on a computer or to transfer data or a program to the main memory of a computer. 3. The amount of electrical power that is drawn from a line or source or any device to which electrical power is delivered. 4. The total force or weight that a structure such as a bridge is designed to withstand. 5. The work required of or applied to an engine or machine measured in kilowatts. 6. An Assembly Language/Instruction

Mnemonic Opcode used to copy or move data unchanged from one storage location to another. The source and destination locations may be CPU registers, main memory locations or input-output ports.

LOAD FACTOR: The factor by which the steady-state loads on a structure are multiplied to obtain the equivalent static effect of dynamic loads.

LOADED CONCRETE: Concrete containing elements such as iron or lead with a high mass number. It is used to make radiation shield around nuclear reactors.

LOADER: 1. A machine used to load crops, manure, etc into trailer or spreaders. 2. A program which copies other (object) programs from auxiliary (external) memory to main (internal) memory prior to its execution. In other words, a loader program places a machine language program into main memory for execution. On a Unix System, the operating system kernel loads a program into memory and starts it running.

LOAFING PLATFORM: A nest like structure built by few birds that can float on water. These birds keep their young ones in them to keep them out of water and afloat.

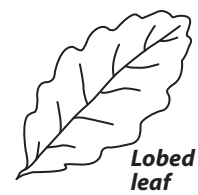
LOAM: A type of fertile soil composed of sand, silt, clay and organic matter in evenly mixed particles of various sizes. It is porous and allows for good aeration and retention of moisture. It is the most fertile soil for proper growth of plants.

LOBACHESKIAN GEOMETRY (HYPERBOLIC GEOMETRY): The branch of non-Euclidean geometry that replaces the parallel postulate of Euclidean geometry with the postulate that two distinct lines may be drawn parallel to a given line through a point not on the given line. In hyperbolic geometry, contrary to the parallel postulate, there exists a line m and a point p not on m such that at least two distinct lines parallel to m pass through p . As a result, the sum of the angles of a triangle is less than 180° and, for a right triangle, the square of the hypotenuse is greater than the sum of the squares of the other two sides.

LOBE: 1. A rounded projection or division of a part in the body or an organ, especially in the liver, brain or lungs or a rounded part that projects from the main body, for example, the ear. 2. A rounded segment on a leaf that is not divided all the way to the midrib.

LOBED: A term used to describe a leaf that is divided into curved or rounded parts connected to each other by an undivided central area.

LOBPOD: 1. A member of the phylum Lobopodia. They have segmented bodies and bear legs with hooked claws 2. A blunt-ended pseudopodium found in certain protozoans.



LOBSTER: Lobsters are a family of large marine crustaceans. They have long bodies with muscular tails and live in crevices or burrows on the sea floor. Three of their five pairs of legs have claws, including the first pair which is much larger and meaty than the others. Spiny Caribbean lobsters have no claws but are preferred mainly for their tails.

LOCAL (RELATIVE): In mathematics, a phenomenon is said to be local, if it occurs only within a small neighbourhood of values rather than for all values. For example, a local maximum of a function is a value of the function greater than any adjacent value, but it may not be the greatest value of the function over its whole range.

LOCAL ACTION: Voltaic action at an electrode of an electrolytic cell as a result of the zinc plate having impurities such as carbon and iron, which sets up tiny local cell at the zinc surface. It is prevented by cleaning the zinc plate in sulfuric acid and rubbing small globules of mercury over the surface with a small piece of cotton wool. The mercury dissolves pure zinc out of the plate and forms a bright coating of zinc amalgam all over the surface. The amalgam covers up the impurities and prevents them from coming into direct contact with the acid.

LOCAL AGENDA 21: The local plans for sustainable environmental protection, development, economic prosperity and community well-being that each local authority is meant to develop through a consultative process with their populations, with particular attention paid to women and youth.

LOCAL AREA NETWORK (LAN): A communications network that serves users within a confined geographical area. It is made up of servers, workstations, a network operating system and a communications link.

LOCAL CONFORMATION: The conformation of a macromolecule at the scale of the constitutional units. In the polymer literature, local conformation is sometimes referred to as micro conformation.

LOCAL GRAVITATIONAL CONSTANT: The acceleration of any particle close to sea level at any place on earth and due to the Earth's gravity alone. Its symbol is g that is equal to γMR^{-2} , where γ is the universal gravitational constant, M is the mass of the Earth and R is the radius of the Earth at that point. Its value at the North Pole is 9.8321 ms^{-2} and 9.7801 ms^{-2} at the equator.

LOCAL GROUP: The small cluster of galaxies containing the Milky Way Galaxy to which Earth and its solar system belong. It consists of some 30-40 known members, the most massive of which are the Milky Way and the Andromeda galaxies. It appears as a faint band of light in the night sky.

LOCAL HORIZON: 1. The apparent or visible horizon. 2. The lower boundary of the observed sky or the upper outline of terrestrial objects, including nearby obstructions or irregularities.

LOCAL MAXIMUM (RELATIVE MAXIMUM): A value of a function $f(x)$ that is greater than the adjacent values of x , but is not the greatest of all values of x .

LOCAL MEMORY: A portion of memory in a computer with multiple processors that is assigned to a single unit or to a single function or program.

LOCAL MINIMUM: A value of a function $f(x)$ that is less than the adjacent values of x but is not the lowest for all values of x .

LOCAL OSCILLATOR: An oscillator in a heterodyne or super heterodyne receiver. It supplies the radio-frequency signal that beats with the incoming signal to produce the intermediate frequency.

LOCAL VERTICAL: The direction in which the force of gravity acts at a particular point.

LOCALHOST: It refers to the local computer that a program is running on. For example, if you are running a Web browser on your computer, your computer is considered to be the Localhost, which always translates to the loopback IP address 127.0.0.1.

LOCALISATION (LOCALIZATION): 1. The confinement of electrons to a particular atom in a molecule or to a particular chemical bond. 2. In the theory of disordered solids, the concept that an electron is concentrated around a specific site and cannot contribute to the solid's electrical conductivity (at absolute zero) by moving through the system. In one dimension any amount of disorder makes all electron states localised. In three dimensions a small amount of disorder makes electron states near the top and the bottom of the energy bands localised. The states in the centre of the bands are called **extended states** because they can propagate through the system and hence contribute to electric conductivity. The dividing energies between localised and extended states are called **mobility edges**. Given sufficient disorder all states become localised. In two dimensions all electron states in disordered solids are thought to be localised, with some states being strongly localised around specific sites while other states are weakly localised around specific sites. Localisation also occurs in disordered solids for other excitations, such as phonons and spin waves.

LOCALISED BOND: A covalent bond between two atoms in which the bonding electrons are confined between just the two atoms. The electrons in such a bond are said to be localised.

LOCALISED ELECTRONS: A pair of electrons in a covalent bond located in the vicinity of the nuclei of the two contributing atoms. Such electrons cannot move beyond this area.

LOCALISED MOLECULAR ORBITALS (LMO):

The molecular orbitals located on certain fragments of a molecular system and spatially separated from each other as much as possible. The LMOs are derived from the electron occupied canonical molecular orbitals by subjecting them to a unitary transformation determined by an appropriate physical criterion, for example, by maximising the sum of squares of the centroids of occupied MOs (the Foster–Boys procedure) or by minimising the sum of the exchange (or Coulomb) repulsion integrals between the occupied MOs (the Edmiston–Ruedenberg procedure).

LOCALISED VECTOR: A vector, as a force that has magnitude and direction and acts through a particular point. An example is the weight of a body, which is a force that acts through the object's centre of gravity.

LOCALLY COMPACT: In mathematics, a topological space is locally compact, if it has the property that every point has a neighbourhood that is a subset of a compact set. Examples include Euclidean space, the unit interval $[0,1]$, Cantor set and Hilbert cube.

LOCALLY CONNECTED: In mathematics, a topological space is locally connected, if it has the property that for every point p and every neighbourhood U of p there is a connected open set V that is a subset of U such that $p \in V$. A space is locally connected if and only if the components of all open subsets are open.

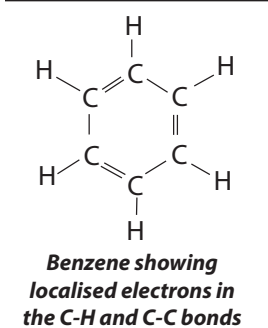
LOCALLY CONVEX: In mathematics, a topological vector space is locally convex, if it has a neighbourhood base consisting of convex sets.

LOCALLY EUCLIDEAN: A topological space is locally Euclidean if each point has a neighbourhood that is homeomorphic to an open set in a Euclidean space of specified dimension, of which the dimension is associated to the space.

LOCALLY FINITE: 1. In mathematics, a collection of subsets of a topological space is locally finite, if each point has a neighbourhood that intersects only finitely many of the sets in the collection. 2. A topological space is locally finite if every point has a finite neighbourhood. 3. A group is locally finite, if it can be studied in ways analogous to a finite group. 4. A measure is locally finite if every point of the measure space has a neighbourhood of finite measure. 5. A linear operator $f: V \rightarrow V$ is locally finite if the space V is the union of a family of finite-dimensional f -invariant subspaces. 6. A locally finite poset is a partially ordered set P such that for all $x, y \in P$, the interval $[x, y]$ consists of finitely many elements.

LOCANT: The number or a letter which indicates the position of attachment of groups on the root or parent carbon chain.

LOCATION PROBLEMS: In mathematics, various generalisations of Fermat's problem, in which one seeks to locate a point in space in such a way as to minimise a weighted sum of distances (in some metric) from a given set of points. For example, these problems arise when trying to place central facilities to serve a number of communities.



LOCK-AND-KEY MECHANISM: A mechanism that explains binding between the active site of an enzyme and a substrate molecule. The active site has a fixed structure, the lock, which exactly matches the structure of a specific substrate, the key. Thus, an enzyme-substrate complex is formed. This substrate is converted into products that no longer fit the active site of the enzyme and are released, liberating the enzyme.

LOCKJAW (TRISMUS): A symptom of tetanus in which spasm of the jaw muscles or muscles of mastication occurs and also inability to open the jaws.

LOCOMOTION: Any of various animal movements that result in progression from one place to another, which is a characteristic of most animals but not plants. This requires propulsive force acting against a supporting structure. Locomotion is classified as appendicular, accomplished by special appendages; or axial, achieved by changing the body shape. Some protists have contractile fibres that exert force on the plasma membrane to change the shape of the cell, such as amoeboid movement and the flagella movement of Paramecium. In animals, the force required to generate locomotion is generated by voluntary muscles which act against a supporting framework provided by a skeleton.

LOCOMOTIVE: Any type of self-propelled vehicle used by railroads to pull or push other types of passenger, freight and work cars. The locomotive differs from other kinds of self-propelled railroad vehicles in that it is used solely as a power unit and is not designed for carrying passengers or freight, as are trolley cars and some electric cars.

LOCOMOTOR DISORDER: A disease of the organs involved in motion such as bones, muscles, joints, synovial membranes, related connective tissues, peripheral blood vessels and nerves. Examples of locomotor disorders include muscular and ligamentous sprains, chronic back pain, different types of arthritis, spondylosis (compression of nerves in the spine by ruptured discs), spondylitis (inflammation of the vertebrae and the attached disks and ligaments), bursitis (inflammation of a fluid-filled sac bursa of the body, particularly at the elbow, knee or shoulder joint), tenosynovitis (inflammation of a tendon sheath, usually in the wrist), fibrositis (pain and stiffness, especially in the back muscles) and myositis (muscle inflammation).

LOCULE (LOCULUS): A small chamber within an organ in a plant or animal body. In plants, the locule of the ovary is the cavity containing the ovules and the locules of the anther contain the ovules or seeds. Carpels and fruits can be classified as uni-locular, bi-locular or multi-locular depending on the number of locules in the ovary. **Locular** means having locules.

LOCULICIDAL: A term used to describe fruit dehiscence in which the slits in the pericarp arise along the dorsal suture of each carpel. On the other hand, **septicidal** means splitting through the septa and between the locules.

LOCUS: 1. The specific location of a gene or DNA sequence on a chromosome where a particular gene occurs. The alleles of a gene occupy the same locus on homologous chromosomes. 2. A set of points whose location is specified by an equation. For example, the locus of a point P which is equidistant from a fixed point O , is a circle with centre O .

LOCUST: 1. Any of several species of grasshoppers (family Acrididae) native to Africa and Asia, which undergo population explosions and migrate long distances in swarms, eating and destroying crops over large areas. Locusts are characterised by antennae shorter than their bodies, song-producing organs on the forewings and hind hind legs well developed for jumping. Locusts lay their eggs in the ground, with wingless nymphs, when hatched. 2. Any of a number of tree species in the genera *Robinia* or *Gleditsia* that include black locust, (*Robinia pseudoacacia*) and honey locust, also known as thorny locust or thorn tree (*Gleditsia triacanthos*), a leguminous tree with pods having a

sweet, edible pulp. Black locust is a tree of the genus *Robinia* of the subfamily Papilionoideae, family Fabaceae (formerly Leguminosae) native to North America. It has been widely planted in North America, Europe, Southern Africa and Asia. It grows up to about 30 metres tall on average at maturity with a trunk diameter of about 1 metre, with thick, deeply furrowed blackish bark. The leaves are pinnate with 9-19 oval leaflets. Black locusts produce pea-like flowers in branched, elongated clusters, as well as long, papery toxic pods that contain hard seeds. The pods are used for making honey. The wood is extremely hard, resistant to rot and durable and used for making furniture, flooring, paneling, fence posts and small watercraft. Honey locust has reddish a strong, coarse grained and moderately durable wood. Its common uses include for making fence post, railroad ties, furniture, interior woodwork and firewood.

LOCUST BEAN: The pod of the carob tree, *Ceratonia siliqua*, used as animal feed. The pod also contains a sweet edible pulp and seeds that yield a gum used as a stabiliser in food products.

LODESTAR (LADSTAR): A relatively bright star, which can be identified easily, used in navigation or astronomy particularly with reference to Polaris (the pole star). Polaris presently indicates the direction to the north within a fraction of a degree. Lodestones can be used to approximately find the northern direction and are particularly convenient when the stars are not visible.

LODESTONE (LOADSTONE): A natural magnetic piece of magnetite, which is a form of iron ore, formerly used as compass.

LODGE, SIR OLIVER JOSEPH (1851-1940): British physicist, who became principal of Birmingham University in 1900. He investigated lightning, the voltaic cell and electrolysis and electromagnetic waves. His best-known work was in radio, where in 1894 he perfected and named the 'coherer', a radio-wave detector used in early radio receivers and invented by the French physicist Edouard Branly. After 1910 he became increasingly interested in spiritualism, reconciling science and religion.

LODICULE: Either of two small scales situated at the base of the ovary of grasses. They may represent the reduced perianth and be involved in the opening of the floret at the time of pollination.

LOESS: Sedimentary deposit composed largely of fine grained silt or clay, thought to have formed as the result of grinding by glaciers and to have been deposited by the wind.

LOG: 1. In mathematics, it is the abbreviation and symbol for logarithm. 2. In computer science, it is a record of computer activities used for statistical purposes as well as backup and recovery. **Log files** are often generated during software installations and are created by Web servers, but they can be used for many other purposes as well. Most log files are saved in a plain text format, which minimises their file size and allows them to be viewed in a basic text editor.

LOG IN (LOGIN): Information such as password, an ID number or security code that authenticates an identity before secure Web sites allow visitors to access certain areas of the site.

LOG ON: To sign in to gain access to a network server, Web server or a computer. When you log on to a system, you provide log in information that authenticates you as a user.

LOG OUT (LOG OFF): To exist a computer system, for example, e-mail account or a website by typing a particular command.

LOG PAPER: A graph paper, with one of the axes having a logarithmic scale. Double log paper has logarithmic scales on both axes.

LOG SUM INEQUALITY: An inequality is used often in information science to prove non-negative of functions, for example, relative entropies. For n positive numbers, $a_1, \dots, a_n$ and $b_1, \dots, b_n$,

$$\sum_{i=1}^n a_i \log \frac{a_i}{b_i} \geq \sum_{i=1}^n a_i \log \frac{\sum_{i=1}^n a_i}{\sum_{i=1}^n b_i}.$$

LOG-LINEAR GRAPH: A graph on which one axis has a logarithmic scale and the other has a linear scale. On a log-linear graph, an exponential function (i.e equation of the form $y = be^{ax}$ where a and b are constants) is a straight line. Values of x are plotted on the linear scale and values of y on logarithmic scale.

LOG-LINEAR MODEL: A mathematical model that takes the form of a function whose logarithm is a first-degree polynomial function of the parameters of the model, which makes it possible to apply (possibly multivariate) linear regression.

LOGARITHM: The power to which a number, called the base, has to be raised to give another number, whose logarithm is being determined. A logarithm to the base a , of a number N , written $\log_a N$, is the power to which the base a must be raised to give the number N (i.e., if x is the logarithm of N to the base a , then $a^x = N$). For example, the logarithm of 1000 to base 10 is 3 ($\log_{10} 1000 = 3$), because 3 is the power to which 10 must be raised to give 1000: $10^3 = 1000$, so $\log_{10} 1000 = 3$. If the base is 10, the logarithm is called **common logarithm**; if the base is e , it is called **natural logarithm**; and if the base is 2, it is called **binary logarithm**. For example, the logarithm of 8 to the base 2 is 3, since $2^3 = 8$. It was discovered by Scottish mathematician John Napier and Swiss mathematician Justus Byrgius. Logarithm is used in statistics, chemistry, physics, astronomy, computer science, economics, music and engineering and helps mathematicians to extract the roots of numbers and simplify many calculations by basing them on addition and subtraction rather than multiplication and division. **Logarithmic** is defined as relating to or containing logarithms; for example, a logarithmic scale. Also, consisting of or using, points or lines whose distances from a fixed point or line are proportional to the logarithms of numbers, such as a logarithmic scale.

LOGARITHMIC CONVEXITY: The convexity of the logarithm of a given function. For example, a function $f(x)$ is logarithmically convex on the interval $[a, b]$ if $f > 0$ and $\ln f(x)$ is convex on $[a, b]$.

LOGARITHMIC DERIVATIVE: In mathematics, the derivative of the logarithmic of a given function, f , defined by the formula $\frac{f'(x)}{f(x)}$,

where f' is the derivative of f . For example, the digamma function is the logarithmic derivative of the gamma function.

LOGARITHMIC DIFFERENTIATION: A method of differentiating algebraically complicated functions, where the usual formulae of differentiation do not apply. An example of such a problem is finding the derivative y' , if $y = x^{\sin x}$. The differentiation is performed after taking the logarithms of both sides.

LOGARITHMIC FUNCTION: 1. The function defined by $\log_x$ or $\ln x$ that is the inverse of an exponential function such as $y = e^x$ or $y = x^t$ as the definite integral from 1 to x of $1/t$ with respect to t . Its value for any argument is its natural logarithm, so that $\ln \exp x = 1 = \exp \ln x$. 2. Any function containing the logarithmic function or the logarithm of a function to any base.

LOGARITHM TABLE (LOG TABLE): A table showing the logarithm of numbers. Before computers and electronic calculators became popular, logarithm tables were used as the only alternative to slide rules in logarithm calculations.

LOGARITHMIC PHASE: The period during the growth of microorganisms in culture when the cells are dividing rapidly and there is a huge increase in cell number. If the logarithm of cell numbers is plotted against time then a straight line is obtained on the graph. This period usually ends quite abruptly when nutrient levels fall, the pH of the medium changes or toxic wastes build up. The logarithmic phase may be maintained by transferring cells to a fresh culture medium or by adding more nutrients to the existing medium.

LOGARITHMIC SCALE: A scale of measurement on the axis of a graph in which an increase of one unit represents a tenfold increase

in the variable quantity or a scale measurement in which an increase or decrease of one unit represents a tenfold increase or decrease in the quantity measured. Examples are equally spaced divisions on the vertical axis of a chart labelled 1, 10, 100, 1000, instead of 1, 2, 3, 4 and decibels and pH measurements. If a curve $y = x^n$ is plotted on graph paper with logarithmic scales on both axes, the result is a straight line of slope n , i.e. $\log y = n \log x$, which enables n to be determined.

LOGARITHMIC SERIES: The expansion of a logarithmic function, such as $\log_e(1+x) = x - \frac{x^2}{2} + \frac{x^3}{3} - \dots \frac{(-1)^n x^n}{n}$ for $-1 < x \leq 1$ or $\log_e(1-x) = -x - \frac{x^2}{2} - \frac{x^3}{3} - \dots - \frac{x^n}{n}$ for $-1 \leq x < 1$.

LOGARITHMIC SPIRAL (EQUIANGULAR SPIRAL; GROWTH SPIRAL; LOGISTIC SPIRAL): A type of spiral that is seen commonly in the natural world. Examples are found in the shells of some molluscs, such as that of the nautilus and the fossil ammonites and also in spider webs. The logarithmic spiral can be constructed from equally spaced rays by starting at a point along one ray and drawing the perpendicular to a neighbouring ray. The logarithmic spiral is an example of a transcendental plane curve.

LOGGING: The job of felling, trimming and transporting timber.

LOGIC: 1. The branch of science concerned with the principles governing correct or reliable inference. 2. The analysis, without regard to meaning or context, of the patterns of reasoning by which conclusions are validly derived from sets of premises. 3. Any particular formal calculus or formal language which define axioms and rules of inference. 4. The system and principles of reasoning used in a particular field of study. 5. Also known as **logic circuit**, a circuit element as in computers, designed to perform complex functions defined in terms of elementary functions of mathematical logic.

LOGIC BOARD (MAINBOARD; MOTHERBOARD): The main circuit board of a computer to which the CPU, ROM, memory RAM expansion slots, PCI slots and USB ports are attached. It also includes controllers for devices like the hard drive, DVD drive, keyboard and mouse. Basically, the motherboard is what makes everything in the computer work together. Each motherboard has a collection of chips and controllers known as the chipset.

LOGIC BOMB: A software program that usually causes harmful effects such as destroying data, at a certain point in time or when certain conditions are met. It can also be used to stop a consumer from using certain software after the trial date has elapsed.

LOGIC CIRCUIT (LOGIC GATE): The basic switching of circuits or gates used in digital computers and other digital electronic devices. Logic circuits can be built from any binary electric or electronic devices, including switches, relays, electron tubes, solid-state diodes and transistors in accordance with the input system. Basic logic circuits include the AND gate, the OR gate and the NOT gate, which perform the logical functions AND or and NOT. The AND circuit gives a binary 1 output if a binary 1 is present on each input circuit; otherwise the output is a binary 0. The OR circuit gives a binary 1 output if a binary 1 is present on at least one input circuit; otherwise the output is binary 0. The NOT circuit inverts the input signal, giving a binary 1 output for a binary 0 input for a 1 input. Often these basic logic circuits are used in combination. For example, a NAND circuit consists of NOT + AND circuits. In terms of electronic equipment, logic circuits are now almost exclusively embodied into integrated circuits.

LOGIC ERROR (LOGICAL ERROR): A bug in a program's source code that results in incorrect or unexpected behaviour. It is a type of runtime error that may simply produce the wrong output or may cause a program to crash while running.

LOGICAL ADDRESS: A reference to a memory location independent of the current assignment of data to memory. A translation must be made to a physical address before the memory access can be achieved.

LOGICAL AXIOMS: The axioms valid in every structure for a given formal language; the axioms for the logical constants of the theory.

LOGICAL CONSEQUENCE: The relation that obtains between the conclusion and the premises of an argument when it is formally valid. In other words, the relation that holds between one statement and a set of statements when it is impossible for the former to be false in any model in which the latter are all true.

LOGICAL CONSTANT: One of the connectives of a given system of formal logic, especially those of the sentential calculus, which are NOT and or and IF ... THEN ... or their formal equivalents negation, conjunction, disjunction and implication.

LOGICAL FORM: The syntactic structure that may be shared by different expressions, as abstracted from their contents and articulated by the logical constants of a particular logical system, especially, the structure of an argument by virtue of which it can be shown to be formally valid. Thus "John is tall and thin, so John is tall" has the same logical form as "London is large and dirty, so London is large", namely $P \& Q$, so P .

LOGICAL PRODUCT (CONJUNCTION; INTERSECTION): In logic, it is the binary truth functional sentential connective that forms a compound sentence from two given sentences and corresponds to the English word "and".

LOGICAL RECORD: A record that is independent of its physical environment. Portions of one logical record may be located in different physical records or several logical records or parts of logical records may be in one physical record.

LOGICAL SUM (ALTERNATION; DISJUNCTION): The proposition that presents two or more alternative terms, with the assertion that only one or at least one is true. It is an OR statement of the form A OR B. It is true if either A OR B is true. Its symbol is "V". The truth set table is given in the diagram.

LOGICAL TOPOLOGY (SIGNAL TOPOLOGY): The way in which data flows within a network from node to node, irrespective of its physical design.

LOGICAL TRUTH: 1. Also known as **tautology**, in logic, a statement that is always true; especially a truth functional expression that takes the value true for all combinations of values of its components, such as "either it is raining or it is not raining". 2. The property of being true under any possible state of affairs.

LOGICISM: The philosophical theory that all of mathematics can be deduced from logic.

LOGISTIC CURVE: An S-shaped curve which depicts the growth of a population in an area of fixed limits. Its equation is $y = \frac{k}{1 + e^{a+bx}}$, where b is less than zero.

LOGISTIC DISTRIBUTION: The limiting probability distribution as n tends to infinity, of the average of the largest to the smallest sample values, of random samples of size n from an exponential distribution.

The distribution is given by $f(x) = \frac{\exp\left[-\frac{(x-\alpha)}{\beta}\right]}{\beta\{1+\exp\left[-\frac{(x-\alpha)}{\beta}\right]\}^2}$, where $-\infty < x$

$< \infty$, $\beta > 0$. The location parameter, α is the mean. The variance of the distribution is $\frac{\pi^2 \beta^2}{3}$.

LOGISTIC GROWTH MODEL: The model appropriate for a growth curve when the rate of growth is proportional to the product of the size at the time and the amount of growth remaining. Specifically the model is defined by the equation, $y = \frac{\alpha}{1 + \gamma e^{-\beta t}}$ where α , β and γ parameters.

LOGISTIC MODEL: In epidemiology, a model of risk as a function of exposure to a risk factor.

LOGISTIC NORMAL DISTRIBUTIONS: A class of distributions that can model dependence more flexibly than Dirichlet distribution.

LOGISTIC POPULATION GROWTH: In biology, a model describing population growth that levels off as population size approaches carrying capacity.

LOGISTIC REGRESSION (LOGISTIC MODEL; LOGIT MODEL):

A form of regression analysis used when the response variable is binary variable. The method is based on logistic transformation or logit of a proportion namely, $\text{logit}(p) = \ln \frac{p}{1-p}$. As p tends to zero, the $\text{logit}(p)$ tends to $-\infty$ and as p tends to one, $\text{logit}(p)$ tends to ∞ . The $\text{logit}(p)$ is sigmoid curve that is symmetric about $p = 0.5$. Applying this transformation, this form of regression is written as;

$\ln\left(\frac{p}{1-p}\right) = \beta_0 + \beta_1 x_1 + \dots + \beta_n x_n$, where $p = p(\text{dependent variable} = 1)$ and $x_1, x_2, \dots, x_n$ are explanatory variables.

LOGISTIC SPIRAL (EQUIANGULAR SPIRAL; GROWTH SPIRAL; LOGARITHMIC SPIRAL): A type of spiral that is seen commonly in the natural world. Examples are found in the shells of some molluscs, such as that of the nautilus and the fossil ammonites and also in spider webs. The logarithmic spiral can be constructed from equally spaced rays by starting at a point along one ray and drawing the perpendicular to a neighbouring ray. The logarithmic spiral is an example of a transcendental plane curve.

LOGISTIC SYSTEM: In mathematics, an uninterpreted formal calculus containing a set of axioms and a set of rules of inference, especially a system of formal logic.

LOGIT CONFIDENCE LIMITS: The upper and lower ends of the confidence interval for the logarithm of the odds ratio, is given by, $\ln \varphi \pm Z_{\frac{\alpha}{2}} \sqrt{\text{Var}(\ln \varphi)}$, where φ is the estimated odds ratio, $Z_{\frac{\alpha}{2}}$ the

normal equivalent deviate corresponding to a value of $\frac{\alpha}{2}$, $(1 - \alpha)$ being the chosen size of the confidence interval. The variance term may be estimated by $\text{var}(\ln \varphi) = \frac{1}{a} + \frac{1}{b} + \frac{1}{c} + \frac{1}{d}$, where a, b, c and d are the frequencies in the two-by-two contingency table from which φ is calculated. Two limits may be exponential to yield a corresponding interval for the odds ratio itself.

LOGNORMAL DISTRIBUTION: The probability distribution of a random variable, X , for which $\ln(x)$ has a normal distribution with mean μ and variance σ^2 . The distribution is given by;

$f(x) = \frac{1}{x\delta(2\pi)^{\frac{1}{2}}} \exp\left[-\frac{1}{2\delta^2}(\ln x - \mu)^2\right]$, $0 \leq x \leq \infty$ where the mean = $\exp\left(\mu + \frac{1}{2}\delta^2\right)$, variance = $\exp(2\mu + \delta^2)(\exp(\delta^2) - 1)$.

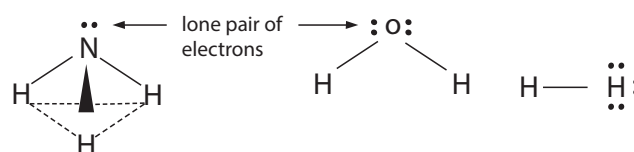
LOMENTUM: A type of dry dehiscent fruit formed from a single carpel but divided into one-seeded compartments by constrictions between the seeds. Examples are sainfoin (*Onobrychis viciifolia*) and Desmodium species of legumes.

LONDON FORCES (DISPERSION FORCES): Attractive forces between polar molecules, due to their mutual polarisability. They are also components of the forces between polar molecules.

LONDON FORMULA: A formula giving an expression for the induced-dipole-induced-dipole interaction between molecules called the dispersion interaction or London interaction. The London formula for the interaction energy V is given by $v = -\frac{C}{r^6}$, where $C = \frac{2}{3}\alpha'_1\alpha'_2I_1I_2/(I_1+I_2)$. Here α'_1 and α'_2 are the polarisability volumes of molecule 1 and 2 respectively, I_1 and I_2 are the ionisation energies of molecules 1 and 2 respectively and r is the distance between the molecules. The London formula is named after Fritz London (1900-54), who derived it. The interaction described by the London formula is usually the dominant term in intermolecular forces unless hydrogen bonds are present.

LONE PAIR (LONE PAIR ELECTRONS): A pair of unshared valence electrons in a molecule that is not involved in bonding in that molecule. Electron pairs are therefore, considered lone pairs if two electrons

are paired but are not used in chemical bonding. For example, in ammonia the nitrogen atom has five electrons, three of which are used in forming single bonds with hydrogen atoms. The other two occupy a filled atomic orbital and constitute a lone pair. The orbital containing these electrons is equivalent to an orientation, accounting for the pyramidal shape of the molecule. In the water molecule, there are two lone pairs on the oxygen atom. A single lone pair can be found with atoms in the nitrogen group such as nitrogen in ammonia, two lone pairs can be found with atoms in the oxygen group such as oxygen in water and the halogens can carry three lone pairs such as chlorine in hydrogen chloride.



The trigonal pyramid shape of ammonia (NH₃) showing lone pair of electrons

LONG BONE: A bone that is significantly longer than it is wide. Examples of long bones are the femur, tibia and fibula of the leg, the humerus, radius and ulna of the arm and the phalanges of the fingers and toes. A long bone consists of a long shaft (diaphysis) with two bulky ends or extremities (epiphyses).

LONG DIVISION: 1. A method of division in arithmetic, usually used when the divisor is a large number that does not use the slash (/) or obelus (÷) signs, but uses the symbol, $\overline{)}$. The process is written down with the dividend under the symbol, the divisor on the left side outside the symbol and the quotient at the top of the symbol. For

example, to divide 67598 by 16, it is written down as $\overline{16)}67598$. The process begins by dividing the first digit of the dividend, in this case, 6 by the divisor, 16. Since 6 cannot be divided by 16, the next digit, 7 is paired with 6, making it 67. Now, 16 can go into 67 four times, leaving a remainder of 3. The 4 is written on top of the symbol and the remainder is paired with the next digit, 5, making it 35. Again, 16 can go into 35 two times leaving a remainder of 3. The process is repeated until the final answer is arrived at, sometimes leaving a remainder, which becomes part of the quotient, sometimes expressed as a fraction of the divisor. 2. Polynomial long division is a method for dividing polynomials, which is essentially the same as long division for numbers.

LONG INTERSPERSED ELEMENT (LINE): 1. Any of a class of dispersed moderately repetitive DNA found in eukaryotes, consisting of numerous copies of relatively long (6-8 kb) sequences scattered throughout the genome. Lines are retro transposons and can spread by reverse transcription: an RNA transcript is formed, then a DNA copy of this, subsequently undergoes insertion into the genome. An example found in humans and other mammals is the L1 element. This is 6kb long, representing up to 21% of the total human DNA, has 50 000 copies in the human genome and contains two large coding regions plus non-coding flanking sequences. One of these encodes both an endonuclease and a reverse transcriptase. 2. A series of adjacent points that extends to infinity in two directions.

LONG RADIUS: In mathematics, the line or the distance, between the centre and a vertex of a regular polygon.

LONG SHORE DRIFT (LSD; LITTORAL DRIFT; LONGSHORE TRANSPORT): The movement of sediments along a coast at an angle to the shoreline, which is dependent on prevailing wind direction, swash and backwash, occurring in or within close proximity to the surf zone.

LONG TERMINAL REPEAT (LTR): Sequences or a length of double stranded DNA, typically 250-600 bp, which is repeated at the end of integrated retroviral DNA and certain retrotransposons. One LTR directs the host cell to initiate transcription of the retroviral DNA, while the other directs cellular enzymes to trim the primary RNA transcript.

LONG-DAY PLANT: A plant that can be induced or enhanced to flower by long days, usually of more than 12 hours of daylight or a plant that flowers in response to a long photoperiod longer than a critical length. Examples of such plants are spinach, lettuce, peas and some varieties of wheat.

LONG-GRAIN RICE: A variety of rice with long and slender grain grown in tropical climate. The grains are separate and fluffy after cooking.

LONG-LIVED COLLISION COMPLEX: When a potential-energy surface is such that the system performs a number of vibrations and rotations in the region of a col, the corresponding intermediate is known as a long-lived collision complex and the reaction is said to be indirect.

LONG-RANGE INTRAMOLECULAR INTERACTION: An interaction between segments widely separated in sequence along the chain that occasionally approaches one another during molecular flexing. This type of interaction is closely related to the excluded volume of a segment, the latter quantity reflecting interactions involving segments and solvent molecules.

LONG-SIGHTEDNESS (HYPERMETROPIA; HYPEROPIA; FARSIGHTEDNESS): An abnormal condition of the eye in which vision is better for distant objects than for near objects. It results from the eyeball being too short from front to back, causing images to be focused behind the retina. As an object moves toward the eye, the eye must increase its optical power to keep the image in focus on the retina. If the power of the cornea and lens is insufficient, the image will appear blurred. People with the condition can experience blurred vision, asthenopia, accommodative dysfunction, binocular dysfunction, amblyopia and strabismus.

LONG-TERM ENVIRONMENTAL CHANGE: Environmental change that occurs slowly over time and effects organisms over generations. Some of the factors that cause such changes include deforestation, urbanization, Sun's intensity, global warming and radioactive waste/pollution. Some long-term effects of climate change include a decrease in sea ice and an increase in permafrost thawing; an increase in heat waves and heavy precipitation; decreased water resources in semi-arid regions due to longer periods of drought in some regions; an increase in the number, duration and intensity of tropical storms; as well as extinction of some species of organisms.

LONG-TERM MEMORY: Memory in a calculator used by keys with an M on them, such as M- and M+. Numbers in long-term memory are not affected by calculations with keys without an M, which use short-term memory.

LONGHORN: A dual-purpose hardy breed of cattle of Spanish origin with long-curving horns. It is usually red or brown in colour with white markings.

LONGITUDE: The angular distance measured in degrees, seconds and minutes west or east of the prime meridian (longitude 0° or Greenwich Meridian) that stretches from the South Pole to the North Pole. Longitudinal refers to any of the following: (i) Of, relating to or occurring in the lengthwise dimension, for example, a longitudinal bone fracture. (ii) Extending along or relating to the anteroposterior axis of a body or part. (iii) Involving the repeated observation or examination of a set of subjects over time with respect to one or more study variables such as general health, the state of a disease or mortality. (iv) In botany, it refers to dehiscing along the long axis of the anther.

LONGITUDINAL WAVES (L-WAVES): Waves in which the displacement of the medium is in the same direction as, or the opposite direction to, the direction of propagation of the wave. Mechanical

longitudinal waves are also called compressional or compression waves, because they produce compression and rarefaction when travelling through a medium, and pressure waves, because they produce increases and decreases in pressure. Examples of L- wave are ultrasound waves, sound waves and seismic Primary waves (P-waves).

LONGITUDE AT THE EPOCH: The longitude that a planet has at a given date. It is the angle measured along the ecliptic from the vernal equinox eastward to the ascending node and then eastward along the orbital plane to the planet's position at that instant.

LONGITUDE OF PERIHELION: The angle measured from the vernal equinox eastward along the ecliptic to the ascending node of a planet's orbit and then continued eastward along the orbital plane to the perihelion. It is one of the orbital elements.

LONGSHORE CURRENT: A current moving parallel to the coast, caused by a wave reaching a beach or coastline that releases a burst of energy that generates a current.

LONGSHORE DRIFT: The movement of sand and pebbles along a seacoast. The material is carried along the beach by the waves hitting the coast obliquely (swash) but is swept back at right angles to the beach by the backwash. If the carrying power of the waves decreases the material may be deposited to form a spit.

LOOK-AND-FEEL: In computing, it is the general appearance and operation of a user interface. In other words, it is the design, presentation and output of a user interface as experienced by the user. The "look" refers to the design such as the display, layout and typefaces and the "feel" refers to characteristics such as menus and dialogue boxes that enable a user to interact with it (how user friendly they are).

LOOK-UP TABLE (LOOKUP TABLE): In computing, it is an array of data stored in a computer that is searched using special look-up functions.

LOOM: Any machine for weaving yarn or thread into cloth.

LOOP: 1. In mathematics, it is an edge of a graph that joins a vertex to itself. 2. A non-associative algebra with one binary operation. It is a quasigroup with an identity. 3. A closed electric circuit. 4. A set of instructions in a computer program that does not terminate and is repeated either a specified number of times or until a particular condition is met.

LOOP OF HENLE (HENLE'S LOOP; ANSA NEPHRONI): The hairpin-shaped section of a kidney tubule situated between the proximal and distal convoluted tubules; named after its discoverer by F. G. J. Henle. The loop of Henle extends from the cortex into the medulla. It consists of a thin descending limb, which is permeable to water and a thick ascending limb, which is impermeable to water. Complex movement of ions and water across the walls of the loop enables it to function as a counter current multiplier, resulting in the production of concentrated urine in the collecting duct. Here, water is reabsorbed according to the body's needs with the involvement of the anti-diuretic hormone (ADH).

LOOP VARIABLE: A quantity that is used to characterise field theories as an alternative to characterising them in terms of fields and potentials. A loop variable is like a line of force of flux line that is closed to form a loop. Loop variables have been used with great success in quantum electrodynamics (QED), quantum chromodynamics (QCD) and quantum gravity.

LOOPBACK: In telecommunications, any system that sends a signal to the system's output system, which then sends a signal back to the source. For example, in telephone systems, loopback test signals are sent to the network output devices that are returned as received to the system's source, which may used to diagnose problems.

LOOSE SMUT: A fungus affecting wheat and barley caused by *Ustilago nuda*. Masses of black spores collect on diseased heads and replace the entire head of plants, except for the rachis. The spores are dispersed in the wind and only a bare stalk is left.

LOOSE-BOX: A stable for accommodating animals that are kept untied. It is also useful for housing sick animals.

LOPHOPHORE: An organ that surrounds the mouth for filter feeding. It is characteristic of aquatic invertebrates of the phyla Bryozoa, Phoronida and Brachiopoda. It is often horseshoe-shaped or coiled and consists of a ridge of hollow tentacles bearing cilia, which waft food particles into the mouth.

LOPHOTRICHOUS FLAGELLATION: A term used to describe a bacterium that has a group of flagella at one end of the cell.

LOPHOTROCHOZOA: A clade or major grouping of protostome animals, comprising two groups, the trochozoans and the lophophorata. It includes the molluscs and annelid worms, together with the nemertins, bryozoans and other wormlike phyla.

LORENTZ FACTOR: A relativistic factor that varies from approximately 1 at low velocities to approaching infinity near the speed of light. Its symbol is γ .

LORENTZ FORCE: The force F exerted on a charge Q that is moving with a velocity v in a magnetic field with a magnetic flux density B given by the vector product: $F = q(E + v \times B)$, where F is the force in newtons, E is the electric field in volts per metre, B is the magnetic field in tesla, q is the electric charge of the particle in coulombs and x is the vector cross product in metres per second.

LORENTZ GROUP: In physics and mathematics, the Lorentz group is the group of all Lorentz transformations of Minkowski spacetime, the classical setting for all (non-gravitational) physical phenomena. If the Lorentz group is combined with translations in space and time the proper Poincaré group is formed. If inversions in space and time are combined with the proper Poincaré group the unrestricted Poincaré group results. Analysis of the Lorentz group in relativistic quantum mechanics is used to classify elementary particles.

LORENTZ TRANSFORMATION: A set of equations for correlating space and time coordinates in two frames of reference. It may also be defined as one of the equations relating space and time that holds between two observers of an event who are moving with uniform velocity relative to each other.

LORENTZ-DIRAC EQUATION (DIRAC EQUATION): An equation for describing the motion of a point electric charge, derived by classical electrodynamics. It was formulated by British physicist Paul Dirac in 1928. This equation is in accord with special relativity theory and avoids the self-energy problem. However, there are a number of difficulties, which are usually attributed to the fact that classical electrodynamics is not applicable to the point electron. The

Lorentz-Dirac equation is given by, $ma^\alpha = F_{ext}^\alpha + \frac{2}{3}q^2(\delta_\beta^\alpha + \mu^\alpha_\beta)\alpha^\beta$, where $a^\alpha = \frac{da^\alpha}{d\tau}$, m is the mass of the particle, q is the charge of the particle and F_{ext}^α is the external force.

LORENTZ-FITZGERALD CONTRACTION (FITZGERALD CONTRACTION): The contraction of a moving body in the direction of its motion. The contraction was given a theoretical background in Einstein's special theory of relativity. In this theory, an object of length l_0 at rest in one frame of reference will appear, to an observer in another frame moving with relative velocity v with respect to the first, to have length $l_0\sqrt{1 - v^2/c^2}$, where c is the speed of light. The original hypothesis regarded this contraction as a real one accompanying the absolute motion of the body. The contraction is in any case negligible unless v is of the same order as c .

LORENTZ-LORENZ EQUATION (CLAUSIUS-MOSSOTTI RELATION; MAXWELL'S FORMULA): A relation between the polarisability α of a molecule and the refractive index, n , of a substance made up of molecules with this polarisability. The Lorentz-Lorenz equation can be written in the form $\alpha = (\frac{3}{4}\pi N) [(n^2 - 1)/(n^2 + 2)]$, where N is the number of molecules per unit volume. The equation provides a link between a microscope quantity (the polarisability) and a macroscopic quantity (the refractive index).

LORENTZ, HENDRICK ANTOON (1853-1928): Dutch physicist and Nobel laureate, who was one of the founders of the electron theory of matter. He developed the electromagnetic theory of light and the electron theory of matter and formulated a consistent theory of electricity, magnetism and light. Lorentz shared the 1902 Nobel Prize with Pieter Zeeman (1865-1943) for discovering the Zeeman Effect. The Zeeman Effect is the splitting of single lines in a spectrum into two, three or more polarised lines when the source of the spectrum is placed in a magnetic field. Lorentz also solved one of the problems raised by the Michelson-Morley experiment and suggested the Lorentz transformations, both of which were used by Einstein in his special theory of relativity.

LOSCHMIDTS CONSTANT (LOSCHMIDT NUMBER): The number of particles per unit volume of an ideal gas at standard temperature and pressure (STP). Its symbol is n_0 and has a value of $2.686.763 (23) \times 10^{25} \text{ m}^{-3}$. It was first worked out by Joseph Loschmidt (1821-1895).

LOSSLESS COMPRESSION: A technique that reduces a file's size with no loss of quality. The decompressed file and the original are identical. For example, a file compressed using **lossy compression** may be one tenth the size of the original, but with the loss of some of the data or quality; while lossless compression is unlikely to produce a file smaller than half of the original size but maintains the original data or quality.

LOTIC: A term used to describe freshwater ecosystem in which there is a continuous flow of water, such as a river.

LOTKA-VOLTERRA MECHANISM: A simple chemical reaction mechanism proposed as a possible mechanism of oscillating reactions. The process involves a conversion of a reactant R into a product P . The reactant flows into the reaction chamber at a constant rate and the product is removed at a constant rate (the reaction is in a steady state but not in chemical equilibrium). The mechanism involves three steps: $R + X \rightarrow 2X$; $X_b + Y \rightarrow 2Y$; and $Y \rightarrow P$. The first two steps involve autocatalysis. The first step is catalysed by the reactant X and the second by the reactant Y . The kinetics of such a reaction can be calculated numerically, showing that the concentrations of both X and Y increase and decrease periodically with time. This results from the autocatalytic action. Initially, the concentration of X is small, but, as it increases, there is rapid increase in the rate of the first reaction because of the autocatalytic action of Y . This lowers the concentration of X and slows down step 1 so the concentration of X falls. Less X is now available for the second step and the concentration of X also starts to fall. With this full amount of Y , less X is removed and the first reaction again begins to increase. These processes are repeated leading to repeated rises and falls in the concentrations of both X and Y . The cycles are not in phase, peaks in the concentration of Y occurring later than peaks in X . In fact, known oscillating chemical reactions have different mechanisms to the above, but the scheme illustrates how oscillation may occur. This type of process is found in fields other than chemistry. They were investigated by the Italian mathematician Vito Volterra (1860 – 1940) in models of biological systems, for example, predator-prey relationships.

LOUDNESS: 1. A measure of the intensity of sound waves reaching the ear. Its unit is the decibel, symbol dB. Over 120 dB can damage human hearing. Loudness is dependent on frequency and duration. 2. The relative level of bass to treble in high-fidelity equipment that is adjusted depending on the overall volume level.

LOUDSPEAKER: An electromagnetic or electronic device that changes an electrical signal into sound waves providing audible sound in equipment such as radios, televisions, public-address systems, CD players, etc. The part of the speaker that converts electrical into mechanical energy is called the motor or voice coil. It vibrates a diaphragm that vibrates the air in immediate contact with it, producing a sound wave corresponding to the pattern of the original speech, music or other acoustic signal.

LOUPING-ILL: A viral disease of sheep carried by tick in hill pasture, occurring chiefly in the British Isles. It is characterised by acute fever and nervous twitch and staggering.

LOUSE: A small wingless insect of the order Anoplura that lives on the body and in the hair of humans and other mammals. There are two species: the body lice and head lice are of the same species, *Pediculus humanus*; and the crab or pubic lice belongs to *Phthirus pubis*. They feed on blood with their piercing and sucking mouthparts. The name also applies to several other small insects, such as the plant lice, book lice and bark lice. Body lice may transmit rickettsial diseases and bacterial infections such as relapsing fever. Infection results from scratching the crushed louse or its faeces into the skin. The chicken louse, *Menopon pallidum*, if left uncontrolled, can be a major problem in poultry production.

LOUSE DISEASE: An external parasitic disease of cattle. Severe infections lead to loss of condition, wasting and anaemia.

LOW BLOOD PRESSURE (HYPOTENSION): In medicine, an abnormally low blood pressure, usually characterised by dizziness, lightheadedness, lack of concentration, blurred vision, rapid and shallow breathing, fatigue, etc. Some of the causative factors include pregnancy, some heart conditions, anaemia, low blood sugar, dehydration, blood loss, severe infection such as septicemia (when an infection in the body enters the bloodstream). **Postural hypotension** refers to the sudden fall in blood pressure that can occur on standing up suddenly.

LOW ENERGY ELECTRON DIFFRACTION (LEED): A technique for the determination of the surface structure of crystalline materials that uses low-energy electrons (20-200eV), which are strongly diffracted by surface layers of atoms.

LOW FREQUENCY, LF (KILOMETRE BAND; KILOMETRE WAVE): Radio frequencies in the range of 30 kHz–300 kHz (30 to 300,000 Hz) used for atomic clock time signals, amateur radio in various parts of the world, submarine communications, ground-based navigation and other radio-based systems. Low frequencies have wavelengths in the range 1-10 Km.

LOW-PRESSURE CENTRE: Counterclockwise circulation centre, formed along a stationary front in the Northern Hemisphere. The centre of the spiral-shaped cloud system is called the 'eye' and usually where the pressure is lowest.

LOW PRESSURE ELECTRICAL DISCHARGE: Low pressure electrical discharge sources are radiation sources in which radiation is produced by electrical discharges in gases at low pressures, i.e. pressures from 10^2 to 10^{-3} Pa.

LOW TEMPERATURE UV-VIS ABSORPTION SPECTROSCOPY: The measurement of the absorption of UV-VIS radiation by a solute in a highly viscous or solid matrix at low temperature.

LOW WATER MARK (LOW-WATER MARK): The minimum point reached by a low tide, caused by the periodic tidal forces and the effects of meteorological, hydrologic and/or oceanographic conditions.

LOW-DENSITY LIPOPROTEIN (LDL): A plasma protein made up of fat (lipid) & protein that is the form in which lipids are transported in the blood and carries cholesterol from the liver to the tissues of the body. High levels of LDL contributes to plaque, a thick, hard deposit that can clog arteries and make them less flexible causing atherosclerosis.

LOW-DIMENSIONAL SYSTEM: A solid-state system in which the capital dimension is less than three. In practice, a two-dimensional system is a thin film or layer and a one-dimensional system is a thin wire. Two-dimensional systems have applications in semiconductor technology in such devices as MOSFETs. The behaviour of low-dimensional systems is of interest because the problems for low-dimensional systems (particularly one-dimensional systems) are much easier to solve than the corresponding problems in three dimensions. Clusters of atoms and very small crystals can be considered as zero-dimensional systems.

LOW-GAIN ANTENNA (LGA): A small antenna that provides low amplification of either transmitted or received radio signals. LGAs have wide antenna patterns and therefore do not require precise pointing.

LOW-INPUT FARMING: A system of farming based on restricted use of chemical fertilisers, pesticides and herbicides.

LOW-LEVEL LANGUAGE: In computing, any programming language such as assembly language that is designed to operate and manipulate the hardware and components of a computer without any interpretation or translation. They are closer to machine language than human language and reflect the machine codes of specific computers and are therefore, described as machine-oriented languages. They cannot be easily converted to run on a computer with a different central processing unit and are relatively difficult to learn because a detailed knowledge of the internal working of the computer is required.

LOW-PRESSURE MERCURY LAMP (ARC): Resonance lamp that contains mercury vapour at pressures of about 0.1 Torr (0.75×10^{-3} Torr; 1 Torr = 133 Pa). At 25 °C, such a lamp emits mainly at 253.7 and 184.9 nm. They are also called germicidal lamps. There are cold- and hot-cathode as well as cooled electrode-less (excited by microwaves) low-pressure mercury lamps. The Wood lamp is a low-pressure mercury arc with an added fluorescent layer that emits in the UV-A spectral region (315 – 400 nm).

LOW-PRESSURE SYSTEM: An area where the atmospheric pressure at the surface of earth is lower than its surrounding environment and from which winds blow in toward the centre to bring about equilibrium. In the Northern Hemisphere these winds blow in an anticlockwise direction. Low pressure produces unstable weather conditions like rain, storms or snow.

LOW-SPIN: In any coordination entity with a particular d_n ($1 < n < 9$) configuration and a particular geometry, if then electrons are distributed so that they occupy the lowest possible energy levels, the entity is a low spin complex. If some of the higher energy d orbitals are occupied before all the lower energy ones are completely filled, then the entity is a high-spin complex.

LOW-SPIN STATE: When the separation between the highest occupied and the lowest unoccupied molecular orbitals is not large, two alternative electronic states may be considered. The state with two electrons paired up in the HOMO is called a low-spin state. The low-spin state is the ground state when the one electron energy needed to promote an electron to the LUMO is larger than the Coulomb and exchange repulsion energies required to pair up two electrons in the HOMO.

LOW-TEMPERATURE PHYSICS: A study of the properties of gross matter at low temperatures, especially at temperatures so low that the quantum character of the substance becomes observable in effects such as superconductivity, superfluid liquid helium, magnetic cooling, and nuclear orientation.

LOW-TEMPERATURE SCANNING ELECTRON MICROSCOPY: The preparation of specimens for the scanning electron microscope and their subsequent examination, at very low temperatures. The specimen is frozen with liquid nitrogen (the melting point of nitrogen is -210 °C), coated at liquid nitrogen temperatures and examined in a special specimen chamber similarly cooled. In this way frozen hydrated material can be examined. The results are similar to those obtained by critical-point drying but with fewer artefacts. The technique is particularly useful for delicate plant specimens.

LOW-VELOCITY STAR: A star whose motion relative to the local standard of rest is small, indicating that it moves in an orbit about the galactic centre similar to that of the Sun.

LOWER BOUND: Minimum extreme value of a particular quantity or a number that is less than or equal to all of the elements of a given set. For example, 5, 6, 7 and 8 are lower bounds of the interval [8,9].

LOWER CRITICAL TEMPERATURE: The minimum body temperature that can be tolerated by an organism. Below this temperature, the biochemical properties of cell structure, especially membranes, are altered and reactions are slowed such that the

organism cannot maintain its usual bodily function and death may ensue. This varies greatly depending on the 'normal' temperature regime to which the organism is adapted. For example, plants native to warm regions such as maize and cotton are much more sensitive to chilling than plants from cooler regions.

LOWER DARBOUX SUM (LOWER SUM): The weighted sum of the products of the infimal values of a function on a succession of subintervals of a given interval with the lengths of the subinterval.

LOWER INTEGRAL (LOWER DARBOUX INTEGRAL): The limit, as the lengths of its subintervals tend to zero, of the lower sums of a function on that interval. It is one possible definition of the integral of a function. If it is equal to the upper integral then the function is Riemann integrable that a function is Darboux-integrable if and only if it is Riemann-integrable and the values of the two integrals, if they exist, are equal.

LOWER INVERSE IMAGE SET (WEAK INVERSE IMAGE SET): The set in which each element has an image under a given correspondence F with a nonzero intersection with a given set B expressed as $\{x:F(x)\cap B\neq\emptyset\}$, and denoted $F^{(B)}$ or $F^{\cap}(B)$.

LOWER LEVEL SET: For a given real-valued function, any lower level set is of the form $L(r) = \{x : f(x) \leq r\}$ or $L(r) = \{x : f(x) < r\}$.

LOWER LIMIT: The smaller of the two limits of integration of an interval over which a definite integral is taken.

LOWER PREDICATE CALCULUS, LPC (FIRST ORDER PREDICATE CALCULUS): A formalisation of predicate calculus in which quantification is only over individuals and not over classes or predicates.

LOWER SUM (LOWER DARBOUX SUM): The weighted sum of the products of the infimal values of a function on a succession of subintervals of a given interval with the lengths of the subinterval.

LOWER TRANSIT (LOWER CULMINATION): The point at which a celestial object reaches its minimum altitude on the observer's meridian.

LOWER TRIANGULAR: A matrix in which all elements above the

main diagonal are zero. For example, $M = \begin{bmatrix} 3 & 0 & 0 \\ 4 & 5 & 0 \\ 1 & 7 & 6 \end{bmatrix}$.

LOWERING OF VAPOUR PRESSURE: A reduction in the saturated vapour pressure of a pure liquid when a solute is introduced. If the solute is a solid, lowering of vapour pressure or decrease in vapour pressure of liquid is proportional to the concentration of particles of solute (the number of dissolved molecules or ions per unit volume). To a first approximation, it does not depend on the nature of the particles.

LORES: The space between the eye and the bill of a bird, or the eyes and tip of the mouth of any animal, especially a snake.

LOWEST COMMON MULTIPLE, LCM (LEAST COMMON MULTIPLE): The lowest number of which two or more positive whole numbers are factors. For example, the lowest common multiple of 2, 4 and 6 is 12.

LOWEST OBSERVED ADVERSE EFFECT LEVEL (LOAEL): Lowest concentration or amount of a substance, found by experiment or observation, which causes an adverse alteration of morphology, functional capacity, growth, development or life span of a target organism distinguishable from normal (control) organisms of the same species and strain under defined conditions of exposure.

LOWEST TERMS: Terms below which a fraction cannot be simplified because it is not possible to find a number that will divide into both the numerator and the denominator.

LOWEST UNOCCUPIED MOLECULAR ORBITAL (LUMO): The orbital in a molecule that has the lowest unoccupied energy level at the absolute zero of temperature. The lowest unoccupied molecular orbital and the highest molecular occupied molecular orbital (HOMO) are the two frontier orbitals of the molecule.

LOWEST-OBSERVED-ADVERSE-EFFECT-LEVEL (LOAEL):

Lowest concentration or amount of a substance, found by experiment or observation, which causes an adverse alteration of morphology, functional capacity, growth, development or life span of a target organism distinguishable from normal (control) organisms of the same species and strain under defined conditions of exposure. The difference between LOAEL and **lowest observed effect level** (LOEL) is based on the definition of adverse effect only, that is, an experimental study that produced a LOAEL will have stated the adverse effect to be observed before initiation.

LOWLANDS: An area of land where conditions are usually good for farming, as opposed to hills and mountains or highlands.

LOXODROME: 1. Also known as **loxodromic spiral**, on a sphere, it is a curve that cuts all parallels under the same angle. On Mercator projection map it appears as a straight line. 2. Also known as **rhumb line**, it is a steady course along one compass setting taken by a ship or aircraft. It is an arc crossing all meridians of longitude at the same angle, i.e. a path with constant bearing as measured relative to true or magnetic north. On a Mercator projection map, a rhumb line is a straight line.

LOZENGE: A rhombus with a 60° angle.

LPI (LINES PER INCH): A measure of the resolution of images printed in halftones. Because halftone images are printed as a series of dots, the higher the LPI number, the denser the dots, resulting in a finer resolution. Newspapers are typically printed in a resolution of 85 lpi, while magazines may use 133 lpi or higher. Because the naked eye can distinguish halftone dots up to about 120 lpi, the dots in a newspaper print may appear to the eyes more than that of magazines.

LU DECOMPOSITION: In mathematics, the LU decomposition of a matrix is the factorisation of a non-singular matrix A as LU where L and U are respectively lower and upper triangular. More generally, for any square matrix A one may write A as LPU with P a permutation matrix.

LUBRICANT: A substance such as grease or oil used between moving parts to reduce friction, wear and tear, overheating and rusting. A lubricant may also be used to prevent the deposition of solids on close-fitting parts. Oils derived from petroleum or made synthetically are the most widely used as they are relatively inexpensive, and are good coolants provided they are of appropriate range of viscosities and have stability.

LUBRICATION: The use of a substances such as grease or oil to reduce friction between surfaces or the use of a substance to prevent contact between solid surfaces in relative motion in order to reduce friction, wear, overheating and rusting.

LUCAS NUMBER: A member of the integer sequence 2, 1, 3, 4, 7, 11, 18, ..., where each number is the sum of the previous two numbers. This is similar to the Fibonacci rule of adding the latest two numbers to get the next, but the Lucas numbers start from 2 and 1 (in this order) instead of 0 and 1 for the (ordinary) Fibonacci numbers. It was named after the French mathematician Francois Edourd Lucas (1842-1891).

LUCERNE (ALFALFA): A perennial drought-resistant leguminous plant, *Medicago sativa*, belonging to the subfamily Papilionoideae, family Fabaceae (formerly Leguminosae). It is very nutritious and rich in proteins, minerals and vitamins and grown for fodder. It is mainly used for either green feed for animals or for hay or silage in temperate regions. It is also used as a soil-conditioning crop.

LUCITE: A trademark name of the synthetic organic compound polymethyl methacrylate, also known as perspex and plexiglass. Lucite is an extremely tough, lightweight, transparent thermoplastic used for auto and aircraft windshields, contact lenses, watch glasses, etc.

LUMBAR VERTEBRAE: The largest segments of the movable part of the vertebral column. They are vertebrae of the waist region, without ribs between the ribs, bearing thoracic and the sacral vertebrae. In mammals, they bear processes for the attachment of the back muscles. They are designated L1 to L5, starting at the top.

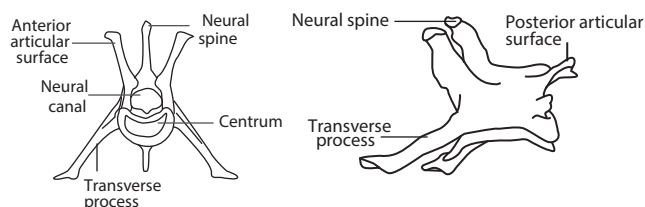


Fig. I: Lumbar vertebra of a mammal (anterior view)

Fig. II: Lumbar vertebra of a mammal (lateral view)

LUMBER: 1. Timber felled and sawn into planks and boards. 2. Situated in or relating to the loins.

LUMEN: 1. Space within a tube, sac or empty cell. 2. The hollow interior of the gut and the site of digestion in all vertebrates and many invertebrates or the space inside any tubular structure in the body such as an artery, vein or intestine. 3. The SI unit of luminous flux. It is equal to the amount of light crossing a unit area at a unit distance from a light source of luminous intensity of one candela. Its symbol is lm.

LUMINANCE: The measure of the brightness of a surface, which is equal to the amount of luminous flux arriving at or passing through a unit area of surface. It is measured in candelas per square metre. Its symbol is L.

LUMINESCENCE: The emission of light by sources other than a rise in its temperature or light emitted by sources other than a hot, incandescent body, such as a black body radiator. Luminescence is caused by the movement of electrons within a substance from more energetic states to less energetic states. In general, atoms of substances emit photons of electromagnetic energy when they return to the ground state after having been in an excited state. The causes of the excitation are various. If the excitation is caused by a photon, the process is called **photoluminescence**; if it is an electron, it is called **electroluminescence**. **Chemiluminescence** is luminescence resulting from a chemical reaction such as the slow oxidation of phosphorus; **bioluminescence** is the luminescence produced by a living organism such as glowworms, fireflies and various fungi and bacteria found on rotting wood or decomposing flesh. If the luminescence persists significantly after the exciting cause is removed it is called **phosphorescence**, if it does not it is called **fluorescence**. This distinction is arbitrary since there must always be some delay; in some definitions a persistence of more than 10 nanoseconds (10⁻⁸ s) is treated as **phosphorescence**.

LUMINOL TEST: A presumptive test for blood. The reagent is a mixture of 3-aminophthalhydrazide, sodium carbonate and sodium perborate. When sprayed with the reagent, traces of blood, even old blood, emit a faint chemiluminescence.

LUMINOSITY: 1. The brightness of a celestial body such as a star, galaxy, or other astronomical object defined as the amount of energy an object emits in a fixed time or the total energy radiated per unit time. Its S.I. units are joules per second or watts. A star's luminosity depends on its size and its temperature, varying as the square of the radius and the fourth power of the absolute surface temperature. 2. The luminous intensity in a particular direction or the apparent brightness of an image.

LUMINOUS: Describing something such as the Sun or a light bulb that gives out light or gives off light.

LUMINOUS EXITANCE: The ability of a surface to emit light expressed as the luminous flux per unit area at a specified point on the surface. Its symbol is M_v .

LUMINOUS FLUX: A measure of the rate of flow of light. Its symbol is Φ_v and is expressed in lumens. It is radiant flux in the form of electromagnetic waves which affects the eye or the time rate of flow of radiant energy evaluated according to its capacity to produce visual sensation.

LUMINOUS INTENSITY: A measure of the light emitting ability of a light source either generally or in a particular direction. It may be considered as the luminous flux on a small surface normal to the given direction, divided by the solid angle (in steradians) which the surface subtends at the source of light. Its S.I. base unit is candela (cd). Its symbol is I_v .

LUMINOUS MOSS (ELFIN-GOLD): A light-reflecting plant of the subclass Bryidae. It is native to the Northern Hemisphere and occurs in caves and dark holes in the woods. Its binomial name is *Schistostega pennata*.

LUMINOUS OBJECT: An object that produces its own visible light. An example is the sun.

LUMINOUS PAINT: Paint that glows in the dark because it contains a phosphor such as zinc sulfide and calcium sulfide. Phosphor is a substance that emits light for a certain length of time after exposure to an energy source, such as ultraviolet radiation. Some luminous paints contain a source of radiant energy, such as salts of radium. These paints glow indefinitely and are used for painting dials and signs to be read at night.

LUMINOUS QUANTITIES: A group of quantities characterising visible light in terms of the appearance to the human eye. The symbols of luminous quantities may be distinguished by adding a subscript v. For example, the symbol of luminous intensity is represented I_v . The units for luminous quantities are derived from the SI base unit candela.

LUMIPHORE (LUMINOPHORE): A part of a molecular entity (or atom or group of atoms) in which electronic excitation associated with a given emission band is approximately localised.

LUNAR ALPS: A mountain range on the Moon that is cut by a valley 150 km long; and named after the Alps of Europe. It can be viewed by using just a small backyard telescope. The highest peak in the range is Mont Blanc, which is about 3,660 metres high.

LUNAR ECLIPSE: A phenomenon in which the Sun, Earth and Moon are in a straight line and the shadow of the Earth falls on the Moon.

LUNAR PARALLAX: The difference, averaging 3422.5 arc-seconds, in the Moon's apparent celestial position as seen from the centre of Earth and when it lies on the horizon from an observer at the surface. Lunar parallax is directly determined from observations made at two places, such as G, Greenwich in England and C, the Cape of Good Hope, that are nearly on the same meridian.

LUNAR PHASES: Cyclically recurring appearances of the Moon due to differing amounts of sunlight reflected by the Moon toward Earth during the Moon's orbit. The eight phases are: **new moon, waxing crescent, first quarter, waxing gibbous, full moon, waning gibbous, last quarter and waning crescent**. **New moon** is when the sunlit (daytime) side of the Moon is facing away from Earth. **Full moon** is when it is facing toward the Earth and thus directly opposite the Sun in the sky.

LUNATE: Having a crescent shape.

LUNATION: The time that elapses between successive new Moons, averaging 29.530589 days or 29 days, 12 hours, 44 minutes, called a lunar month. In a lunar calendar each month corresponds to a lunation. Individual lunations vary in length by several hours, because of the eccentricity of the orbits of the Moon and Earth. However, the lunation is never more than 15 hours behind or ahead of a mean lunation.

LUNE: 1. The portion of a sphere between two great semicircles having common endpoints. 2. A crescent-shaped figure formed on a plane surface by the intersection of the arcs of two circles.

LUNG: The spongy sac-like elastic respiratory organ in the chest cavity, which together with the heart, work to remove carbon dioxide from the blood and provide it with oxygen. They are found in most vertebrates excluding most fish, which use gills and a few amphibian species that respire through the skin. In mammals they are two organs in the thorax, one on each side of the heart which can be expanded or compressed by movements of the thorax in such a way that air is continually taken in and expelled. Each lung consists essentially of a thin moist membrane that is folded to increase its surface area.

Exchange of oxygen and carbon dioxide occurs between blood capillaries on one side of the membrane and air on the other. Air enters the body through the nose or mouth and travels to the chest cavity through the trachea. The trachea divides into two bronchi, each of which enters a lung. In mammals and reptiles the membrane of the lung takes the form of numerous air sacs that are connected to the bronchia via bronchioles. The lungs themselves contain no muscular tissue and are ventilated by respiratory movements, the mechanism which varies with the species.

LUNG BOOK: The respiratory organ of some arachnids such as scorpions and primitive spiders, which occurs in pairs on the ventral side of the abdomen. They consist of leaf-like folds of ectoderm sunk into pockets having slit-like opening at the abdominal surface. The outer surface is surrounded by air while blood circulates within the inner layers. Gaseous exchange occurs by diffusion of air through the ectodermal folds.

LUNG CANCER: Cancer that affects the lung; it is the abnormal growth of cells in one or both lungs. It is caused mainly by smoking cigarettes over a long period; although exposure to carcinogenic substances such as asbestos and radiation is also a causative factor. Treatment is by chemotherapy, radiotherapy and surgery.

LUNGWORM: A parasitic nematode worm of the order Strongylida, which infests the lungs, causing bronchitis or pneumonia.

LUPIN (LUPINES): A genus of perennial leguminous plant, a member of genus *Lupinus*, family Papilionaceae. It is grown as a crop for protein and seed oil.

LUPUS: A chronic inflammatory disease that is caused by autoimmunity. Patients with lupus have in their blood unusual antibodies that are targeted against their own body tissues causing damage to joints, skin, blood vessels and organs. There are many kinds of lupus. The most common type, systemic lupus erythematosus (SLE), affects many parts of the body and is most prevalent among young women. It causes a characteristic skin rash with butterfly distribution over the face. It also causes lung, kidney and blood disease, largely due to abnormal immunity directed against substances in cell nuclei. Treatment is with steroids and immunosuppressives.

LUSH VEGETATION: Vegetation which is thick and green.

LUSIN'S THEOREM (LUZIN'S THEOREM): Named after the Russian mathematician, Nikolai Nikolaevitch Luzin (1883- 1950), it is the theorem that every measurable function is a continuous function on nearly all its domain except on a set of arbitrarily small measure: given such a function f and $\delta > 0$, there exists a continuous function g such that $f(x) = g(x)$ except on a set of measure less than δ .

LUSTRE: 1. The amount and quality of light reflected from the surface of a mineral. 2. A fabric with a glossy surface or sheen. 3. A thin coating containing unoxidised metal which gives an iridescent glaze to ceramics.

LUTEAL PHASE (SECRETORY PHASE): The phase of the oestrus cycle that occurs after ovulation. It begins with the formation of the corpus luteum and ends in either pregnancy or luteolysis. During this phase the corpus luteum secretes oestrogen and progesterone which prepare the uterine wall to receive a fertilised egg.

LUTEINIZING HORMONE, LH (INTERSTITIAL-CELL-STIMULATING HORMONE; ICSH): A hormone, secreted by the anterior pituitary in mammals that stimulates Leydig cell production of testosterone by the interstitial cells of the testes in males. In females, an acute rise of LH triggers ovulation and development of the corpus luteum.

LUTESCENT (LUTEOUS): Yellowish in colour.

LUTETIUM: A silvery metallic element belonging to the lanthanoids (rare-earth metals) and the last member of the lanthanide series in Group 3 (IIIB) of the Periodic Table. Its symbol is Lu, atomic number 71, atomic weight 174.967, melting point about 1663 °C (1936 K) and boiling point about 3395 °C (3668 K). There are two natural isotopes, lutetium-175 (stable) and lutetium-176 with a half life of 2.2×10^{10} years. The metal may be prepared by reduction of the chloride or fluoride with an alkali or alkaline earth metal. The element is used as a catalyst. The irradiated metal is beta-particle emitter and it is used in catalytic processes in the nuclear industry.

LUX: The SI unit of illuminance. It is equal to the illumination produced by a luminous flux of 1 lumen distributed uniformly over an area of 1 square metre. Its symbol is lx.

LUXURY UPTAKE: The absorption of nutrients in excess of what is required by an organism for its normal growth.

LYASES: Enzymes that catalyse either the cleavage of a double bond and the addition of new groups to a substrate or the formation of double bond. In other words, they are enzymes that catalyse the removal of a group to form a double bond or the reverse reaction. Lyases differ from other enzymes in that they require only one substrate for the reaction in one direction, but two substrates for the reverse reaction.

LYATE ION: The conjugate ion formed by removing a hydrogen ion or proton called **deprotonation** from a molecule of a solvent. For example, a hydroxide ion is formed from the deprotonation of water. **Methoxide** (CH_3O^-) is the anion formed by the deprotonation of methanol.

LYCOPENE: A red carotenoid pigment with the molecular formula $\text{C}_{40}\text{H}_{56}$, molar mass $536.89 \text{ g}\cdot\text{mol}^{-1}$, density 0.889 g/cm^3 , melting point $172\text{-}173^\circ\text{C}$ ($445\text{-}446 \text{ K}$) and a boiling point of 660.9°C (934.0 K) at 760 mmHg. It is not soluble in water and polar solvents such as methanol and ethanol; but soluble in solvents such as acetic acid, acetone, benzene and vegetable oil. It occurs naturally in tomatoes, in many berries and other red fruits and vegetables, such as red carrots, watermelons and papayas and is responsible for their red colour. It is the main source of all the natural carotenoids, including beta-carotene, which is responsible for yellow orange or red pigmentation, photosynthesis and photo-protection in plants, photosynthetic bacteria, fungi and algae.

LYCOPERDALES: An order of basidiomycetes comprising fungi such as the puffballs having a fleshy often globose fruiting body filled at maturity with a mass of dustlike spores.

LYCOPHYTA (LYCOPODOPHYTA): A phylum of Tracheophyta containing the club mosses and similar genera, including numerous extinct forms, which reached their peak in the carboniferous period with giant coal-forming tree species. Lycophytes have roots and their stems are covered with numerous small leaves. Reproduction is by means of spore and, the sporangia are usually grouped into cones.

LYDDITE: A powerful explosive used in shells. It consists mainly of picric acid mixed with 10 percent nitrobenzene and 3 percent petroleum jelly.

LYME DISEASE: A form of arthritis that may affect (if not treated) all the systems of the body such as the joints, nervous system, heart, liver, kidneys and eyes. It is caused by the bacterium *Borrelia burgdorferi* and transmitted by the bite of deer ticks (black-legged ticks) - type of ticks that are parasitic on deer.

LYMPH: The colourless interstitial fluid found within the lymphatic system of the human body. It enters the lymph vessels by filtration through pores in the walls of capillaries. It resembles blood plasma, consisting mainly of water and dissolved salts and proteins. The lymph travels to at least one lymph node, eventually entering the blood stream near the heart. It transports nutrients and white blood cells to the tissues and waste matter away from them; picks up bacteria and brings them to lymph nodes to be destroyed; and transports fats from the digestive system. The lymph that leaves a lymph node is richer in lymphocytes and the lymph formed in the digestive system called **chyle** is rich in triglycerides (fat) and looks milky white.

LYMPH GLAND: Glands that secrete disease-fighting materials for the body.

LYMPH HEART: Any of a series of muscular pumping organs occurring in the major lymphatic vessels of certain lower invertebrates such as lungfishes, all amphibians, reptiles and flightless birds, which maintain a flow of lymph around the body by pumping leaked lymph back into the circulatory system. In animals without lymph heart, the flow of lymph is maintained by a series of muscle contraction and valve.

LYMPH NODE: A small circular ball-shaped organ of the immune system, found throughout the body, many of them occurring at intervals along the lymphatic system. They filter out bacteria and other foreign particles, preventing them from entering the blood stream and causing infection. The lymph nodes also produce lymphocytes, which are important in the proper functioning of the immune system. In humans, major lymph nodes occur in the neck, under the arms and in the groin.

LYMPH VESSEL: A vessel that carries fluid only away from tissues, unlike a blood vessel. This fluid is a pale, watery substance known as lymph.

LYMPHATIC SYSTEM: A system of fluid-containing vessels including the lymphoid tissue and lymphatic vessels in vertebrates, which return lymph to the general blood circulation. It also includes all the structures dedicated to the circulation and production of lymphocytes, including the spleen, thymus, bone marrow and the lymphoid tissue associated with the digestive system. In most vertebrates lymph is pumped by cycles of contraction and relaxation of the lymphatic vessels and also by the action of adjoining muscles. **Lymphatic organ** is an organ such as the thymus, spleen, tonsils and lymph nodes that is characterised by clusters of lymphocytes and other cells, such as macrophages, enmeshed in a framework of short, branching connective tissue fibres.

LYMPHOCYTE: A type of white blood cell (leucocyte) in the vertebrate immune system with a large nucleus and little cytoplasm produced in the bone marrow. It makes antibodies and helps to protect against infection. The three major types of lymphocytes are B lymphocytes (B cells), which produce circulating antibodies and are responsible for humoral immunity; T lymphocytes (T cells), which are responsible for cell-mediated immunity; and NK (natural killer) cells that attack and destroy invading microbes and tumour cells, without requiring prior exposure to them.

LYMPHOCYTOPENIA (LYMPHOPENIA): A condition in which the blood has an abnormal low number of white blood cells or lymphocytes. Low levels of lymphocytes resulting in lymphocytopenia may be caused by acquired diseases such as infectious diseases, for example, AIDS, viral hepatitis, tuberculosis and typhoid fever; autoimmune disorders such as lupus; blood cancer and other blood diseases such as Hodgkin's disease and aplastic anaemia; radiation and chemotherapy. It may also be caused by certain hereditary diseases such as lymphocytopenia DiGeorge anomaly, Wiskott-Aldrich syndrome, severe combined immunodeficiency syndrome and ataxia-telangiectasia.

LYMPHOID TISSUE: Tissues, cells and organs, which compose the immune system found in the bone marrow, lymph nodes, tonsils, spleen and thymus. They are responsible for producing lymphocytes contributing to the body's defence against infection.

LYMPHOKINE: Any of a group of cytokines that are secreted by type of immune cells known as lymphocytes, typically T helper cells, which play a role in cell-mediated immunity by influencing the activity of other cell types. Examples include the interleukins (2, 3, 4 and 5), macrophage-activating factor and macrophage migration inhibition factor.

LYMPHOMA: A malignant proliferation of lymph tissue, usually in the lymph nodes, spleen or gastrointestinal tract. There are two basic categories of lymphomas: Hodgkin lymphoma, which is marked by the presence of a type of cell called the Reed-Sternberg cell; and non-Hodgkin lymphoma, which includes a large, diverse group of cancers of immune system cells.

LYNCHET (STRIP LYNCHETS): A strip of land formed as a result of the movement of soil down a slope due to ploughing over a long period of time. The disturbed soil slips down the hillside to create a positive lynchet and the area reduced in level becomes a negative lynchet.

LYONIUM ION: The ion or cation formed by the protonation or the addition of a proton (H^+) to a solvent molecule. For example, a hydronium or hydroxonium ion (H_3O^+) is the cation formed from the protonation of water and protonated methanol ($CH_3OH_2^+$) is the cation formed by the protonation of methanol.

LYOPHILIC: The tendency of a surface or functional group to become wetted or solvated or having an affinity for a solvent. If the liquid medium is aqueous such as water, the term **hydrophilic** is used.

LYOPHILIC SOLS: Association colloids in which aggregates of small molecules are formed reversibly or macromolecules in which the molecules themselves are of colloidal size. Lyophilic sols may be prepared by warming the solid with liquid dispersion medium, for example, starch with water.

LYOPHILISE: To preserve food by freezing it rapidly and drying it in a vacuum to remove moisture before returning it to room temperature. This is done by freezing the material below its eutectic point and then providing the latent heat of sublimation. The low processing temperature and absence of liquid water help to retain colour, flavour and texture.

LYOPHOBIC: Not having affinity for liquid or solvent. The term is used to describe a finely dispersed solid colloid that forms an unstable dispersion. Lyophobic colloids are irreversible in nature. Once precipitated, they do not give back the colloidal sol by simple addition of the dispersion medium. Lyophobic sols may be prepared by any of the following two methods: condensation or aggregation and dispersion method. **Hydrophobic** is not having affinity for water.

LYOTROPIC MESOMORPH: An arrangement taken by micelles formed from surfactant molecules in concentrated solutions. It consists of long cylinders in a fairly close-packed hexagonal arrangement. Lyotropic mesomorphs are sometimes called **liquid crystalline phases** for micelles.

LYRATE LEAF: A simple leaf that is pinnately divided into several lobes with an enlarged terminal lobe and smaller lateral lobes. The smallest lobes are at the base.



Lyrate leaf

LYSENKOISM: The official Soviet science policy governing the work of geneticists in the (USSR) from about 1940-1960, which opposed traditional biology. It was founded by T. D. Lysenko, of the Institute of Genetics of the Soviet Academy of Science. This policy dismissed all the advances that had been made in classical genetics; denying the existence of genes and held that the variability of organism was produced solely by environmental changes. For example, he declared that domesticated plants are transformed into weeds by the hostile environment of fields, which had not been properly cultivated.

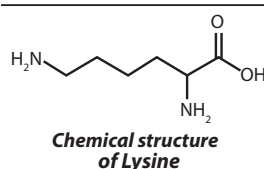
LYSERGIC ACID DIETHYLAMIDE (LSD; LSD-25; LYSERGIDE):

A chemical derivative of lysergic acid that has psychological effects and also provokes extreme mental disturbance when taken, even in extremely small quantities. Its chemical formula is $C_{20}H_{25}N_3O$. It occurs in the cereal fungus ergot and is classified as an ergot alkaloid. LSD acts as an antagonist at serotonin receptors. It is non-addictive.

LYSIGENY: The formation of a space in which secretions accumulate by the destruction of plant cells by enzymatic dissolution. Examples are the oil cavities in the leaves of citrus trees.

LYSIMETER: A device consisting of a vessel containing local soil placed with its top flush with the ground surface for the study of several phases of the hydrological cycle, such as infiltration, runoff, evapotranspiration, soluble constituents removed in drainage, the rate of drainage of water through soil and the soluble particles removed in the process. Also, water evaporated from the plants can be determined by the use of the lysimeter.

LYSINE (DIAMINOCAPROIC ACID): An essential amino acid with the molecular formula $C_6H_{14}N_2O_2$ that occurs in proteins. It is responsible for the basic properties of proteins because of its two $-NH_2$ groups. It is not synthesised in animals and must be ingested as lysine or lysine-containing proteins. In plants and bacteria, it is synthesised from aspartic acid.



LYSIS: The destruction of a living cell caused often by viral, enzymic or osmotic mechanisms. This may be effected by lysosomes or lymphocytes, either as part of the normal metabolic process or as a reaction against invading cells.

LYSOGENIC BACTERIA: Bacteria carrying a bacteriophage integrated into the bacterial chromosome. The virus may subsequently set up an active cycle of infection, causing lysis of the bacterial cells.

LYSOGENIC CYCLE: The period in which a bacteriophage enters and replicates within a host cell such that the host is not destroyed until the lytic cycle is activated. It is one of the two cycles of viral reproduction, the other being the lytic cycle. In the lysogenic cycle, the bacteriophage lies dormant within the host's chromosome until certain stimuli such as environmental conditions revive it and initiate the lytic cycle. In the lytic cycle, the infection of the bacteria is immediately followed by the taking over of the bacterium's machinery to synthesise capsule's proteins and replicate the virus genome and lyse of the bacterium.

LYSOGENY: The biological process in which a bacterium is infected by a bacteriophage that integrates its DNA into that of the host such that the host is not destroyed until the lytic cycle is activated. The bacteriophage lies dormant within the host's chromosome until the host is exposed to certain stimuli such as environmental conditions

revive it and initiate the lytic cycle. A bacterium whose chromosome has been integrated with the DNA of a bacteriophage is called a **lysogene**. The latent form of a bacteriophage in which the viral genes are incorporated into the bacterial chromosomes without causing disruption of the bacterial cell is known as a **prophage**.

LYSOSOME: A membrane-enclosed organelle found mainly in animals that contains several dozen different species of digestive enzymes, each of which can sever particular chemical bonds found in natural materials. They play a part in digestion and in the breakdown of the cell's structure when it dies.

LYSOZYME: An antibacterial enzyme found in body fluid and secretions such as tears, saliva, egg-white and mucus. It catalyses the destruction of bacterial cell walls by hydrolysis and thus has a bactericidal effect. It was first identified and named by Sir Alexander Fleming (1881-1955), British bacteriologist and Nobel laureate, who is best known for the discovery of penicillin.

LYSSAVIRUS: Any virus of the genus Lyssavirus in the family Rhabdoviridae that includes the rabies virus. Hosts include Humans, mammals and vertebrates.

LYTIC CYCLE: A process in which a bacteriophage enters and replicates within a host cell to produce new viral particles eventually causing lysis of the cell. It is one of the two cycles of viral reproduction, the other being the lysogenic cycle. In the lytic cycle, the infection of the bacteria is immediately followed by the taking over of the bacterium's machinery to synthesise capsule's proteins and replicate the virus genome and lyse of the bacterium. In the lysogenic cycle, the virus leaves its genetic material dormant inside the cell as the cell reproduces until environmental conditions revive it and initiate the lytic cycle.

LYTIC INFECTION: An infection of a bacterium by a bacteriophage yielding progeny virus particles that leads to the lysis of the host cell.

LZW COMPRESSION (LEMPER-ZIF-WELSH COMPRESSION): Data compression system used to reduce the size of data files by using a lossless look-up algorithm and thus reducing the time needed to transmit them. It was developed in 1977 by Abraham Lempel (1936-), an Israeli computer scientist, Jacob Ziv (1931) an Israeli electrical engineer and Terry Welch, American computer scientist; and named after them. LZW compression is used in the graphics interchange format (GIF), the tagged image file format (TIFF), portable document file (PDF), etc.

ADVERT SPACE

Contact

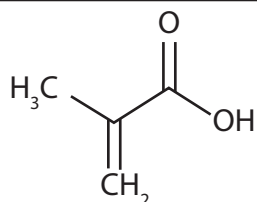
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M

2-METHYL-4-CHLOROPHENOXYACETIC ACID (MCPA): A synthetic auxin of the phenoxyacetic acid group that is used as a selective weedkiller.

2-METHYLPROPENOIC ACID, MAA (METHACRYLIC ACID):

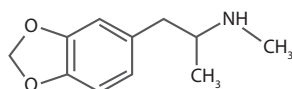
White viscous liquid crystalline unsaturated carboxylic acid with the chemical formula $C_4H_6O_2$, density 1.015 g/cm^3 , melting point about 15°C (287 K) and boiling point 161°C (434 K). It is soluble in warm water and miscible with most organic solvents. It occurs naturally in small amounts in the oil of Roman chamomile; and produced industrially on a large scale as a precursor to its esters, especially methyl methacrylate. It is used in making methacrylate resins.



Chemical structure of
2-methylpropenoic Acid

3,4-METHYLENEDIOXY-METHAMPHETAMINE, MDMA;

(ECSTASY; TENAMFETAMINE): A synthetic drug, with the chemical formula $C_{11}H_{15}NO_2$, initially intended as an appetite suppressant, which produces a feeling of euphoria and hallucination. It is derived from the amphetamine family and a relative of the stimulant methamphetamine. Apart from increasing the amount of the neurotransmitter dopamine, it also increases the production of the neurotransmitter serotonin and blocks its re-absorption in the brain. It stimulates the central nervous system and gives users a feeling of increased energy and lower social inhibition. It is now an illegal drug used as a stimulant and relaxer of inhibitions.



Chemical Structure
of Ecstasy

m: 1. Symbol for milli, which is a prefix in the S.I. system for a factor of one thousandth (10^{-3}). 2. Symbol for metre, the S.I. unit for length.

M: 1. The symbol for Mega, a unit prefix in the metric system denoting a factor of one million (10^6 or 1 000 000). 2. The symbol for the number 1000 in the Roman numerals.

M MATRIX: A matrix M of the form $M = \lambda I - P$, where P is a matrix with positive entries for $\lambda \geq \sigma$, the spectral radius of P .

M PHASE: The cell division phase of the cell cycle, which includes mitosis and meiosis.

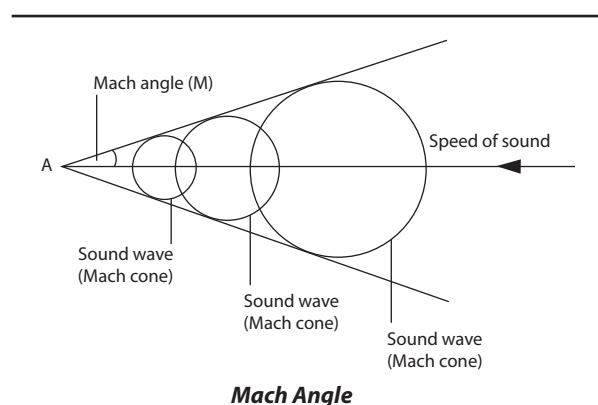
M-COMMERCE (MOBILE COMMERCE): Conducting commercial activities such as buying and selling of goods and services, online banking and bill payment through wireless handheld devices such as laptops, cellular telephone and personal digital assistants (PDAs).

MAC ADDRESS (MEDIA ACCESS CONTROL ADDRESS): A hardware identification number that uniquely identifies each device on a network. The MAC address is manufactured into every network card, such as an Ethernet card or Wi-Fi card and therefore cannot be changed.

MAC OS: The operating system that runs on Apple's Macintosh line of personal computers and servers. It has been continually updated and many new features have been added to it. Each major OS release is signified by a new number, for example, Mac OS 8, Mac OS 9, etc. Mac OS X is the current version of the operating system used on Apple Macintosh computers.

MACERATION: 1. The softening of a solid by soaking. 2. In histology, it refers to the softening of a tissue by soaking, especially in acids, until the connective tissue fibres are dissolved so that the tissue components can be teased apart. 3. In obstetrics, it refers to the degenerative changes with discolouration and softening of tissues and eventual disintegration, of a foetus retained in the uterus after its death.

MACH ANGLE: The angle made by a shock wave with the direction of motion as determined by the velocity of the object and the velocity of the shock propagation. It is the sine ratio of the speed of sound to the speed of a moving body: $\sin \mu = \frac{a}{v} = \frac{1}{M}$, where a is the speed of the sound wave, v is the speed of the moving body and M is the mach angle.



MACH NUMBER: A dimensionless number, which is the ratio of the relative speeds of an object through a fluid such as gas (for example, air) or liquid and a rigid body to the speed of sound in that fluid under the same conditions of temperature and pressure. It was named after Austrian physicist and philosopher Ernst Mach. It can also be defined as the speed of a projectile or aircraft with reference to the ambient atmosphere, divided by the speed of sound in the same medium and under the same conditions; expressed as: $M = V_s / u$, where M is the Mach number, V_s is the speed of the object relative to the medium and u the speed of sound in the medium. If the mach number is less than 1 the speed is **subsonic**; if it exceeds 1 the speed is **supersonic**; and if it exceeds 5 it is said to be moving at **hypersonic** speed. Mach number is a dimensionless quantity rather than a unit of measure. Expressing speeds of an aeroplane moving in air by Mach number rather than by kilometres per hour, is a more accurate representation of the actual conditions encountered in flight can be obtained.

MACH'S PRINCIPLE: A hypothesis that the inertia of any particular piece of matter is determined by the interaction between that piece of matter and the rest of the universe. A body in isolation would have zero inertia. This principle was stated by Ernst Mach in the 1870s and was made use of by Einstein in his general theory of relativity.

MACHAERANTHEROID: 1. Involucral bracts with recurved tips. 2. Belonging to the genus Machaeranthera.

MACHINE: A device that performs useful work by transmitting, modifying or transforming motion, forces and energy or a device that allows a small force called the **effort** to overcome a larger one called the **load**. A force applied at one point gives an output force elsewhere. A simple machine is a device that transforms the direction or magnitude of a force without consuming any energy. Examples of